# The Digestive System and Metabolism



College of Nursing
President Ramon Magsaysay State University

# PRAYER BEFORE THE START OF THE CLASS

We praise you Lord and we thank you
for the opportunity to study
and prepare for a good life in the future.
Bless our teachers who inspire and guide us.
Bless our parents who work hard to support us.
Bless our classmates and all the people
who care for us.
Bless us, help us to be more attentive, patient and diligent to
understand the lessons that our teachers teach us.
This we pray in the name of Jesus Christ our Lord.
Amen.

# COPYRIGHT

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ANATOMY AND PHYSIOLOGY
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# OUTLINE OF TOPICS:



- Organs and Functions of the Digestive System
- Trunks or Layers of the Digestive Tract
- Activities and Movements in the Digestive System
- Secretions in the Digestive System
- Metabolism

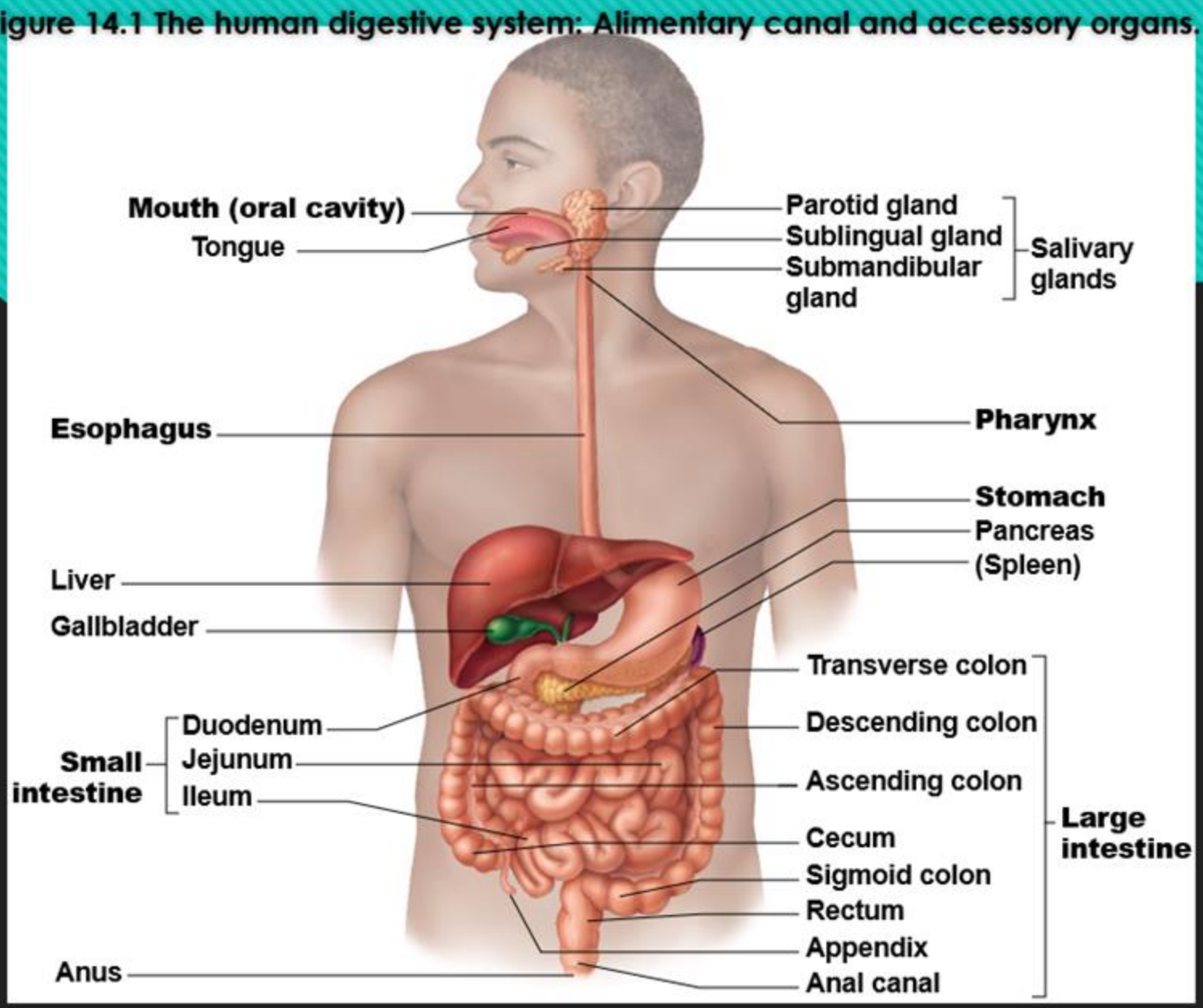
# Anatomy of the Digestive System

- Two main groups of organs
  - Alimentary canal /Gastrointestinal(GI) tract
    - continuous, coiled, hollow tube that runs through the ventral cavity from stomach to anus
    - these organs ingest, digest, absorb, defecate
  - Accessory digestive organs
    - Include teeth, tongue, and several large digestive organs
    - Assist digestion in various ways

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# Organs of the Alimentary Canal

- Mouth
- Pharynx
- Esophagus
- Stomach
- Small intestine
- Large intestine
- Anus




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# Mouth

## Anatomy of the Mouth

- Mouth (oral cavity)—mucous membrane–lined cavity
- Lips (labia)—protect the anterior opening
- Cheeks—form the lateral walls
- Hard palate—forms the anterior roof
- Soft palate—forms the posterior roof
- Uvula—fleshy projection of the soft palate

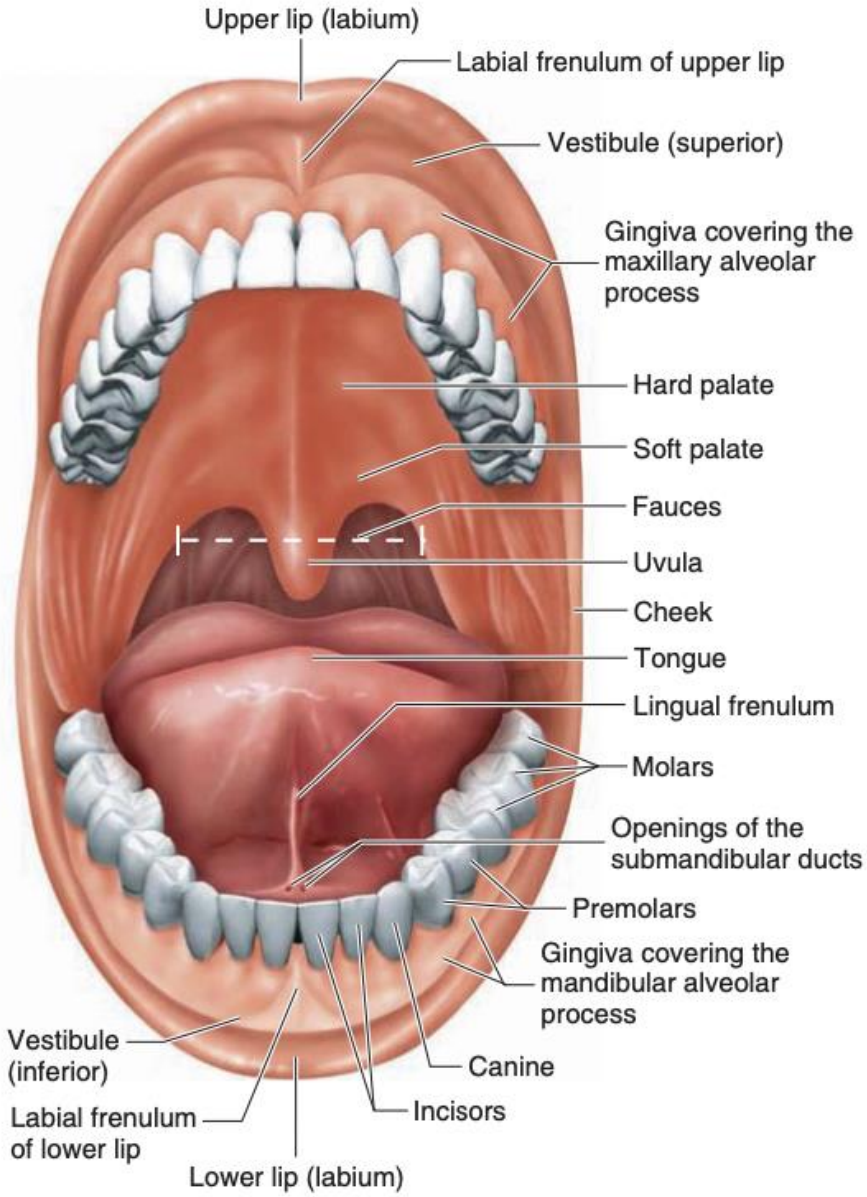


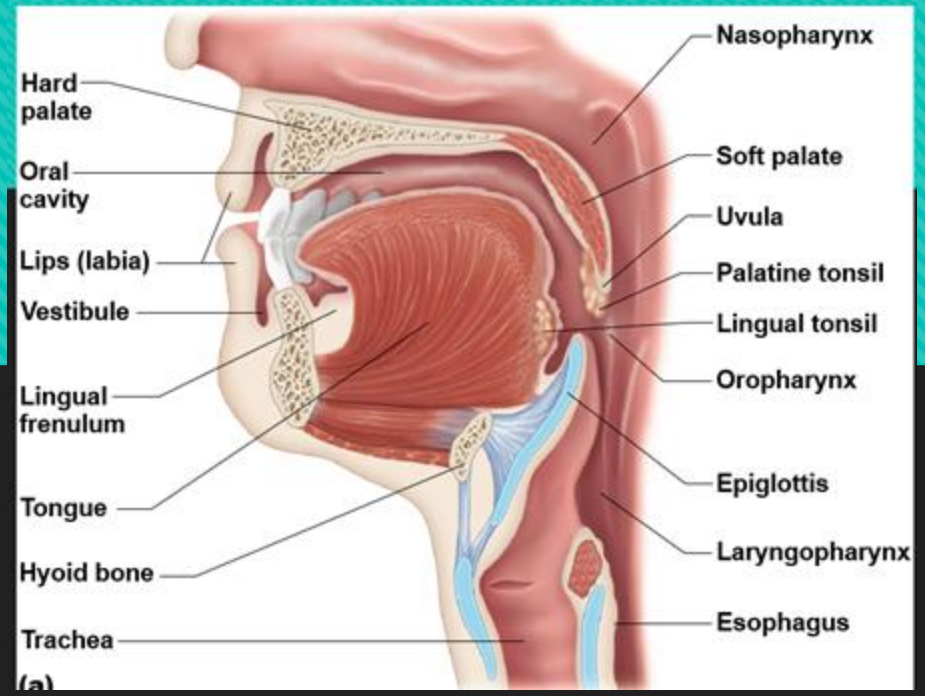

**FIGURE 24.6** Oral Cavity AP|R

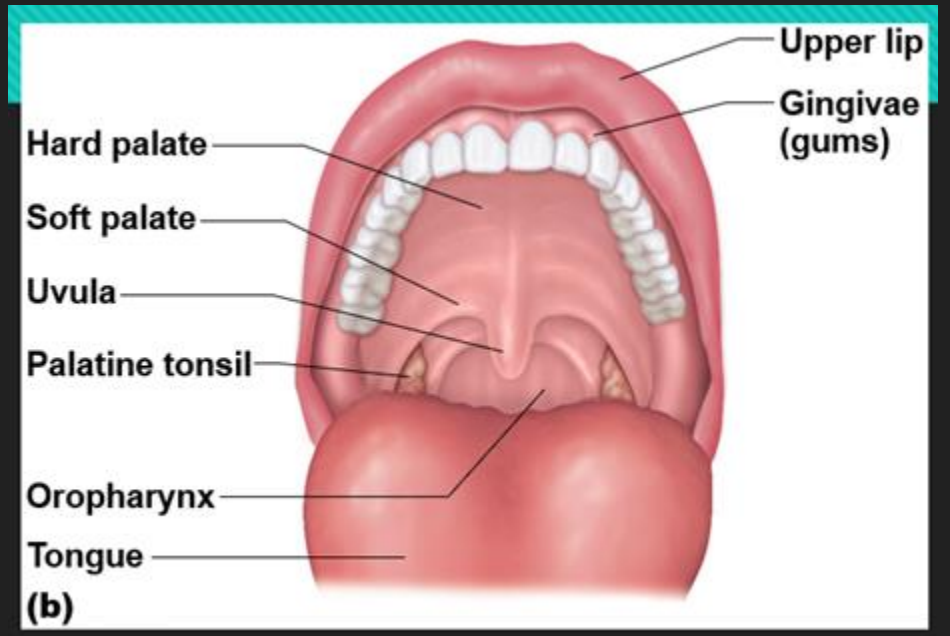
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# Mouth
## Anatomy of the Mouth

- Vestibule—space between lips externally and teeth and gums internally
- Oral cavity proper—area contained by the teeth
- Tongue—attached at hyoid bone and styloid processes of the skull, and by the lingual frenulum to the floor of the mouth
- Tonsils
  - Palatine—located at posterior end of oral cavity
  - Lingual—located at the base of the tongue

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# Mouth
## Functions of the Mouth

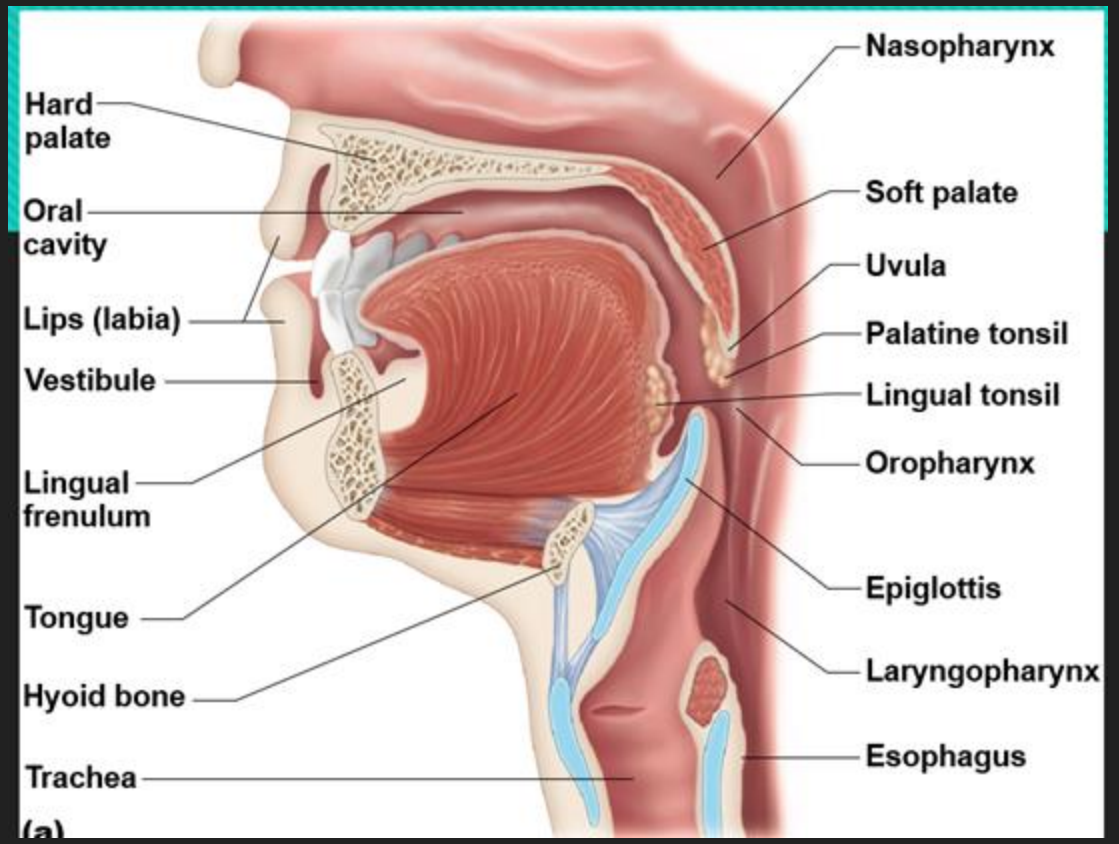
- Mastication (chewing) of food
- Tongue mixes masticated food with saliva
- Tongue initiates swallowing
- Taste buds on the tongue allow for taste – papillae on taste buds

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# Pharynx

- Serves as a passageway for foods, fluids, and air
- Food passes from the mouth posteriorly into the:
  - Oropharynx—posterior to oral cavity
  - Laryngopharynx—below the oropharynx and continuous with the esophagus

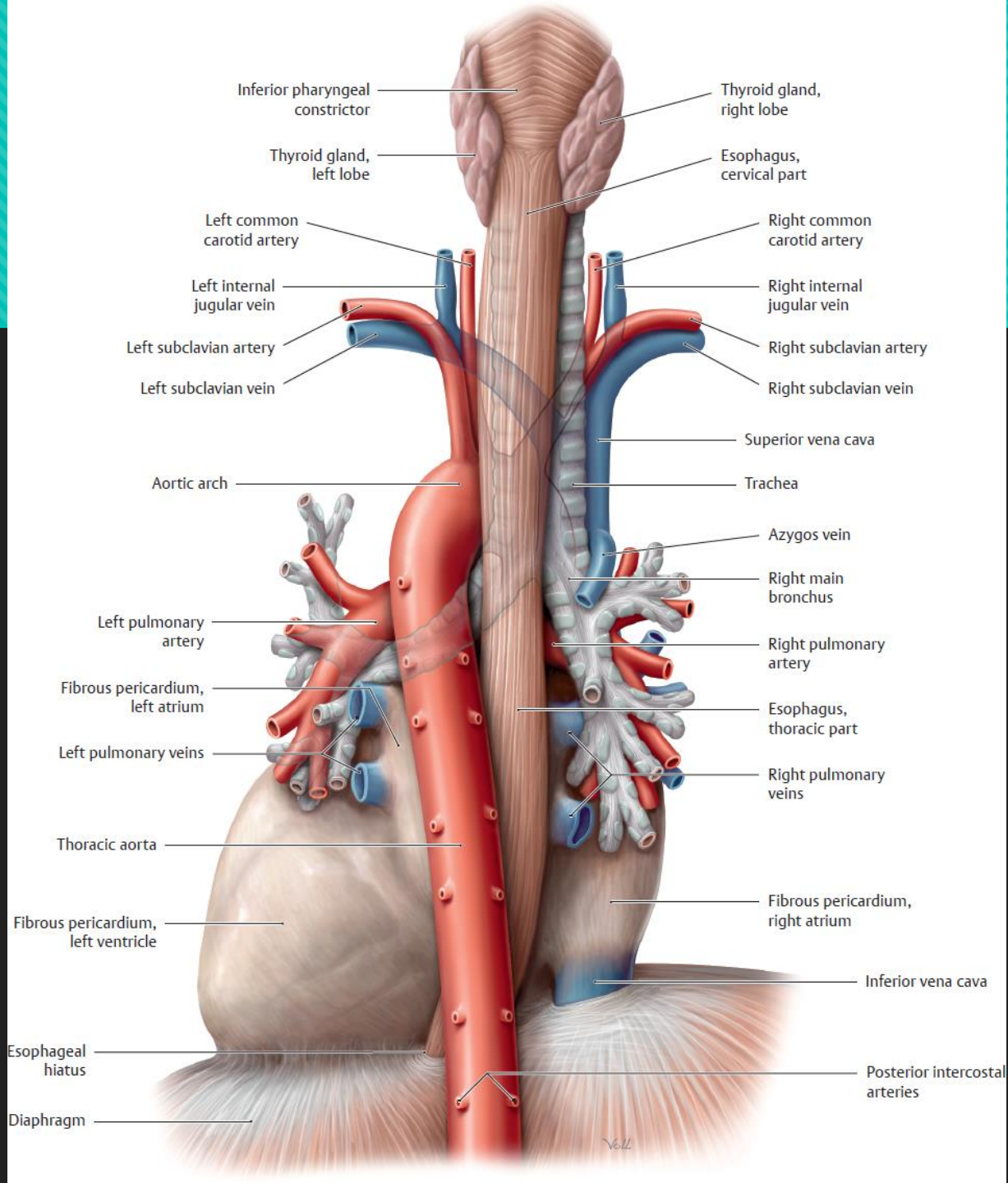
- Food is propelled to the esophagus by two skeletal muscle layers in the pharynx
  - Longitudinal outer layer
  - Circular inner layer
- Alternating contractions of the muscle layers and slow rhythmic squeezing (peristalsis) propel the food



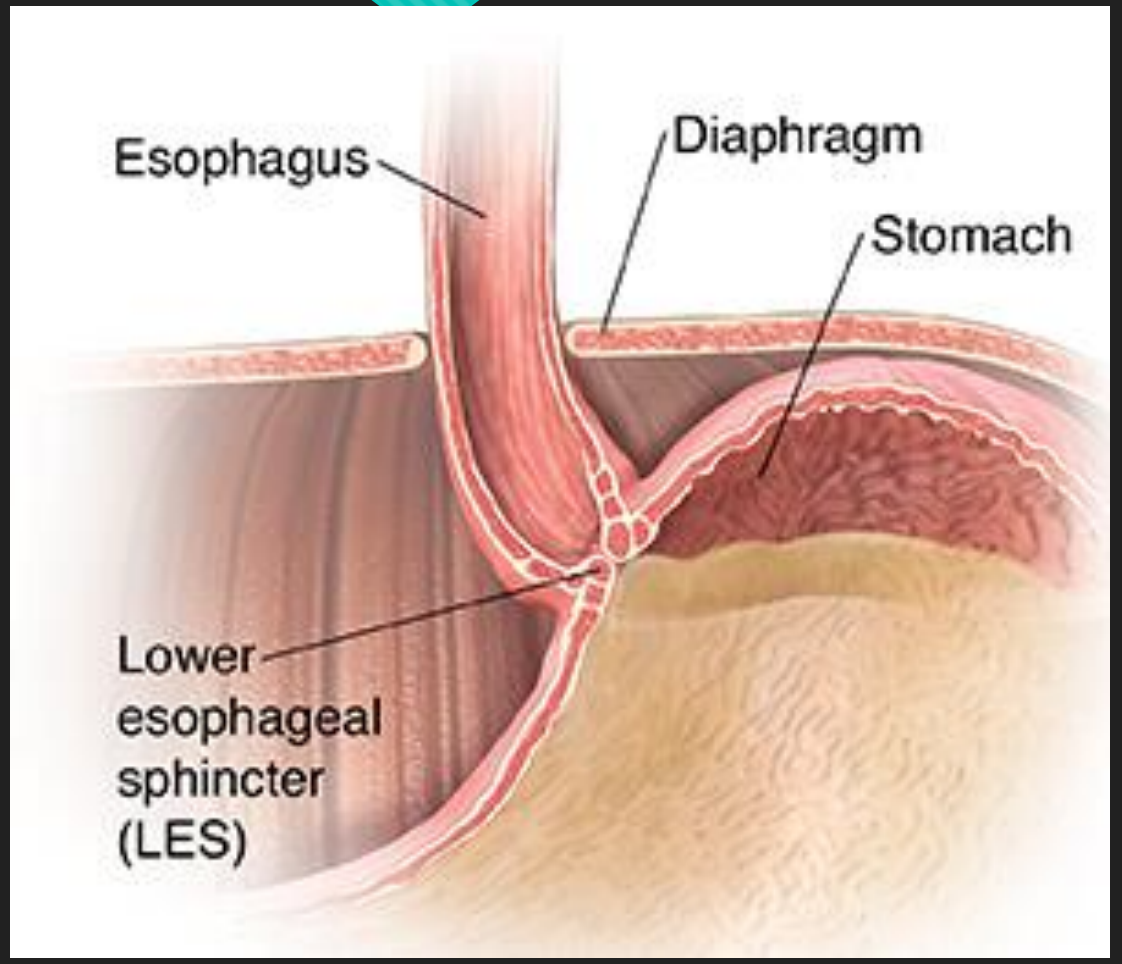

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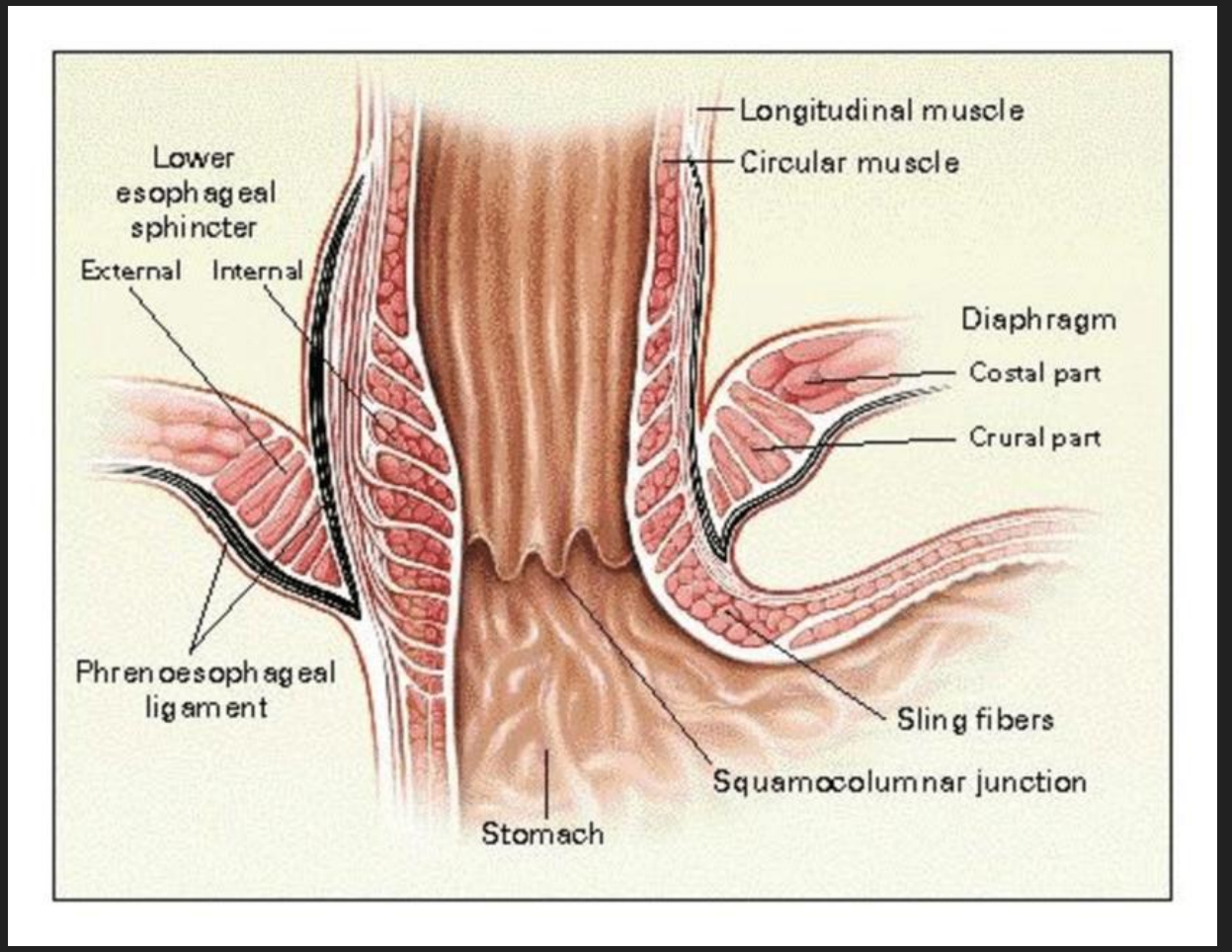
# Esophagus

- Anatomy
  - About 10 inches long
  - Runs from pharynx to stomach through the diaphragm
  - **Lower Esophageal Sphincter (LES)** – a valve of smooth muscles at the base of the esophagus; prevents swallowed food from moving back up
- Physiology
  - Conducts food by peristalsis to the stomach
  - Passageway for food only



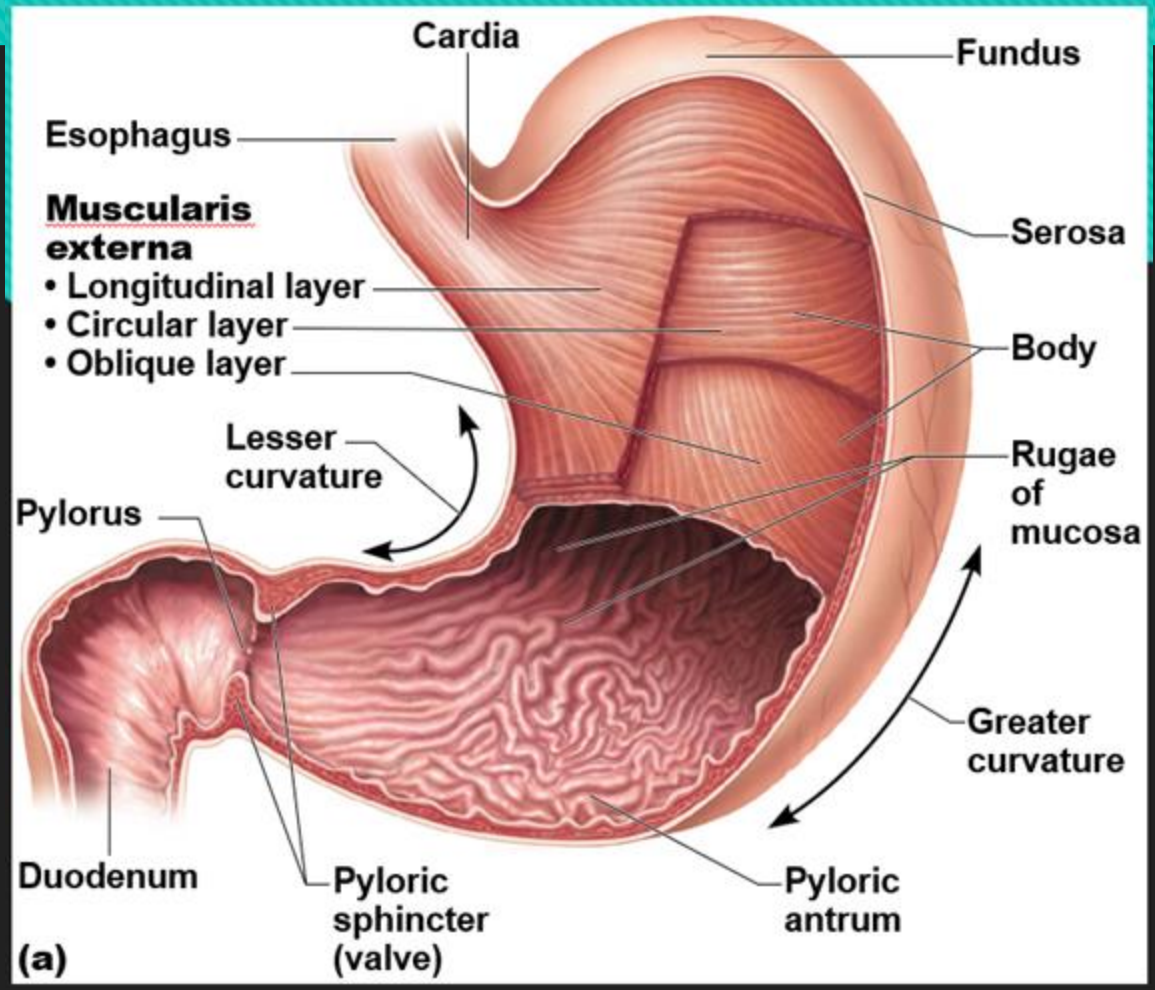
# Esophagus (LOWER ESOPHAGEAL SPHINCTER



# Stomach

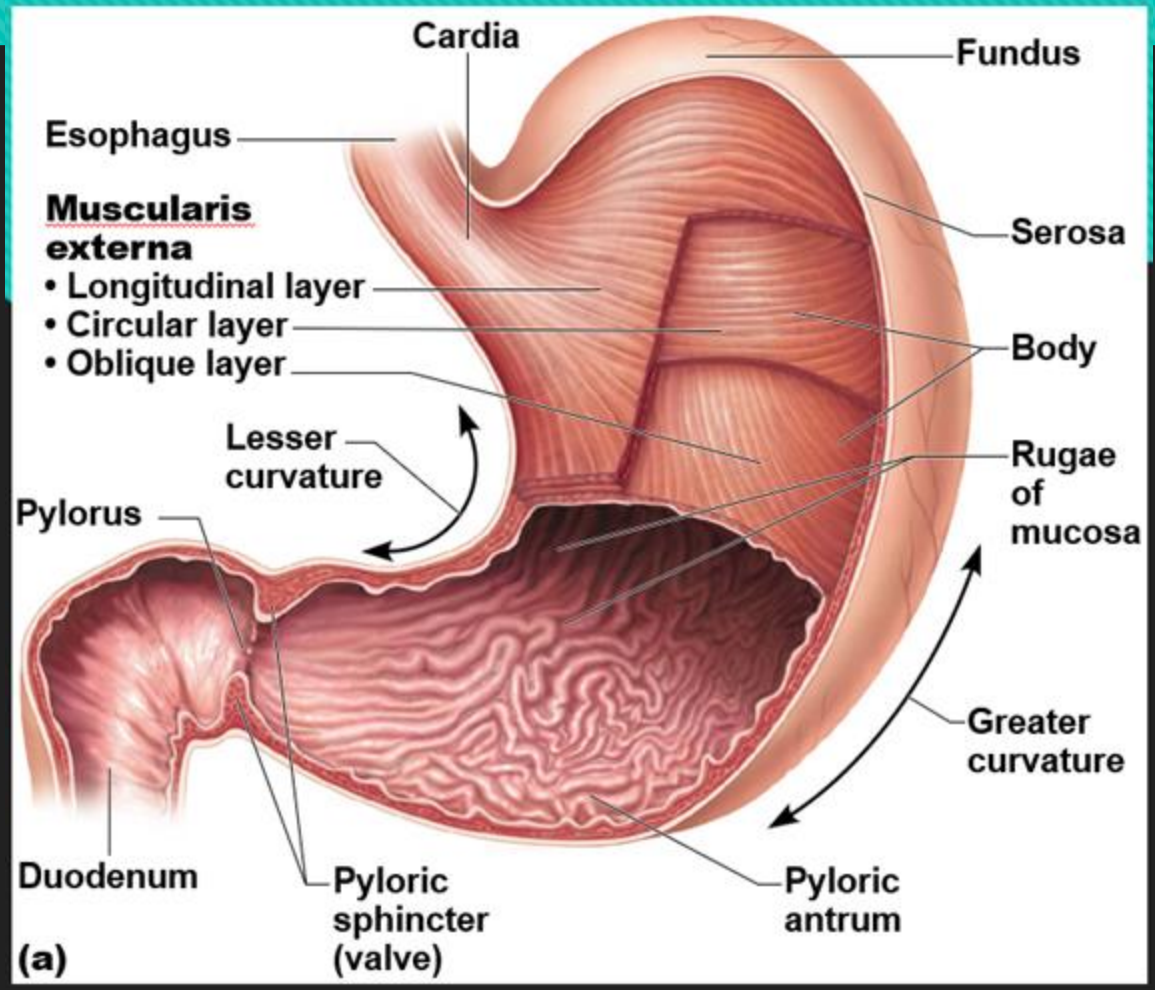
- C-shaped organ located on the left side of the abdominal cavity
- Food enters at the cardioesophageal sphincter/LES from the esophagus
- Food empties into the small intestine at the pyloric sphincter



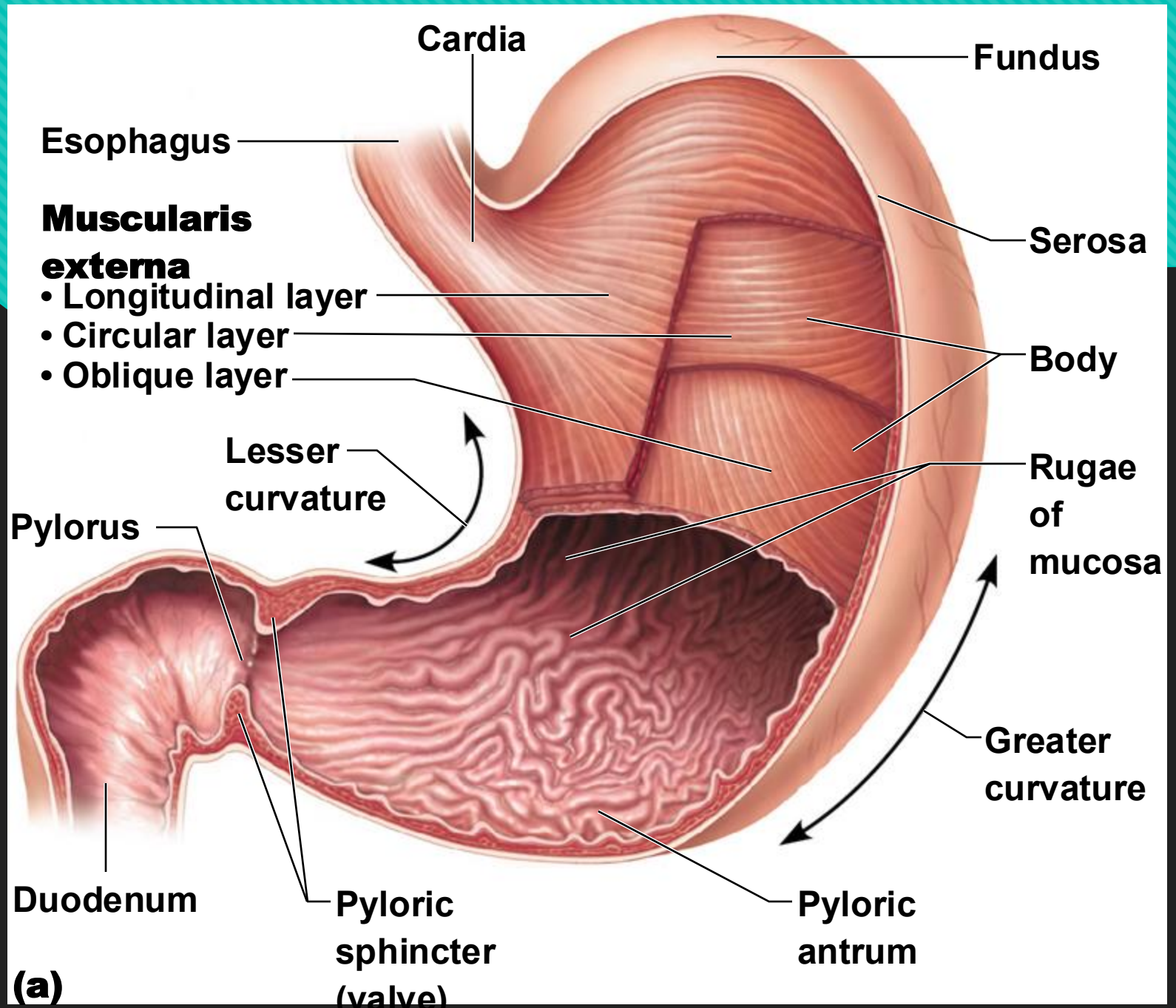

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# Stomach

- Regions
  - Cardial (cardia)—near the heart and surrounds the cardioesophageal sphincter
  - Fundus—expanded portion lateral to the cardiac region
  - Body—midportion
    - Greater curvature is the convex lateral surface
    - Lesser curvature is the concave medial surface
  - Pylorus—funnel-shaped terminal end




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Cardia
Fundus
Esophagus
Muscularis externa
• Longitudinal layer
• Circular layer
• Oblique layer
Serosa
Body
Lesser curvature
Rugae of mucosa
Pylorus
Greater curvature
Duodenum
Pyloric sphincter (valve)
Pyloric antrum
(a)

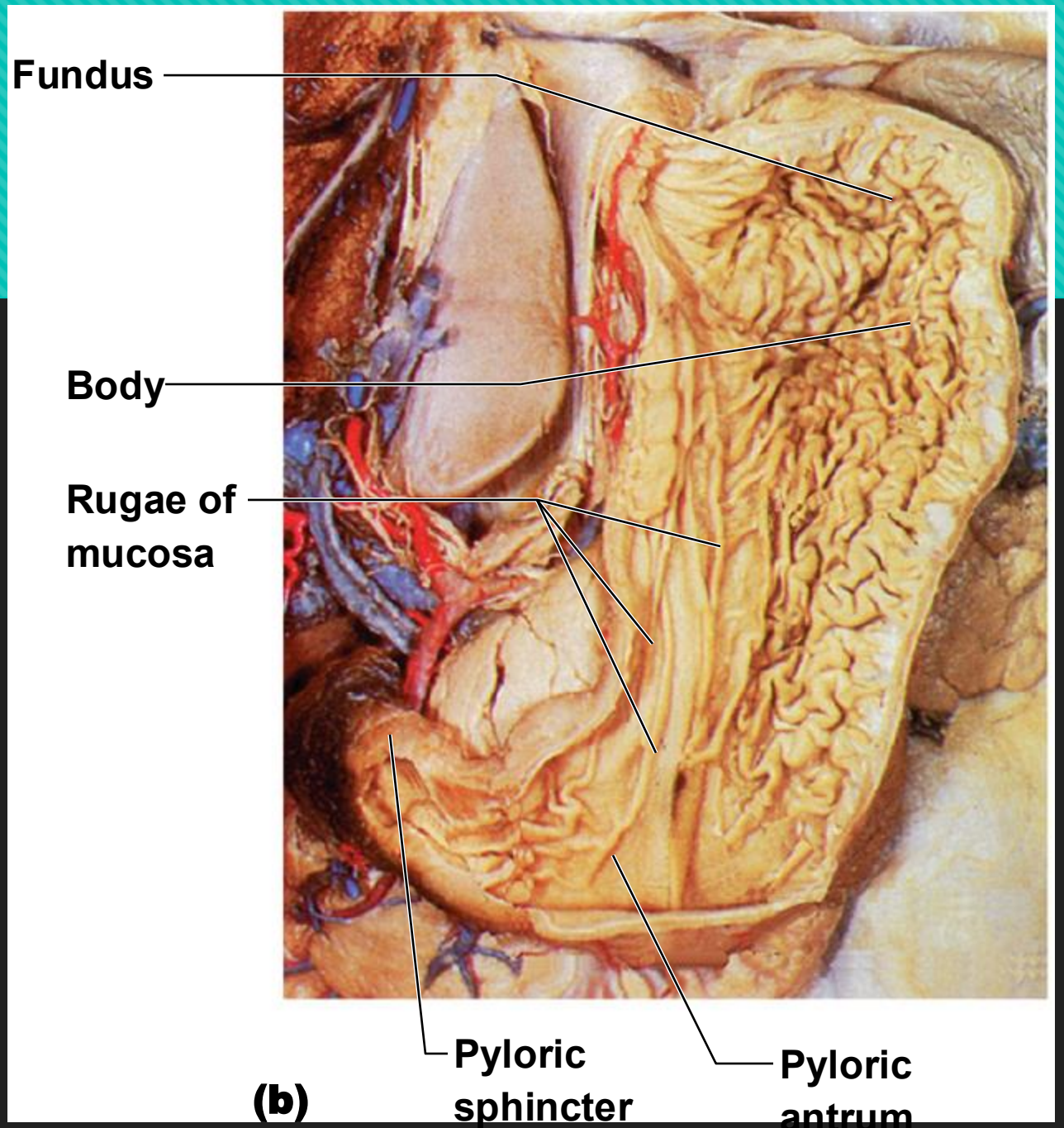



Figure 14.4b Anatomy of the stomach.

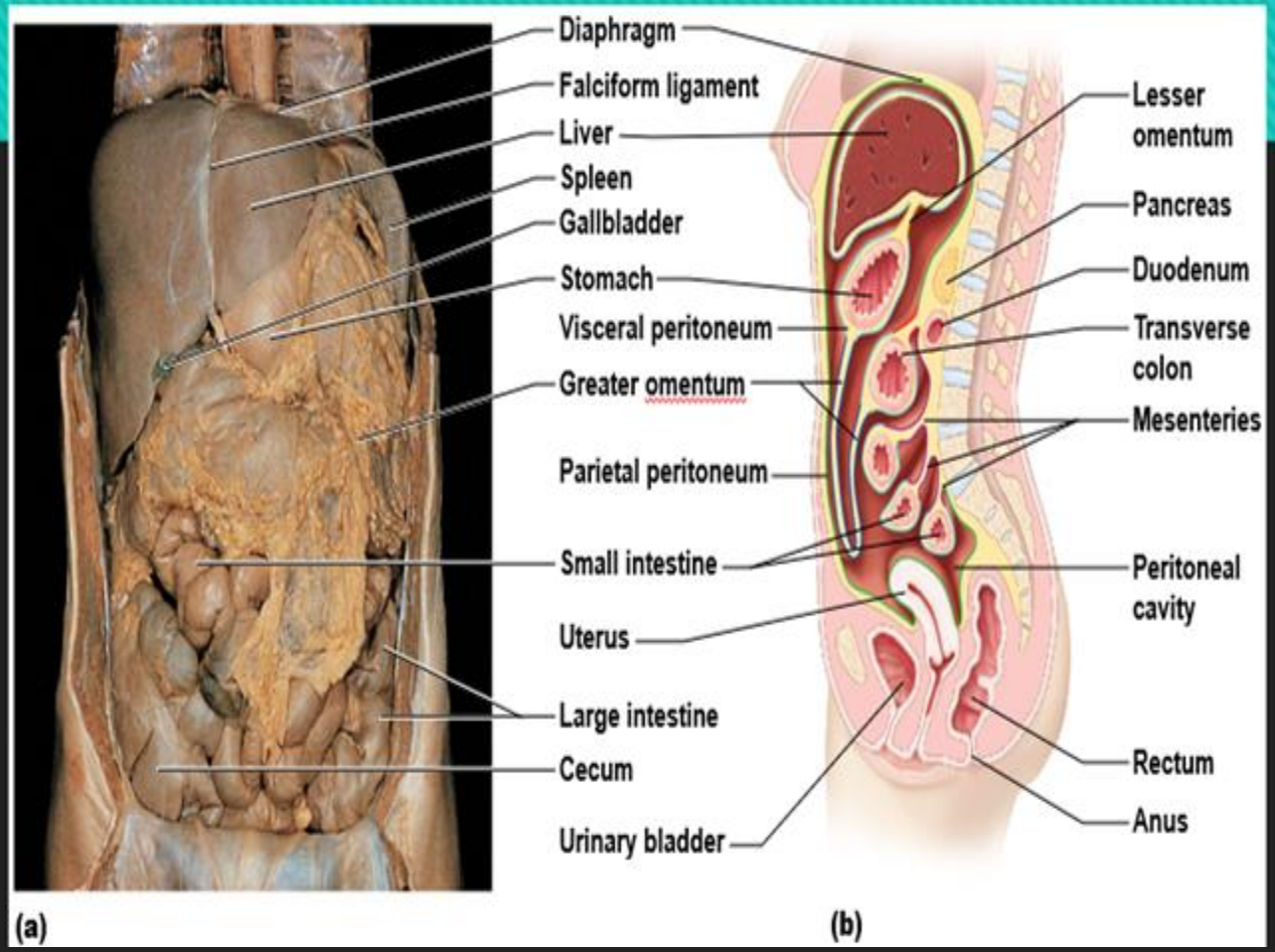
# Stomach

- Stomach can stretch and hold 4 L (1 gallon) of food when full
  - **Rugae**—internal folds of the mucosa present when the stomach is empty

**MESENTERIES**

- connective tissue sheets that hold the organs in place within the abdominal cavity

- **Lesser omentum**
  - Double layer of the peritoneum
  - Connecting the lesser curvature of the stomach and the proximal end of the duodenum to the liver and diaphragm
- **Greater omentum**
  - Another extension of the peritoneum
  - Covers the abdominal organs
  - Extending as a fold from the greater curvature and then to the transverse colon
  - Fat insulates, cushions, and protects abdominal organs



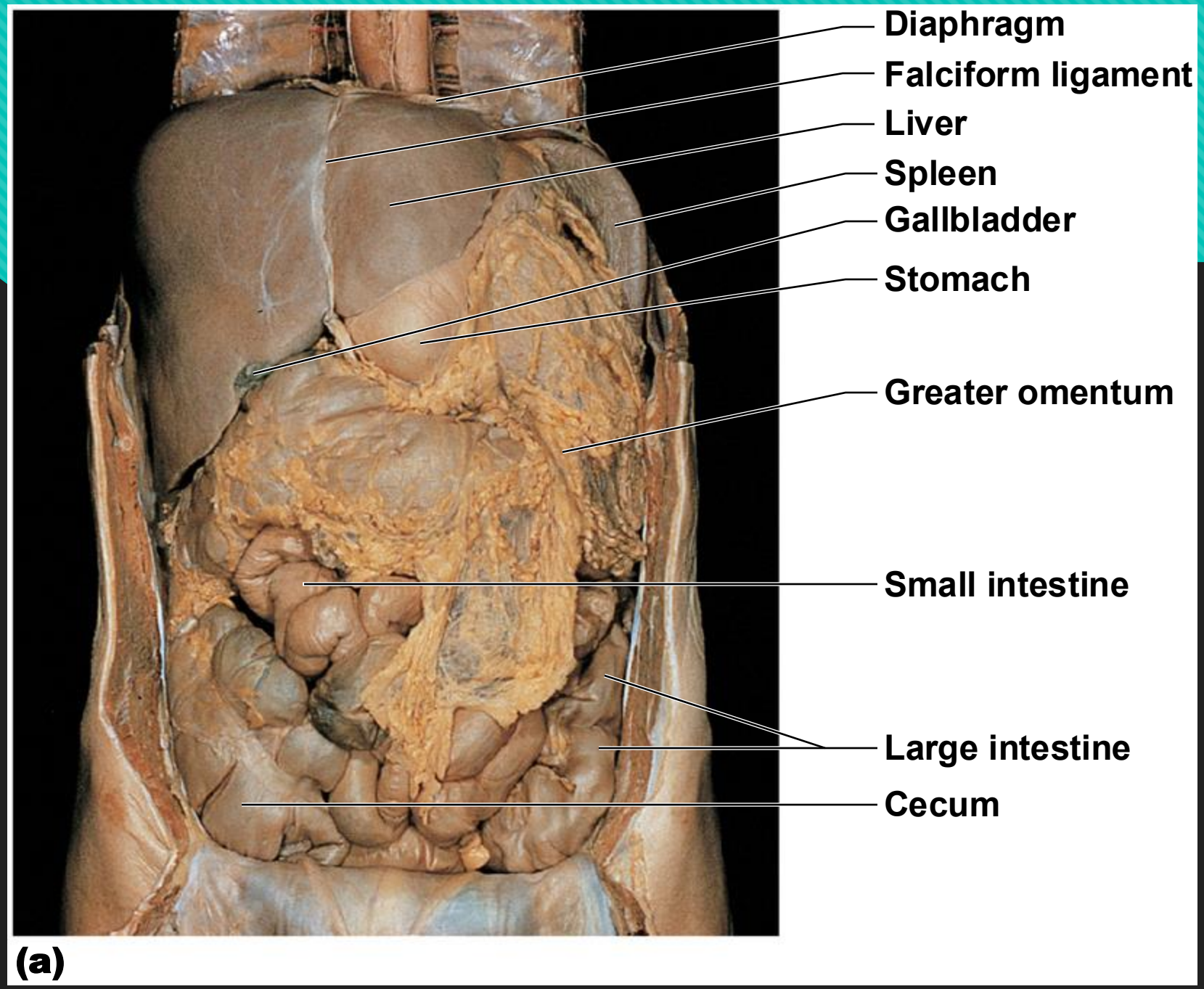
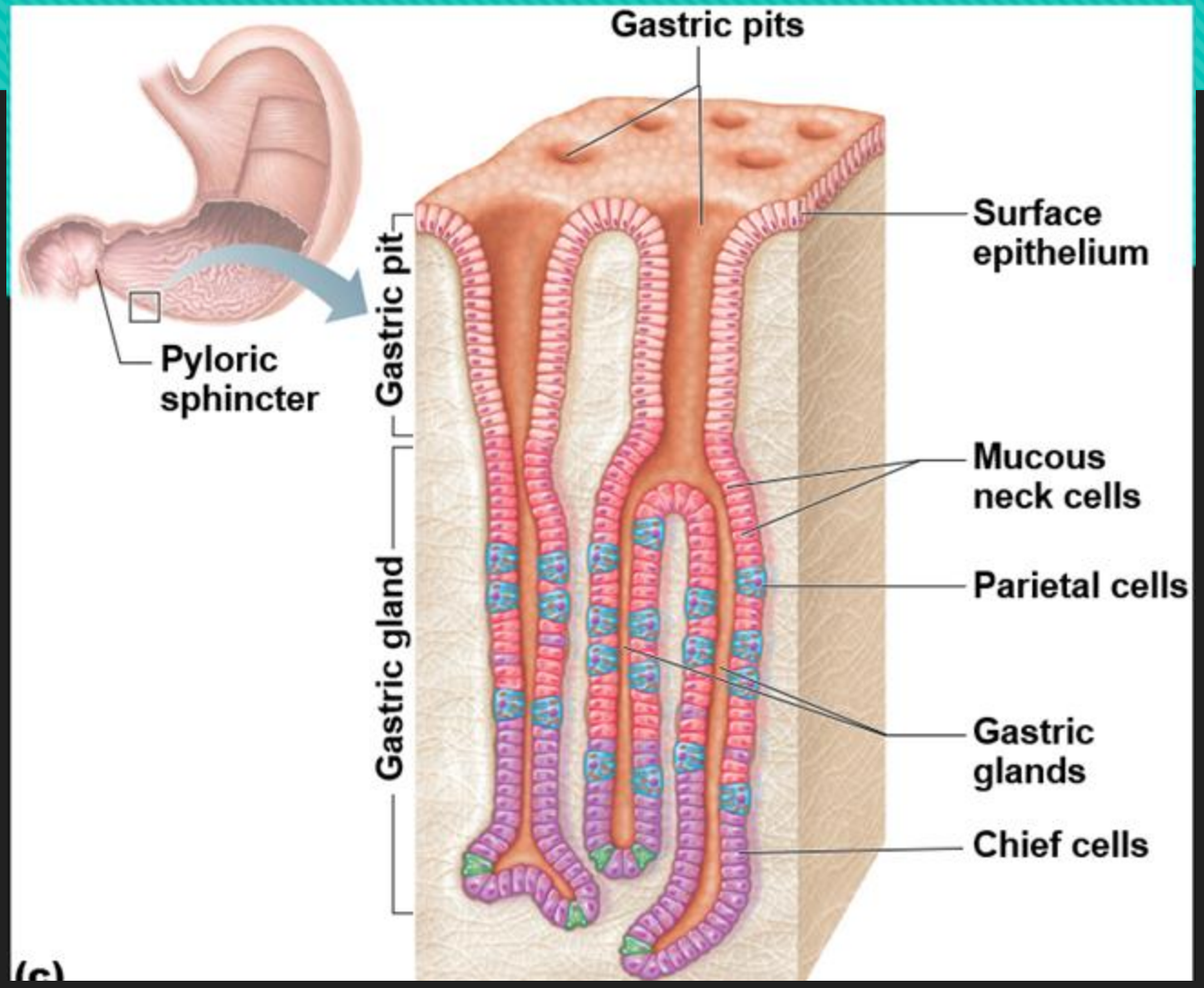



Figure 14.5a Peritoneal attachments of the abdominal organs.

# Stomach

## Structure of the Stomach Mucosa

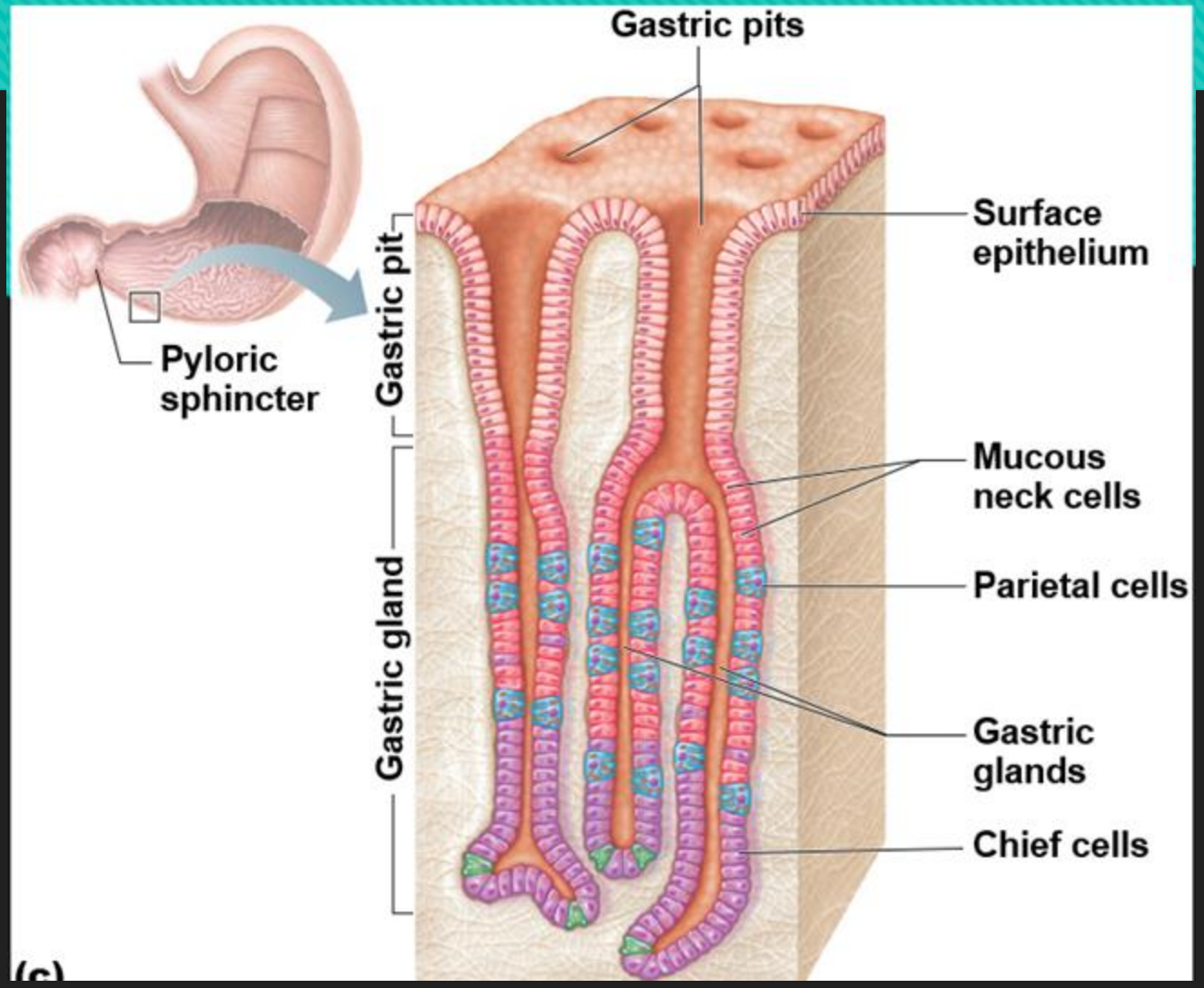
- Simple columnar epithelium composed almost entirely of mucous cells
- Mucous neck cells produce bicarbonate-rich alkaline mucus
- Dotted by gastric pits leading to gastric glands that secrete gastric juice, including:
  - Intrinsic factor, which is needed for vitamin $B_{12}$ absorption in the small intestine




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# Stomach
## Structure of the Stomach Mucosa

- Chief cells—produce protein-digesting enzymes (pepsinogens)
- Parietal cells—produce hydrochloric acid that activates enzymes
- Enteroendocrine cells—produce local hormones such as *gastrin*




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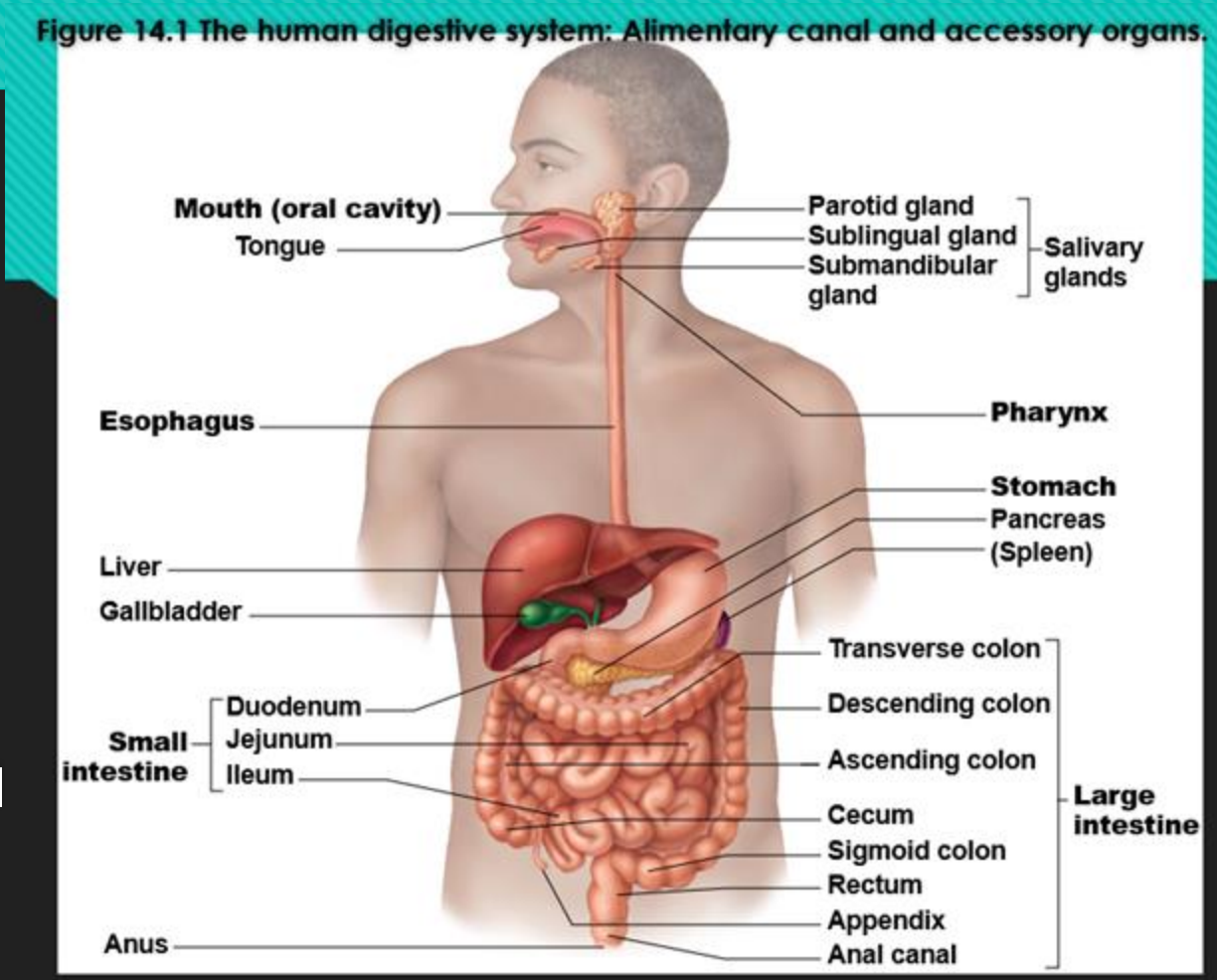
# Stomach
## Functions

- Temporary storage tank for food
- Site of food breakdown
- Chemical breakdown of protein begins
- Delivers chyme (semifluid processed food) to the small intestine

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# Small Intestine

- The body's major digestive organ
- Longest portion of the alimentary tube (4.6-9m)
- Site of nutrient absorption into the blood
- Muscular tube extending from the pyloric sphincter to the ileocecal valve
- Suspended from the posterior abdominal wall by the mesentery




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# Small Intestine

## Subdivisions

- Duodenum
- Jejunum
- Ileum

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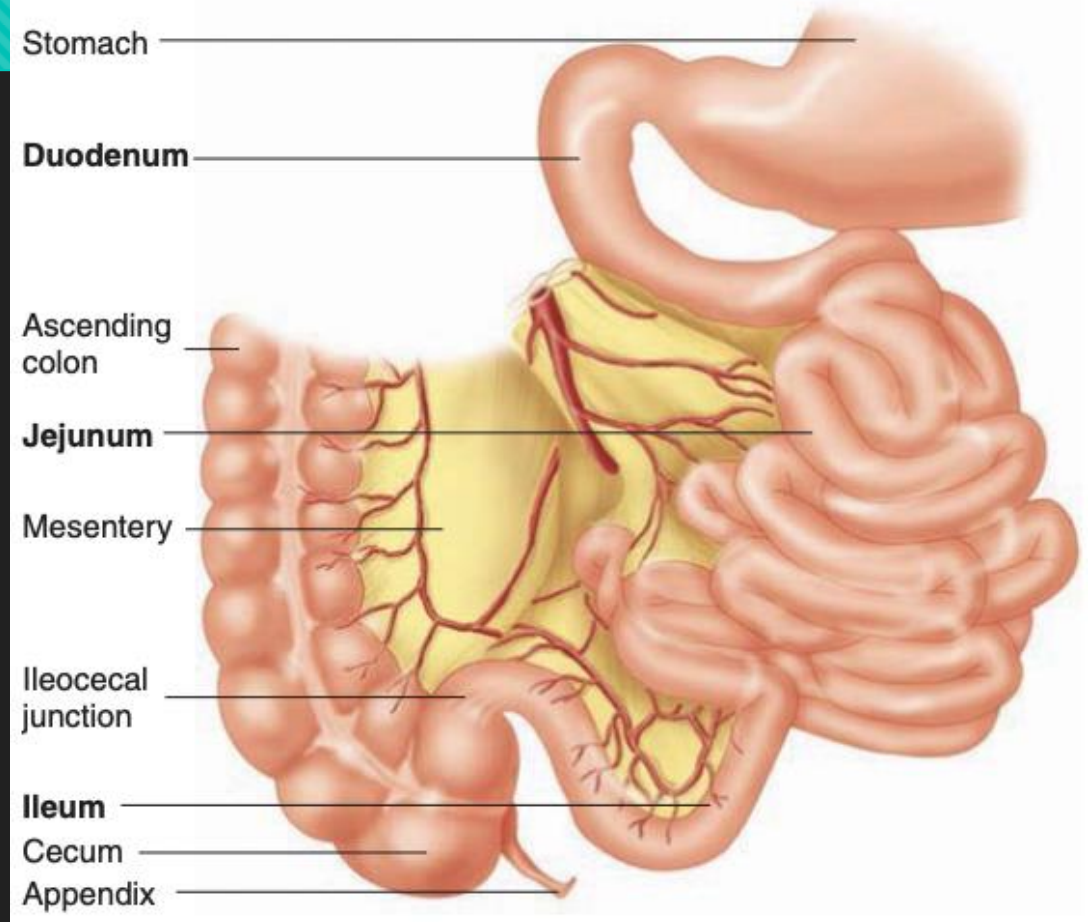
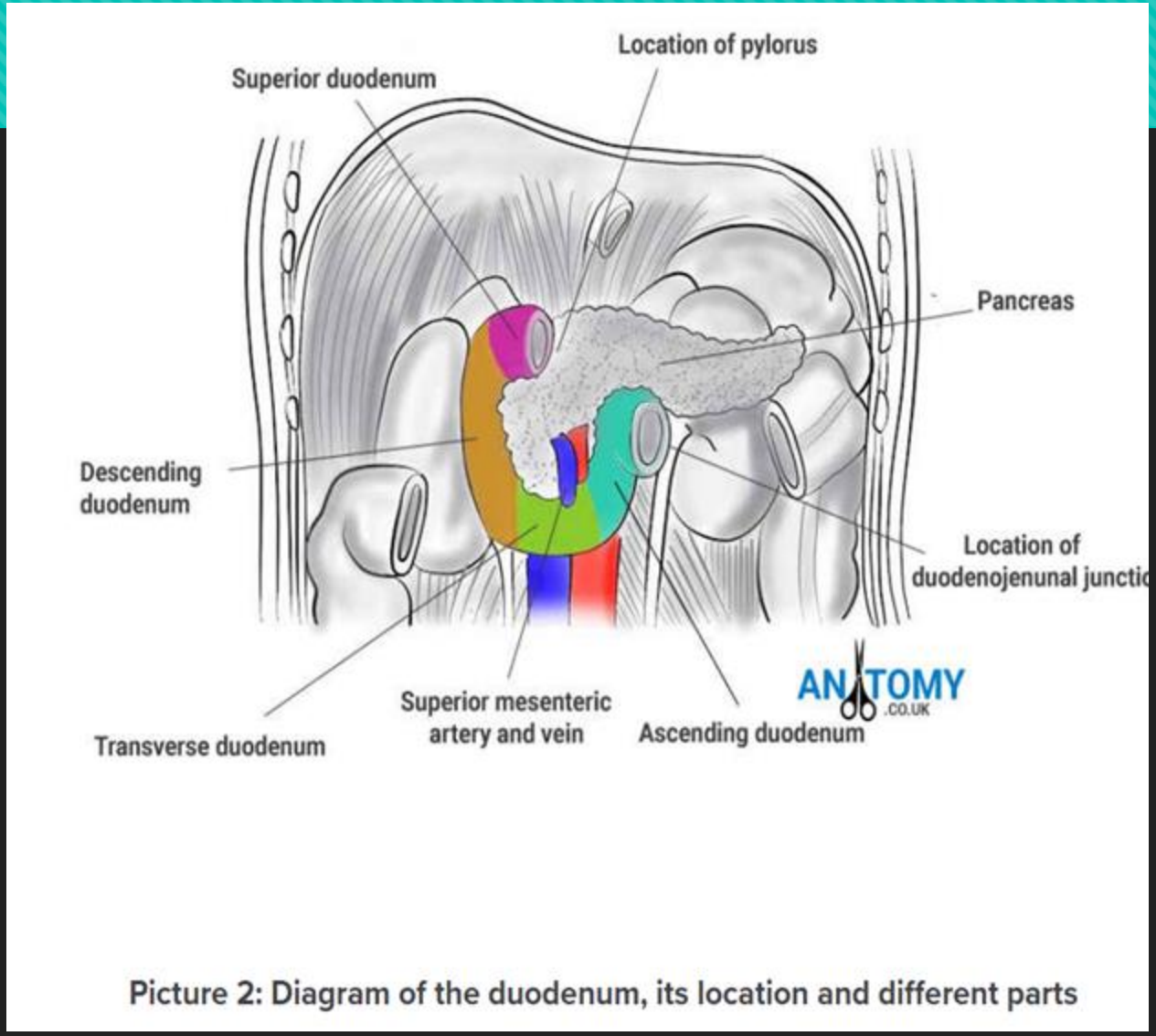



**FIGURE 24.15** Small Intestine AP|R

# Duodenum

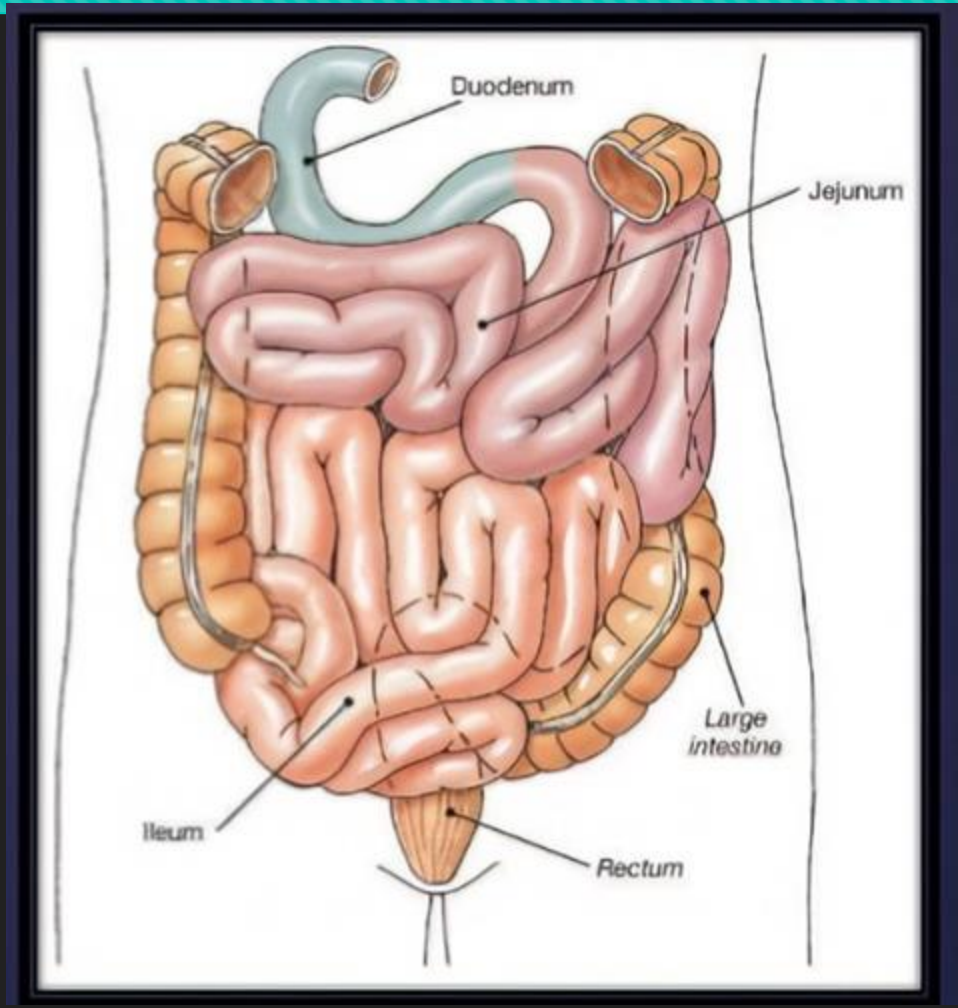
- initial portion of the small intestine
- C-shaped (or horse-shoe shaped) hollow tube that extends from L1 to L3 vertebral levels
- length varies in a range of 10 to 12 inches (25-30 cm)
- receives partially digested food (chyme) with high acidity from the stomach
- involves in **neutralizing the acidity of the food** and propelling it through the rest of the small intestine for further digestion and absorption




Picture 2: Diagram of the duodenum, its location and different parts

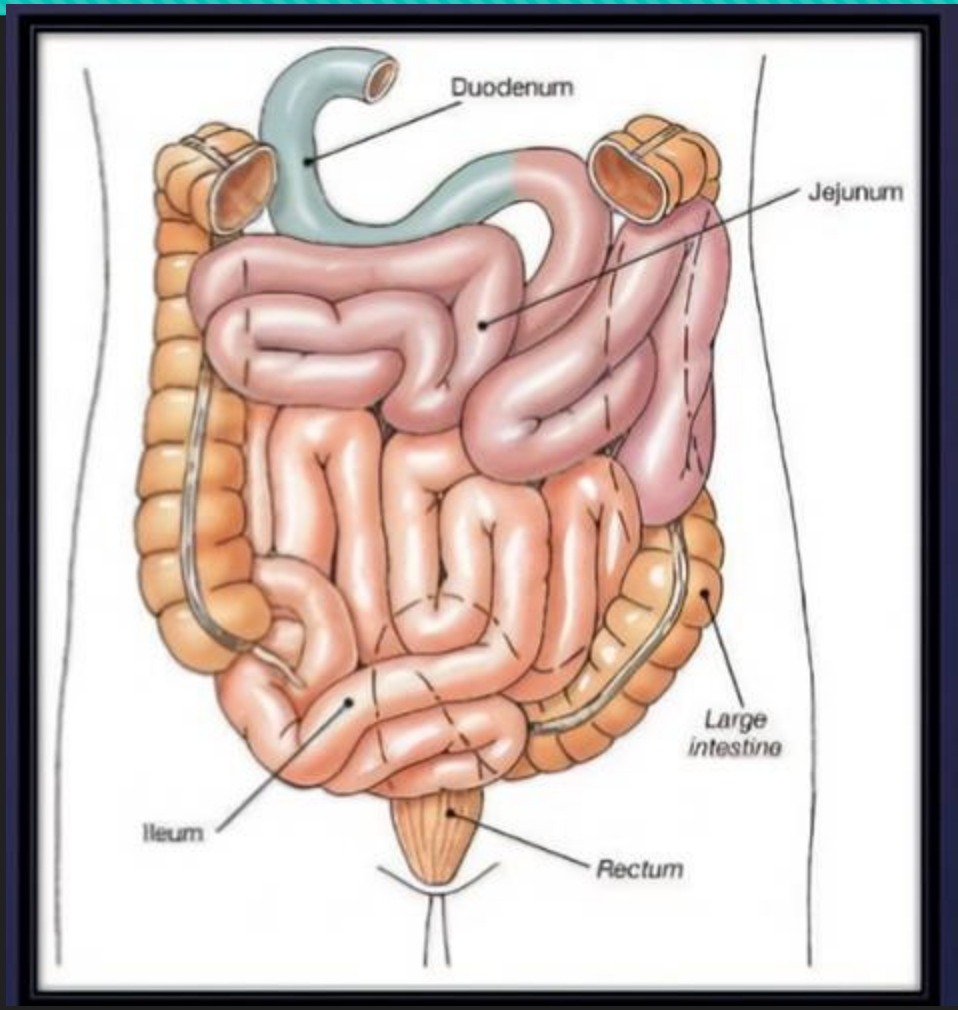
# Jejunum

- located between the duodenum and the ileum
- makes up about two-fifths of the small intestine
- No clear delineanation
- main function of the jejunum is absorption of important nutrients such fatty acids and amino acids
- appears red in color due to numerous blood supply
- held in place in the abdominal cavity by mesentery
- **site of nutrient absorption**



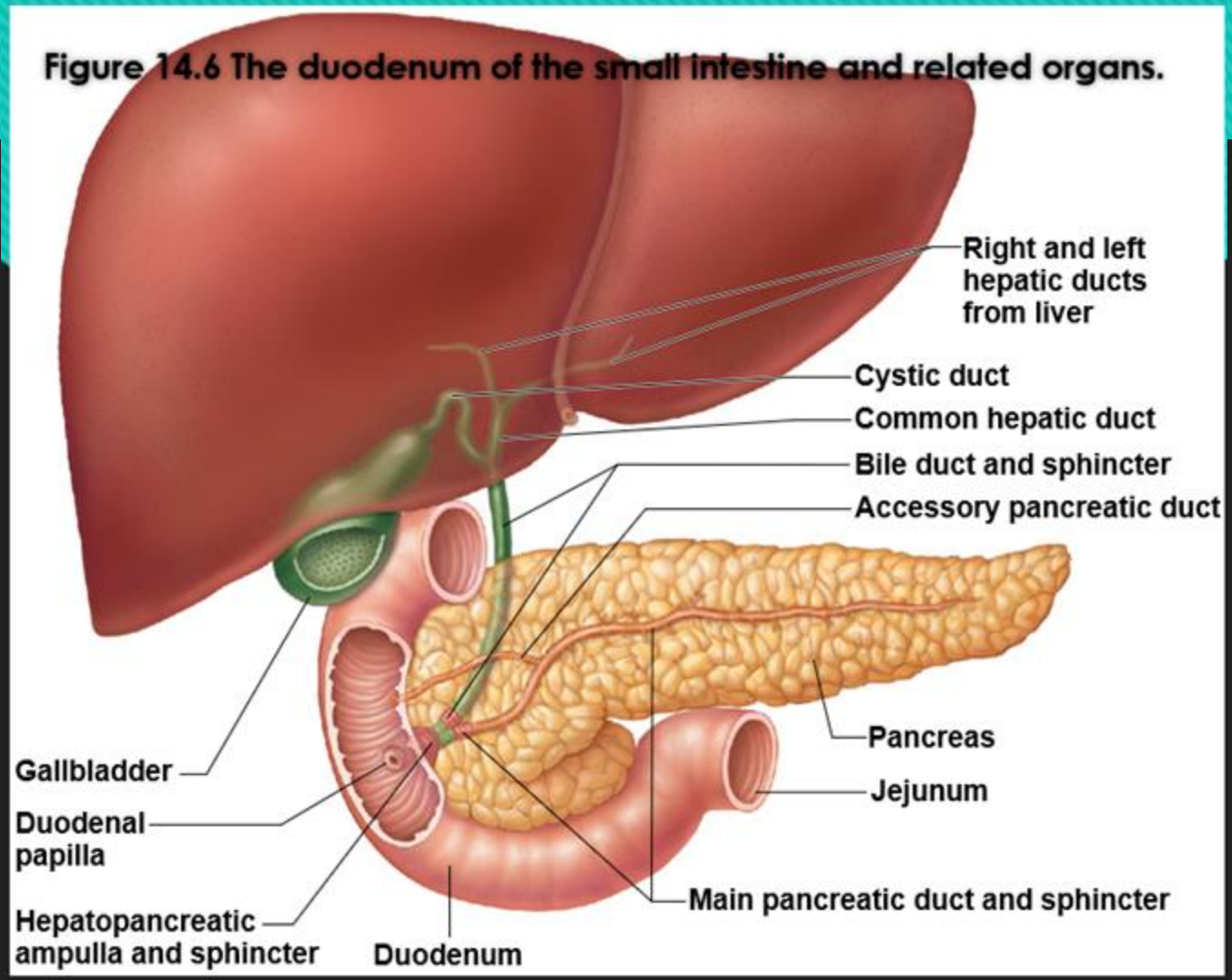
# Ileum

- final and longest segment of the small intestine
- 3.5 metres (11.5 feet) long (or about three-fifths the length of the small intestine)
- extends from the jejunum to the ileocecal valve
- Peyer's patches are embedded in the ileal wall
- **responsible for the absorption of vitamin $B_{12}$** and the **reabsorption of conjugated bile salts**
  - aid digestion by breaking down fats
  - help absorb fat-soluble vitamins
  - eliminate waste products



# Small Intestine

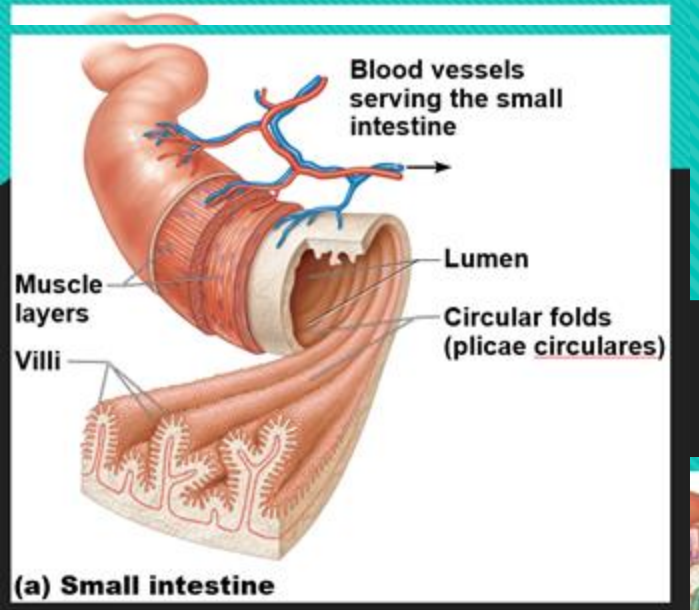
- Chemical digestion begins in the small intestine
  - Enzymes produced by intestinal cells and pancreas are carried to the duodenum by pancreatic ducts
  - Bile, formed by the liver, enters the duodenum via the bile duct
  - Hepatopancreatic ampulla is the location where the main pancreatic duct and bile ducts join

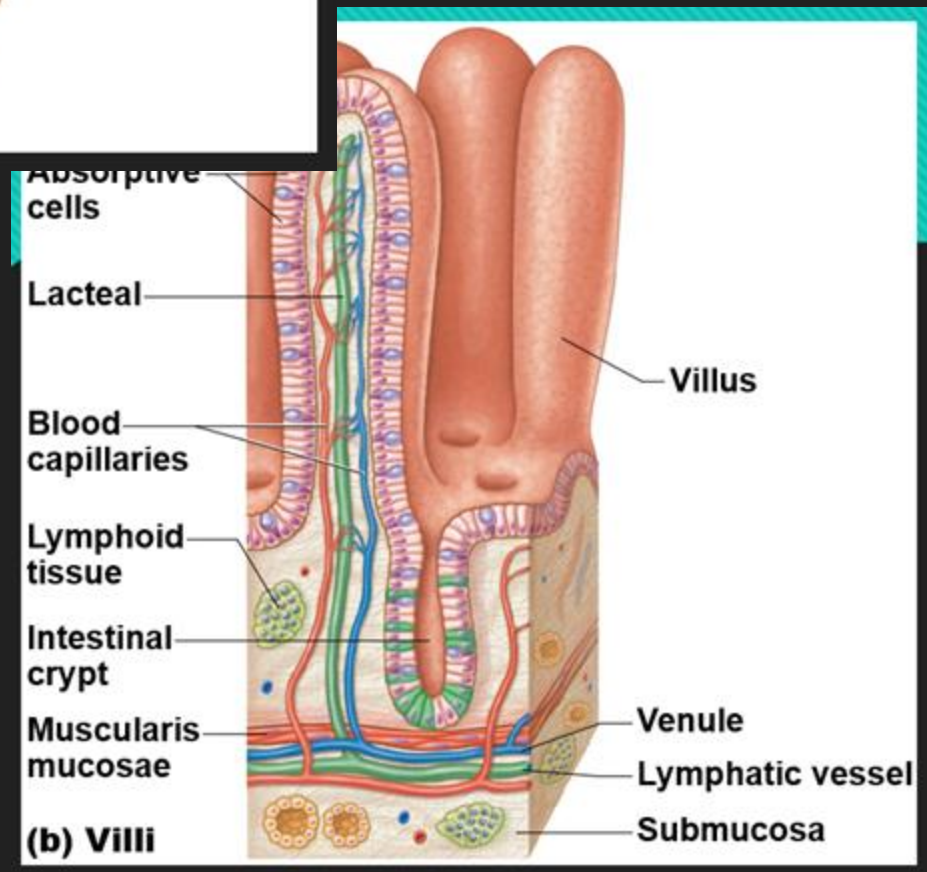



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# Small Intestine

- Structural modifications
  - Increase surface area for food absorption
  - Decrease in number toward the end of the small intestine
    1. Villi—fingerlike projections formed by the mucosa
       - are located within the circular folds
       - house a capillary bed and lacteal (lymph)
         - allow nutrients to be absorbed directly into the bloodstream
       - measure 1 millimeter in length

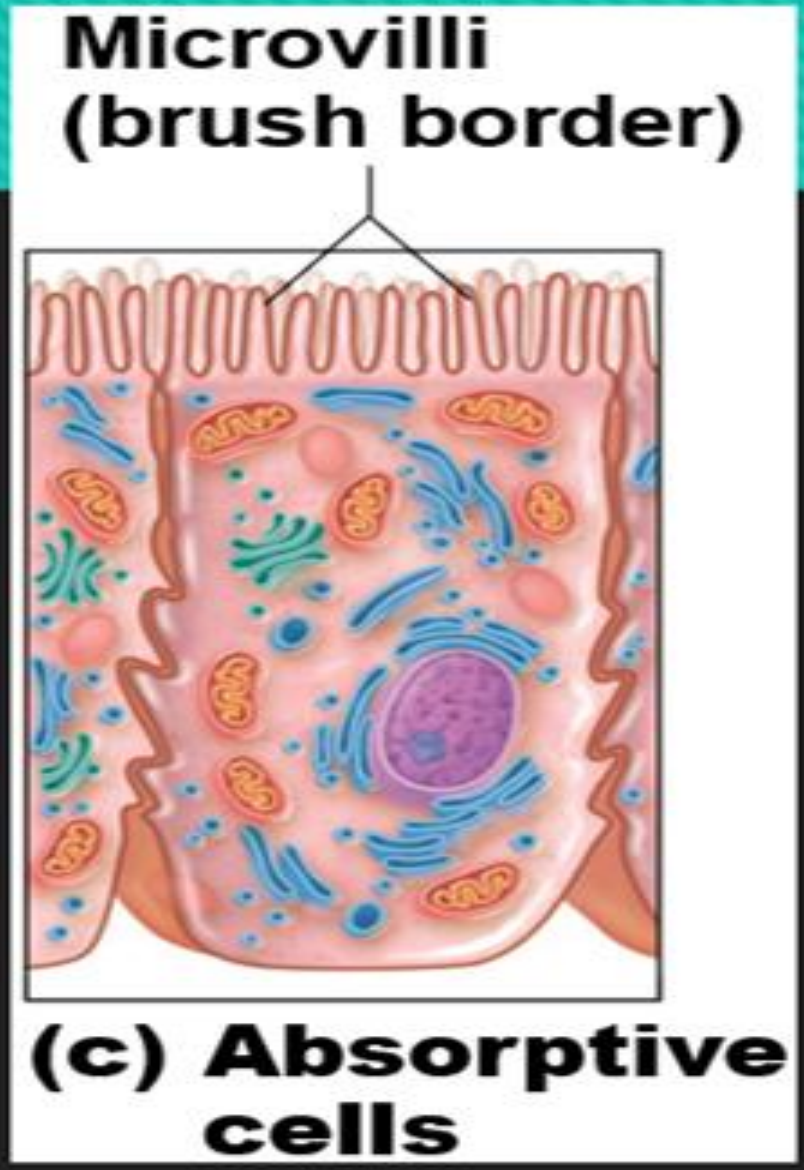



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# Small Intestine

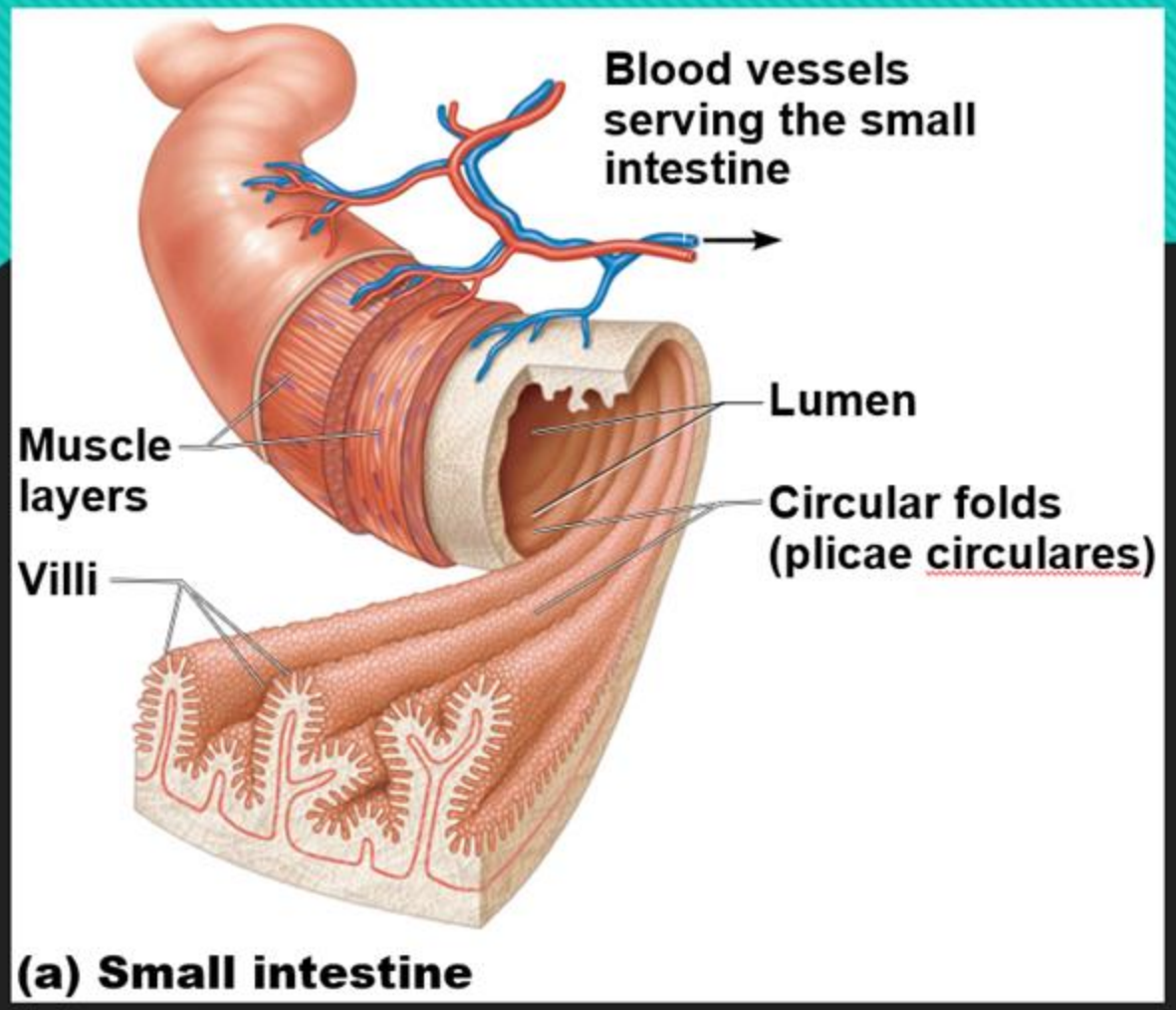
- Structural modifications
  - 2. Microvilli
    - tiny projections of the plasma membrane
    - are even smaller than villi
    - roughly 200 million per square millimeter in the small intestine
    - enzymes on the surface of the microvilli can help to further digest sugars and amino acids




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# Small Intestine

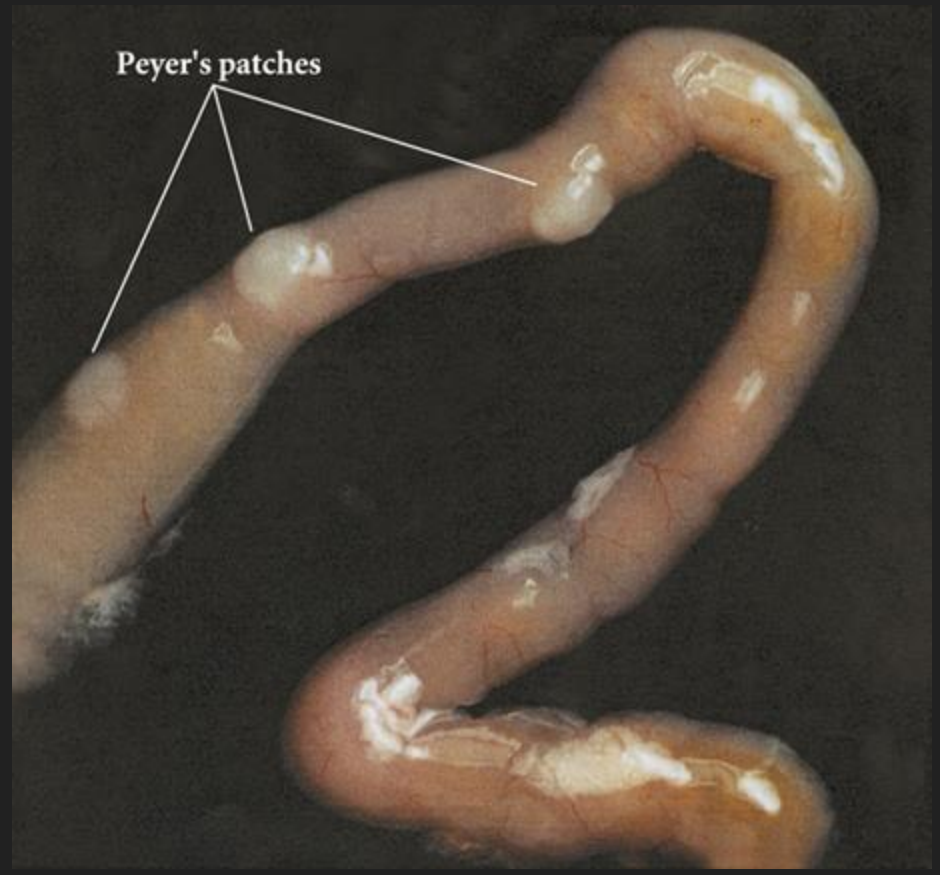
- Structural modifications
  - 3. Circular folds (plicae circulares)
    - deep folds of mucosa and submucosa
    - help to direct and slow the flow of nutrients through the small intestine, allowing for efficient absorption




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# Small Intestine

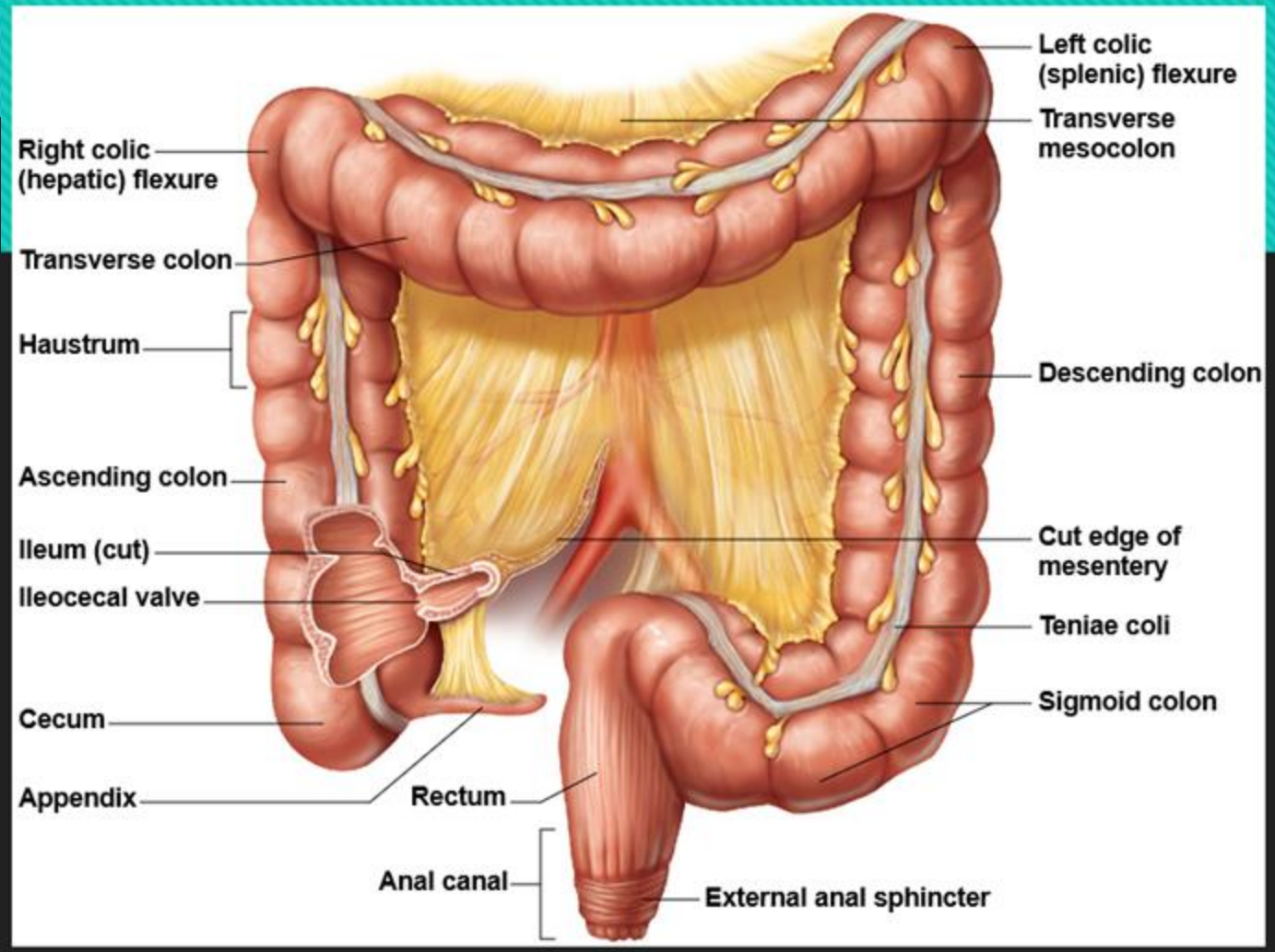
- Peyer's patches
  - Collections of lymphatic tissue
  - Located in mucosa and submucosa
  - More are needed there because remaining food residue contains much bacteria
  - Initiated immune responses against microorganisms that enter the mucosa




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# Large Intestine

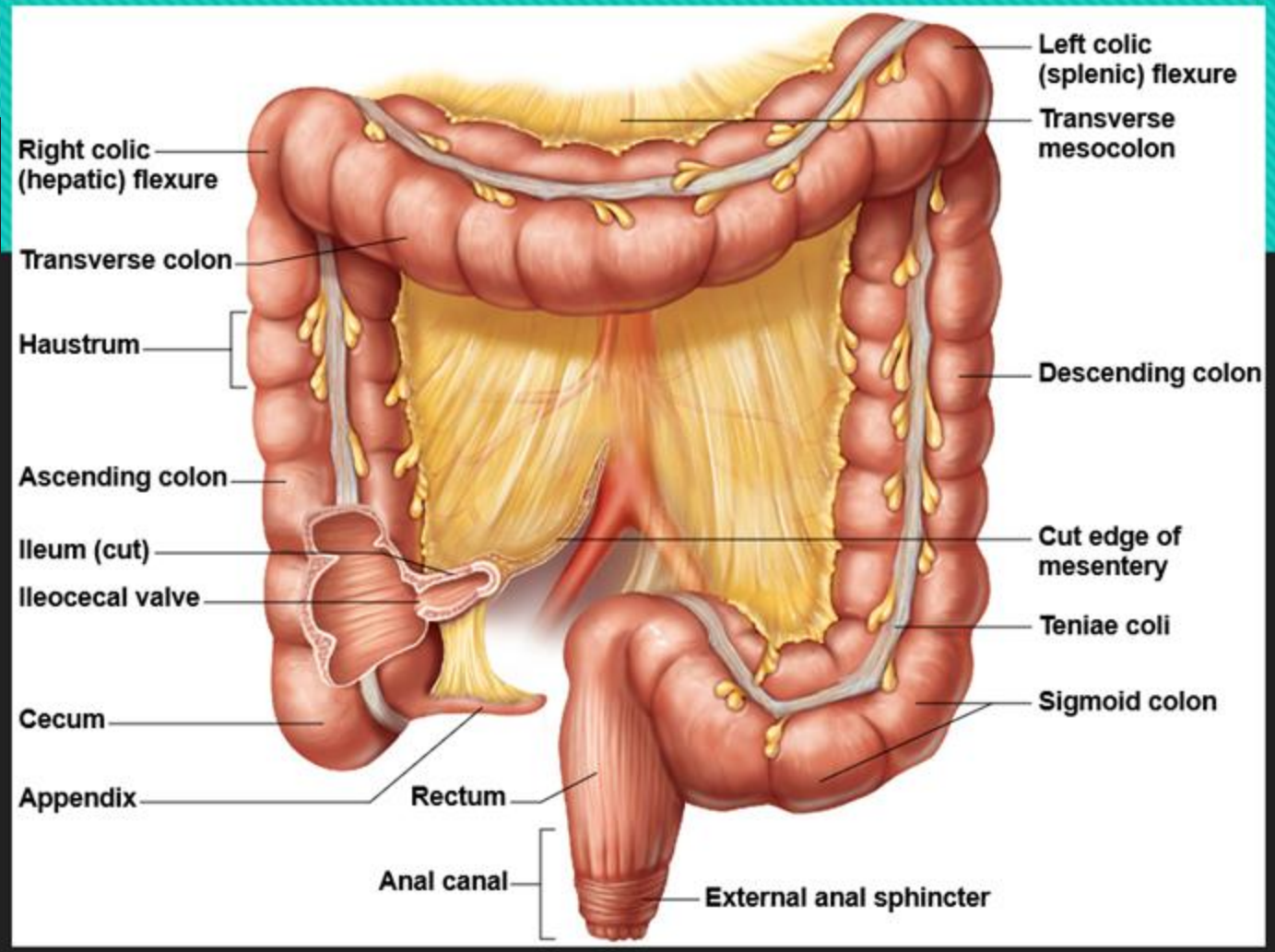
- Larger in diameter, but shorter in length at 1.5 m, than the small intestine
- Extends from the ileocecal valve to the anus
- Subdivisions
  - Cecum
  - Colon
  - Rectum
  - Anal canal




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# Large Intestine

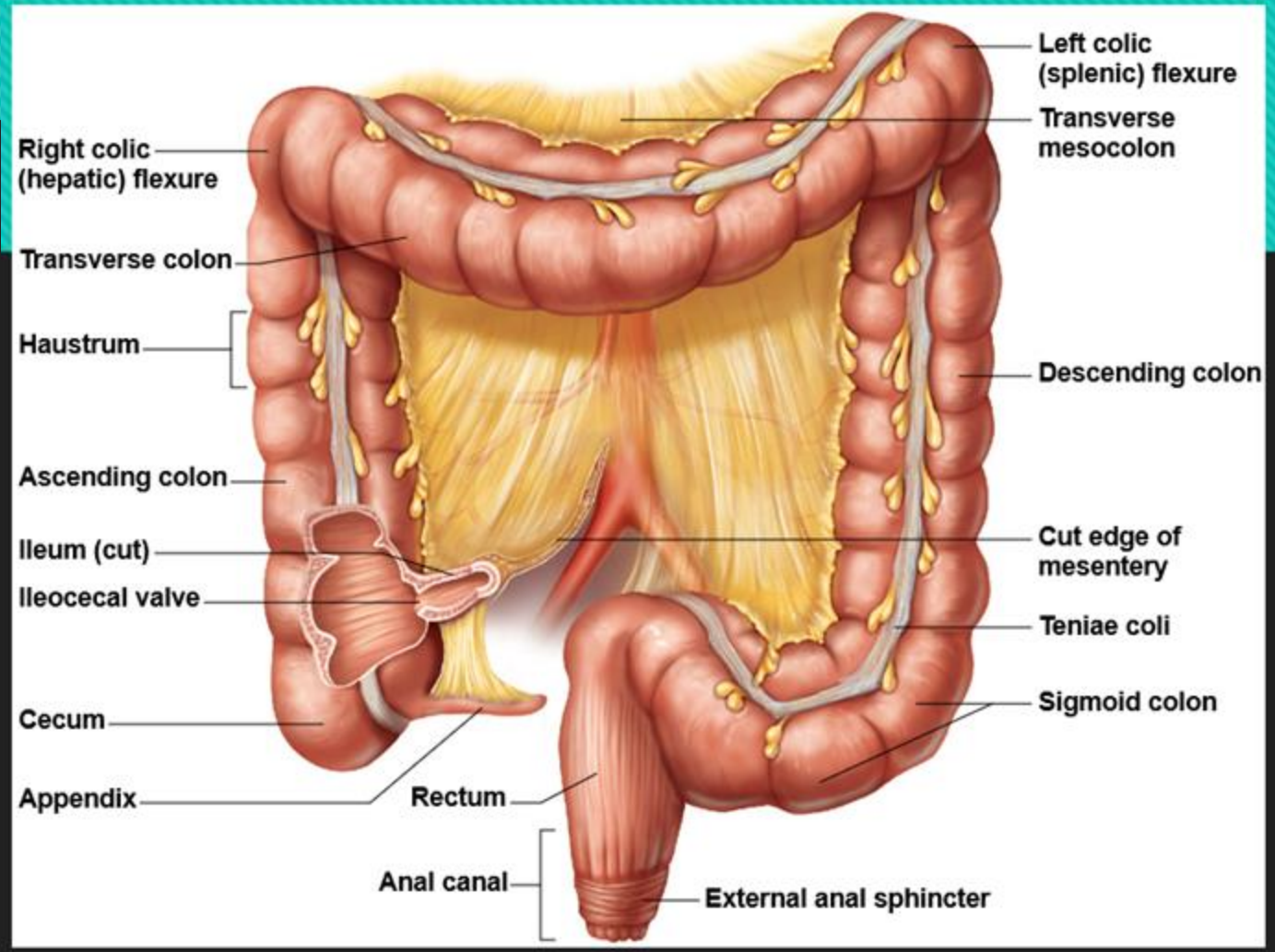
- Cecum—saclike first part of the large intestine
  - Appendix
    - Hangs from the cecum
    - Accumulation of lymphoid tissue




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# Large Intestine

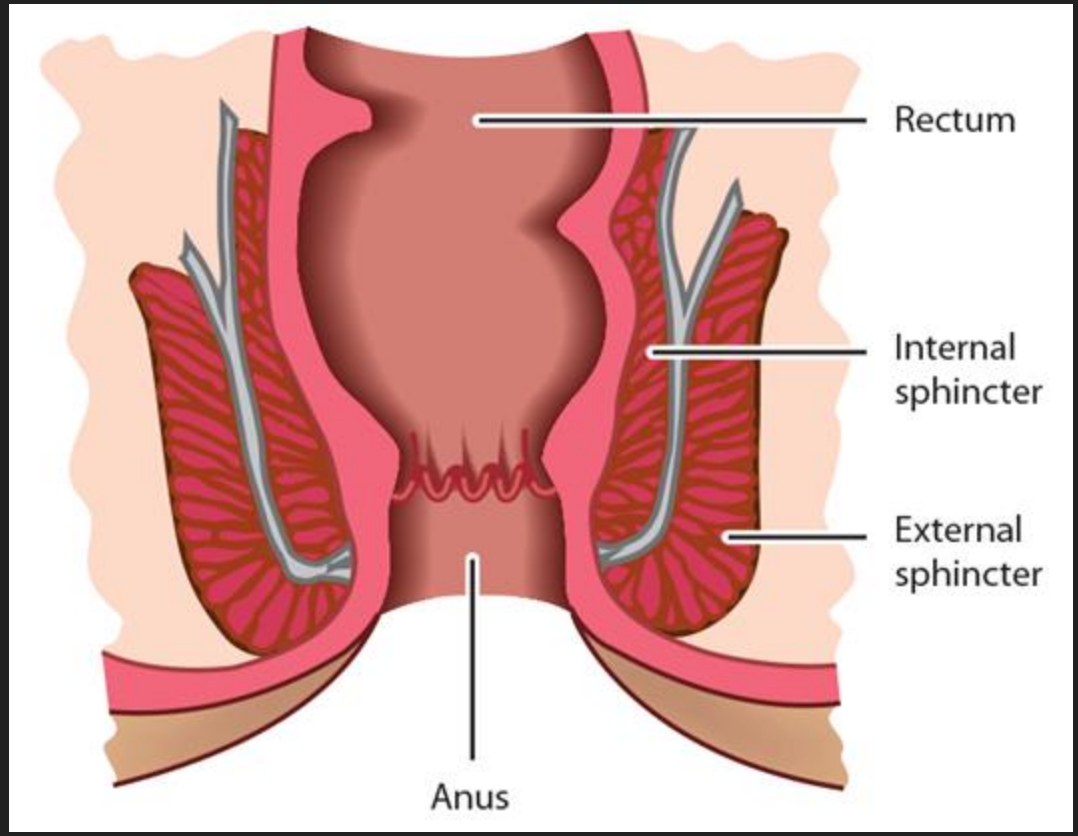
- Colon
  - Ascending—travels up right side of abdomen and makes a turn at the right colic (hepatic) flexure
  - Transverse—travels across the abdominal cavity and turns at the left colic (splenic) flexure
  - Descending—travels down the left side
  - Sigmoid—S-shaped region; enters the pelvis
- Sigmoid colon, rectum, and anal canal are located in the pelvis




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# Large Intestine

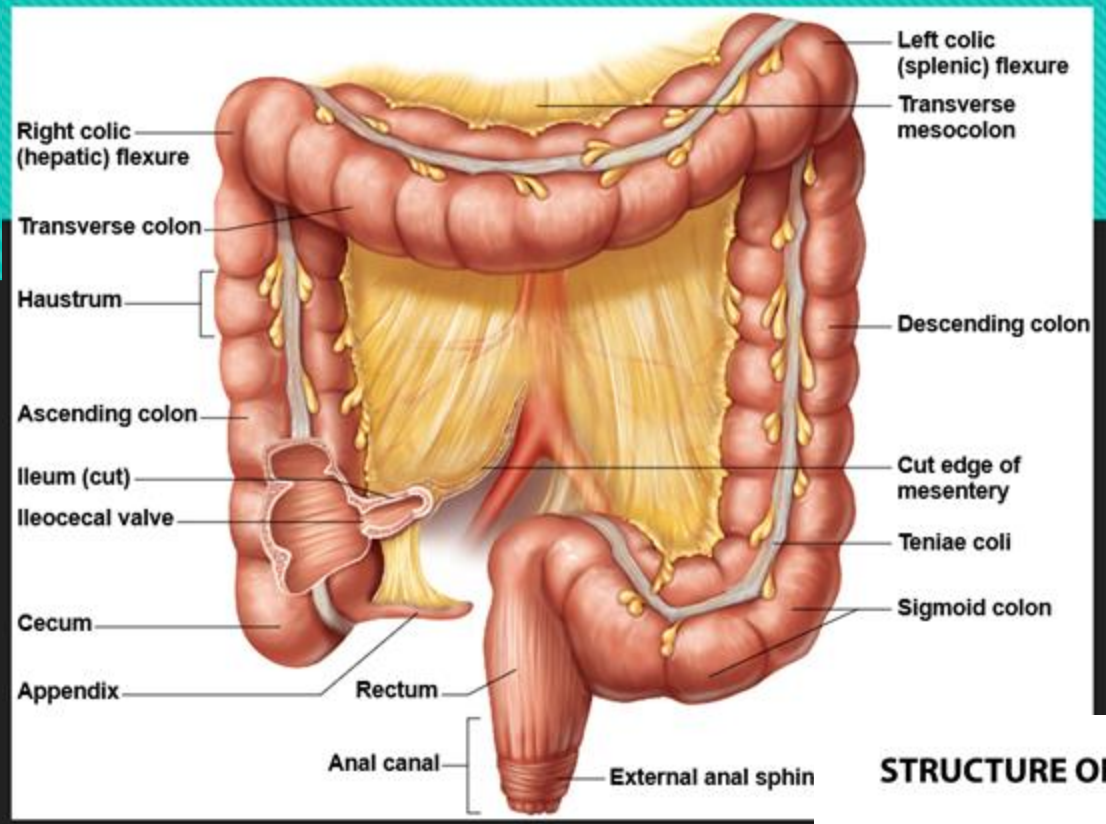
- Anal canal ends at the anus
- Anus—opening of the large intestine
  - External anal sphincter—formed by skeletal muscle and is voluntary
  - Internal anal sphincter—formed by smooth muscle and is involuntary
  - These sphincters are normally closed except during defecation
- The large intestine delivers indigestible food residues to the body's exterior

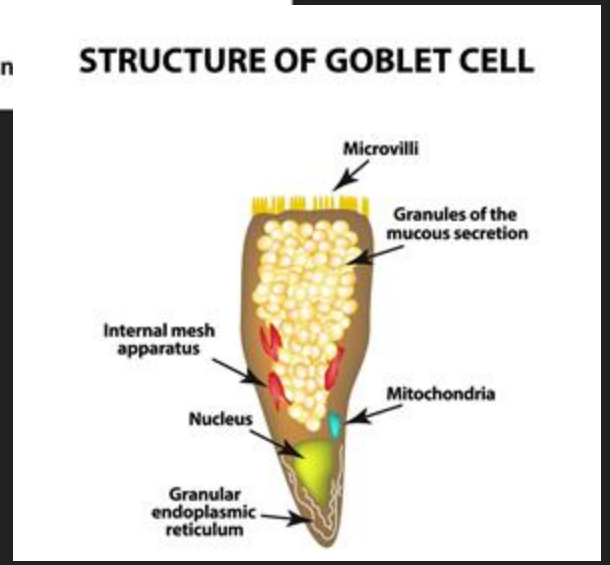



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# Large Intestine

- **Goblet cells** produce alkaline mucus to lubricate the passage of feces
- Muscularis externa layer is reduced to three bands of muscle, called *teniae coli*
- These bands of muscle cause the wall to pucker into haustra (pocketlike sacs)




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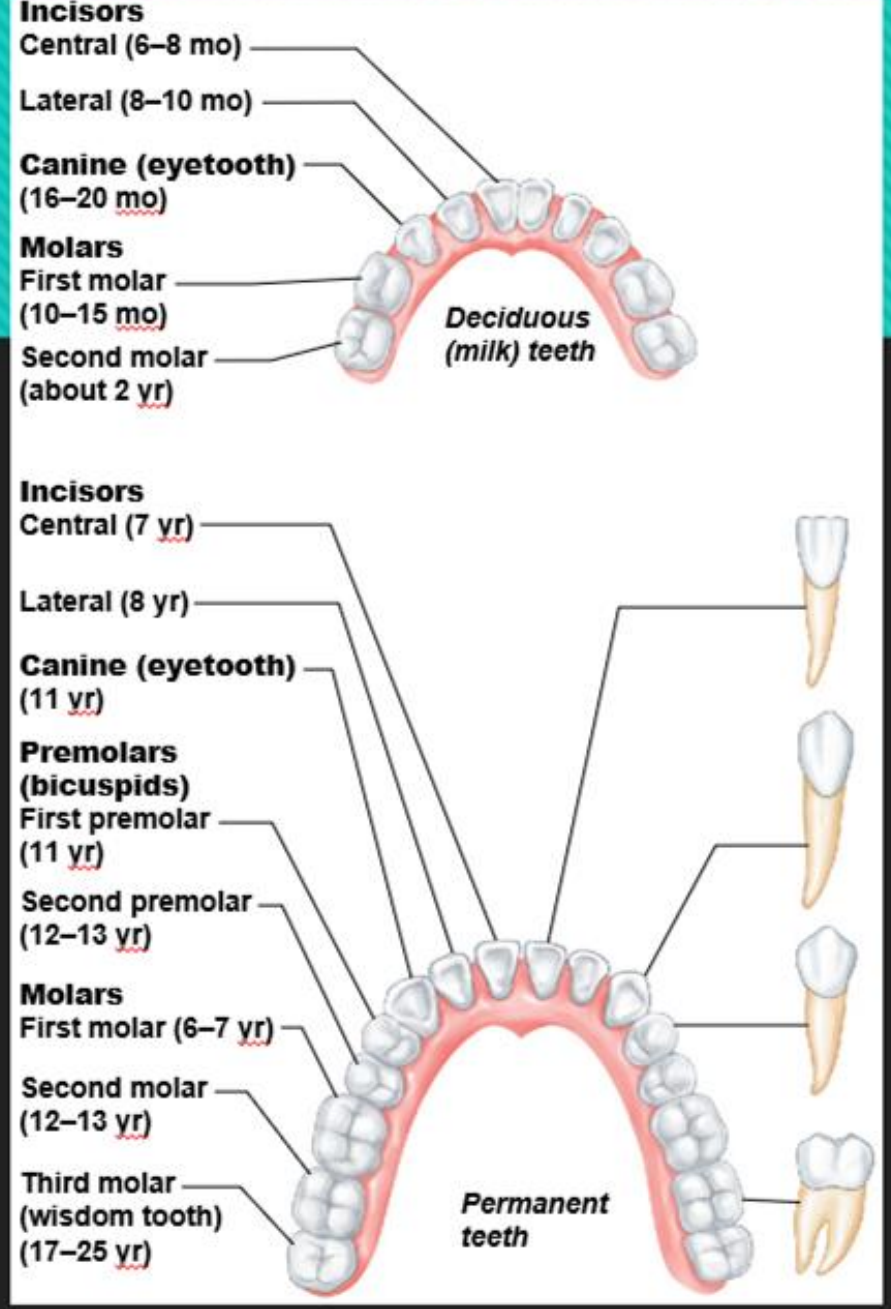
# Accessory Digestive Organs

- Teeth
- Salivary glands
- Pancreas
- Liver
- Gallbladder

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# Teeth

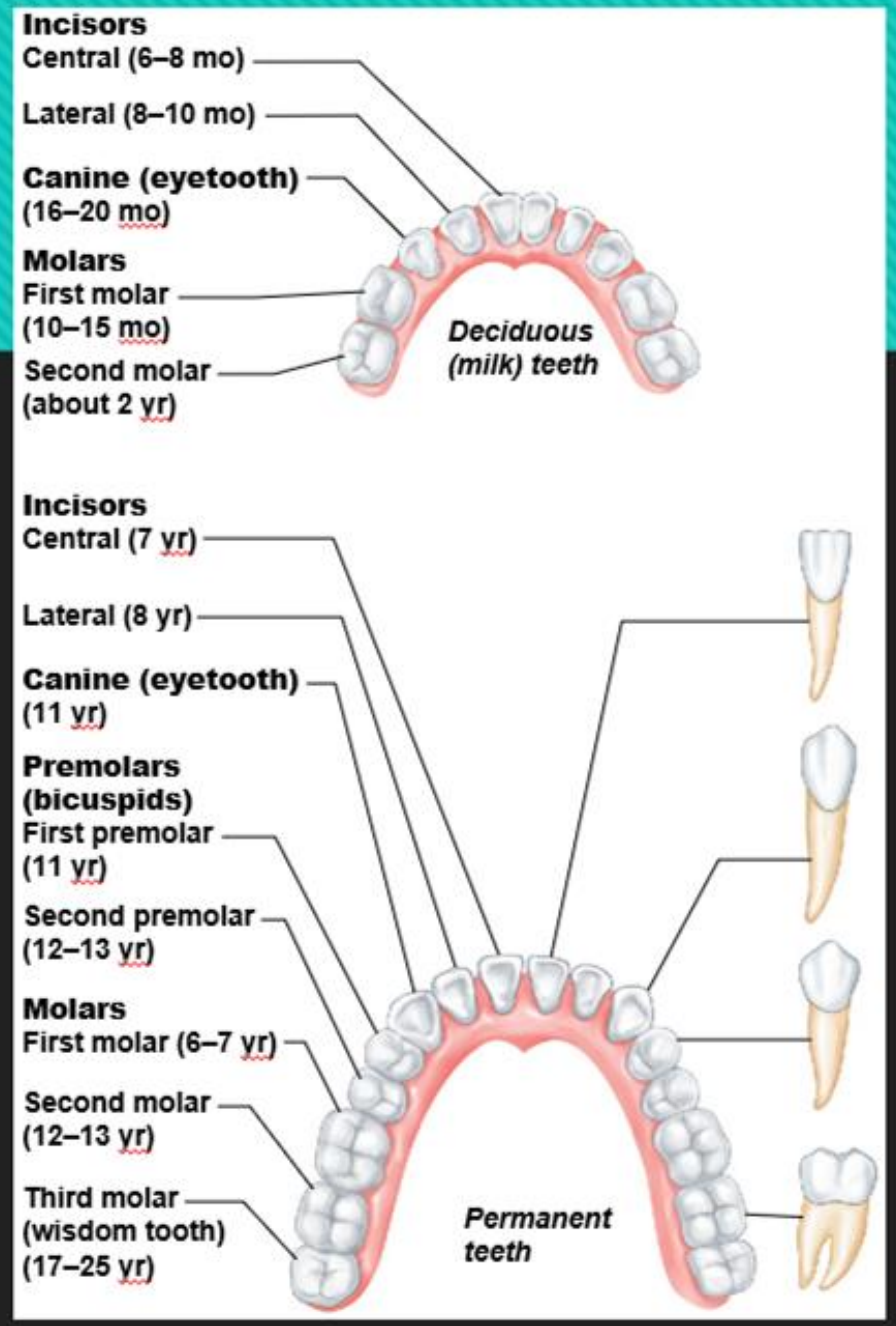
- Teeth masticate (chew) food into smaller fragments
- Humans have two sets of teeth during a lifetime
  1. Deciduous (baby or milk) teeth
     - A baby has 20 teeth by age 2
     - First teeth to appear are the lower central incisors
  2. Permanent teeth
     - Replace deciduous teeth between ages 6 and 12
     - A full set is 32 teeth (with the wisdom teeth)




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# Teeth

- Teeth are classified according to shape and function
  - Incisors—cutting
  - Canines —tearing or piercing
  - Premolars - grinding
  - Molars—grinding

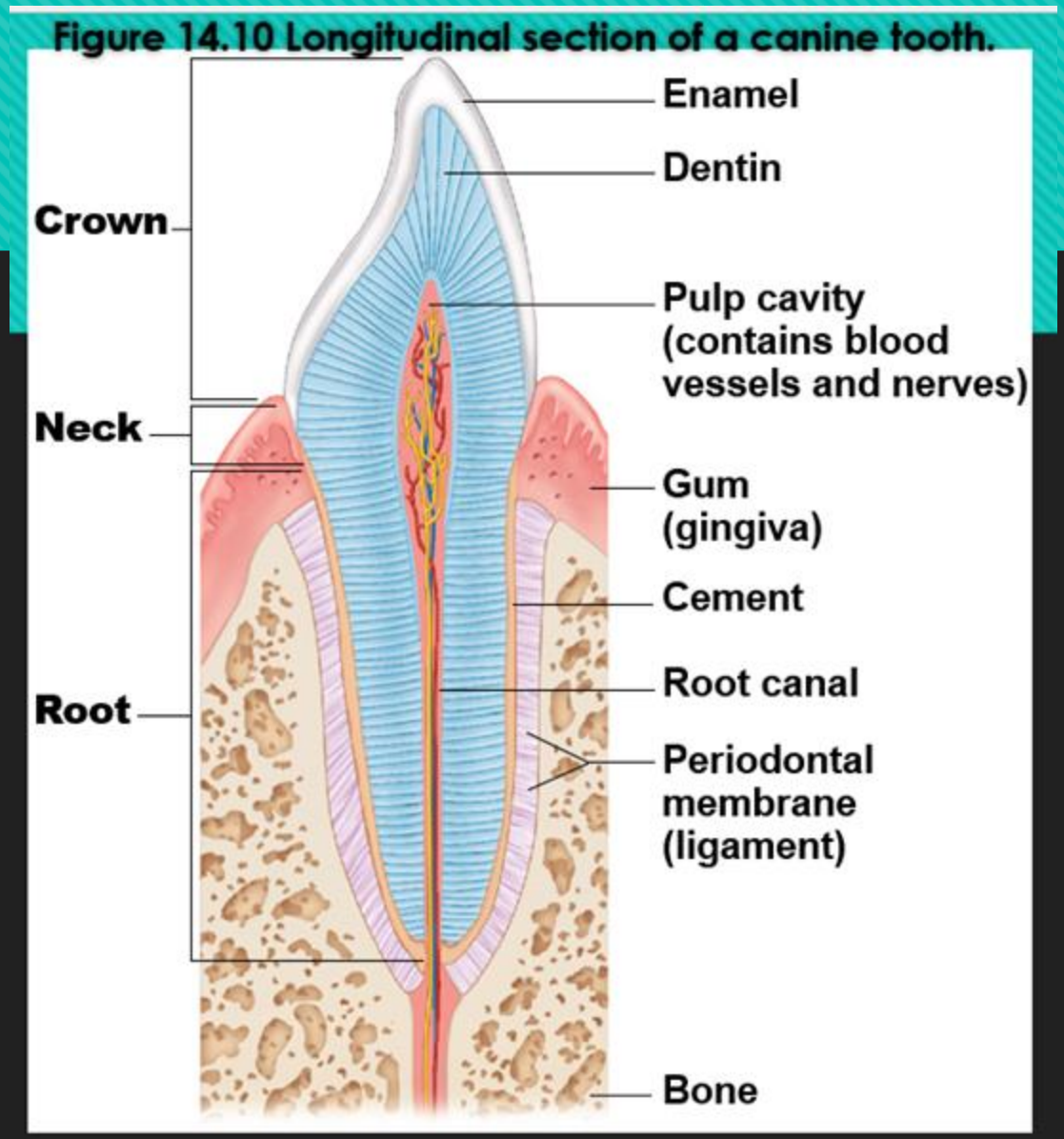



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# Teeth

## Two Major Regions of a Tooth

1. Crown—exposed part of tooth above the gingiva (gum)
   - Enamel—covers the crown
   - Dentin—found deep to the enamel and forms the bulk of the tooth, surrounds the pulp cavity
   - Pulp cavity—contains connective tissue, blood vessels, and nerve fibers (pulp)
   - Root canal—where the pulp cavity extends into the root




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# Teeth

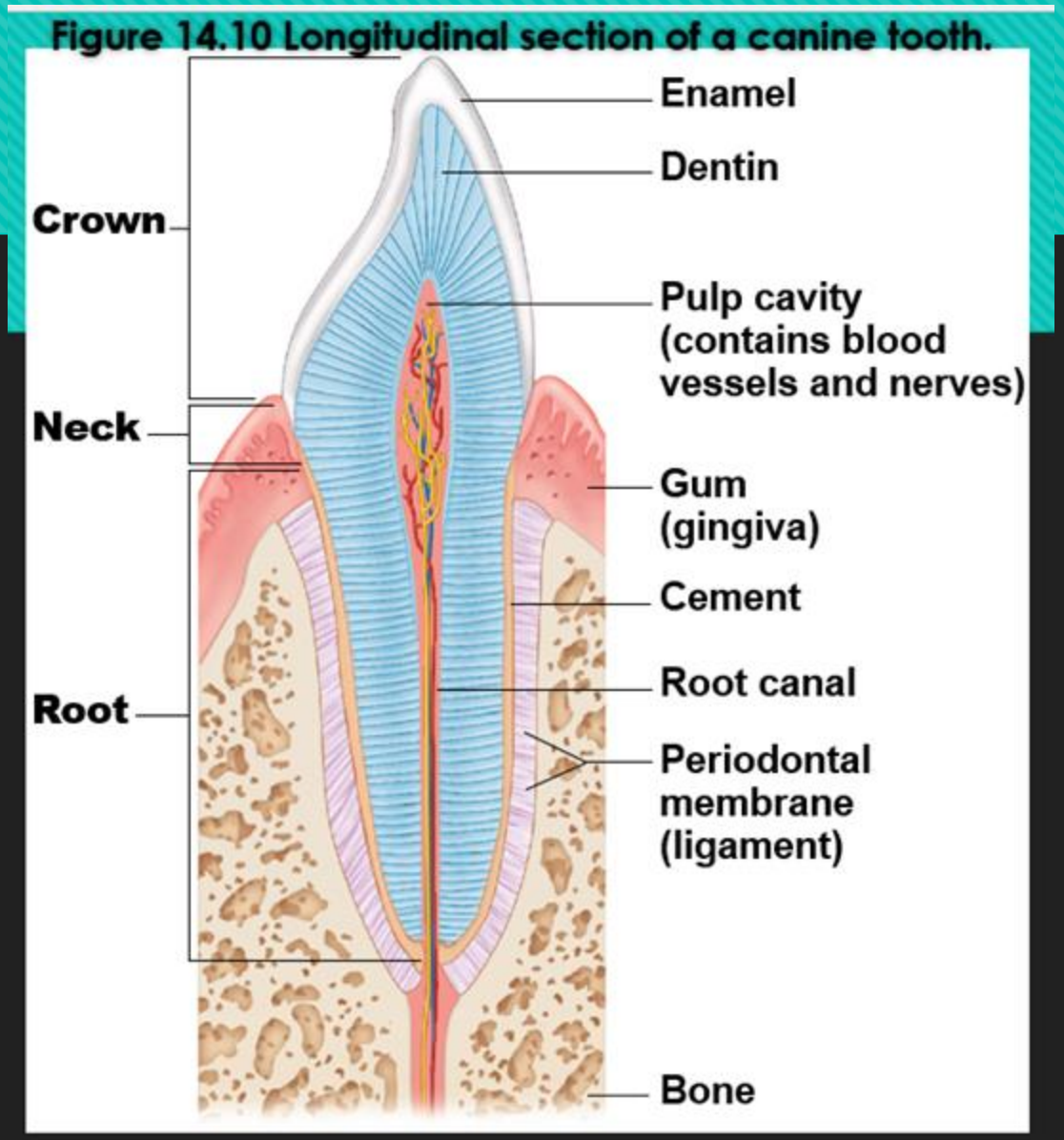
## Two Major Regions of a Tooth

### 2. Root

- Cement—covers outer surface and attaches the tooth to the periodontal membrane (ligament)
- Periodontal membrane holds tooth in place in the bony jaw

*Note*: The neck is a connector between the crown and root

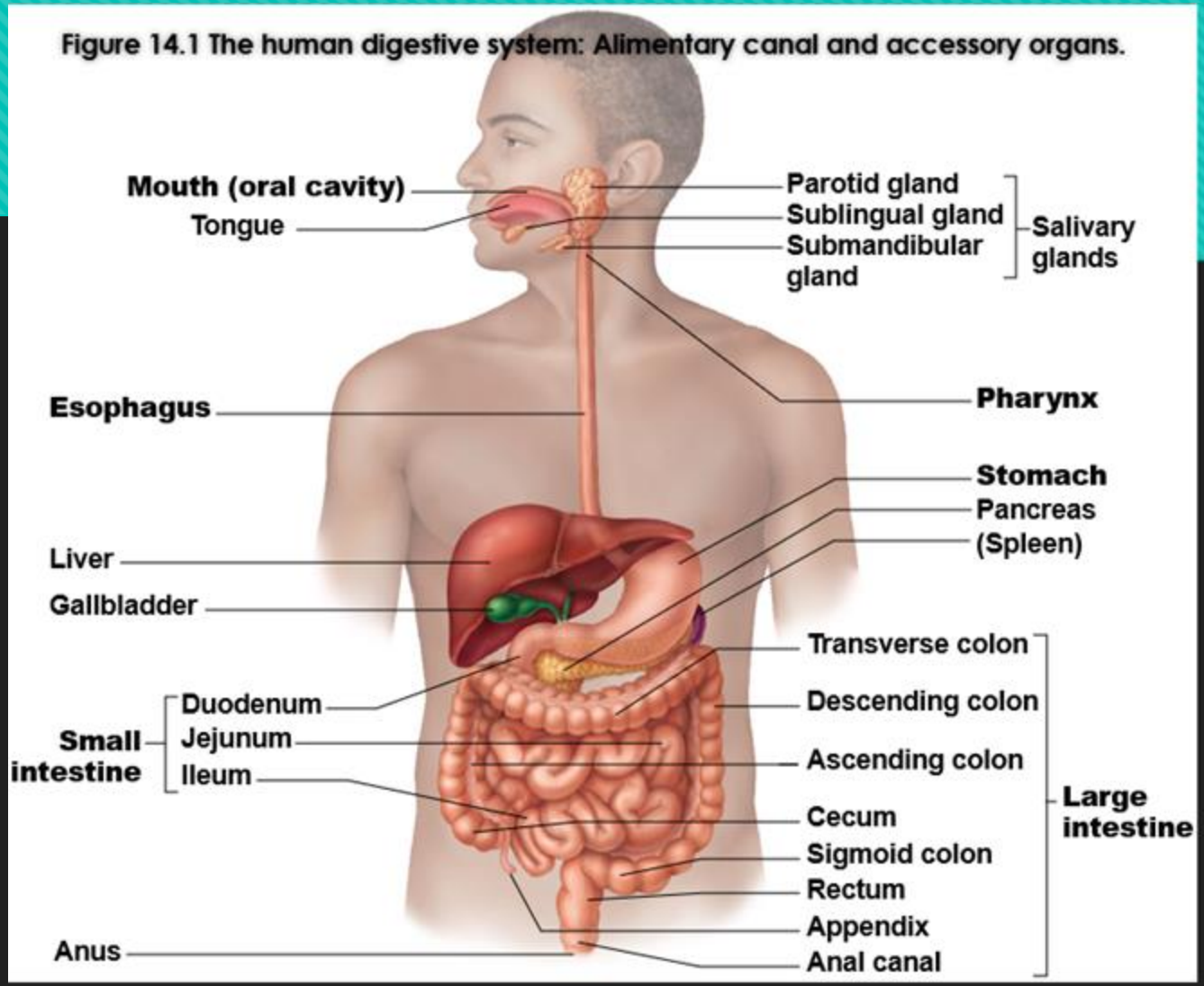
- Region in contact with the gum




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# Salivary Glands

- Three pairs of salivary glands empty secretions into the mouth
  1. Parotid glands
     - Largest; Found anterior to the ears
     - Mumps affect these salivary glands
  2. Submandibular glands
  3. Sublingual glands
     - Both submandibular and sublingual glands empty saliva into the floor of the mouth through small ducts




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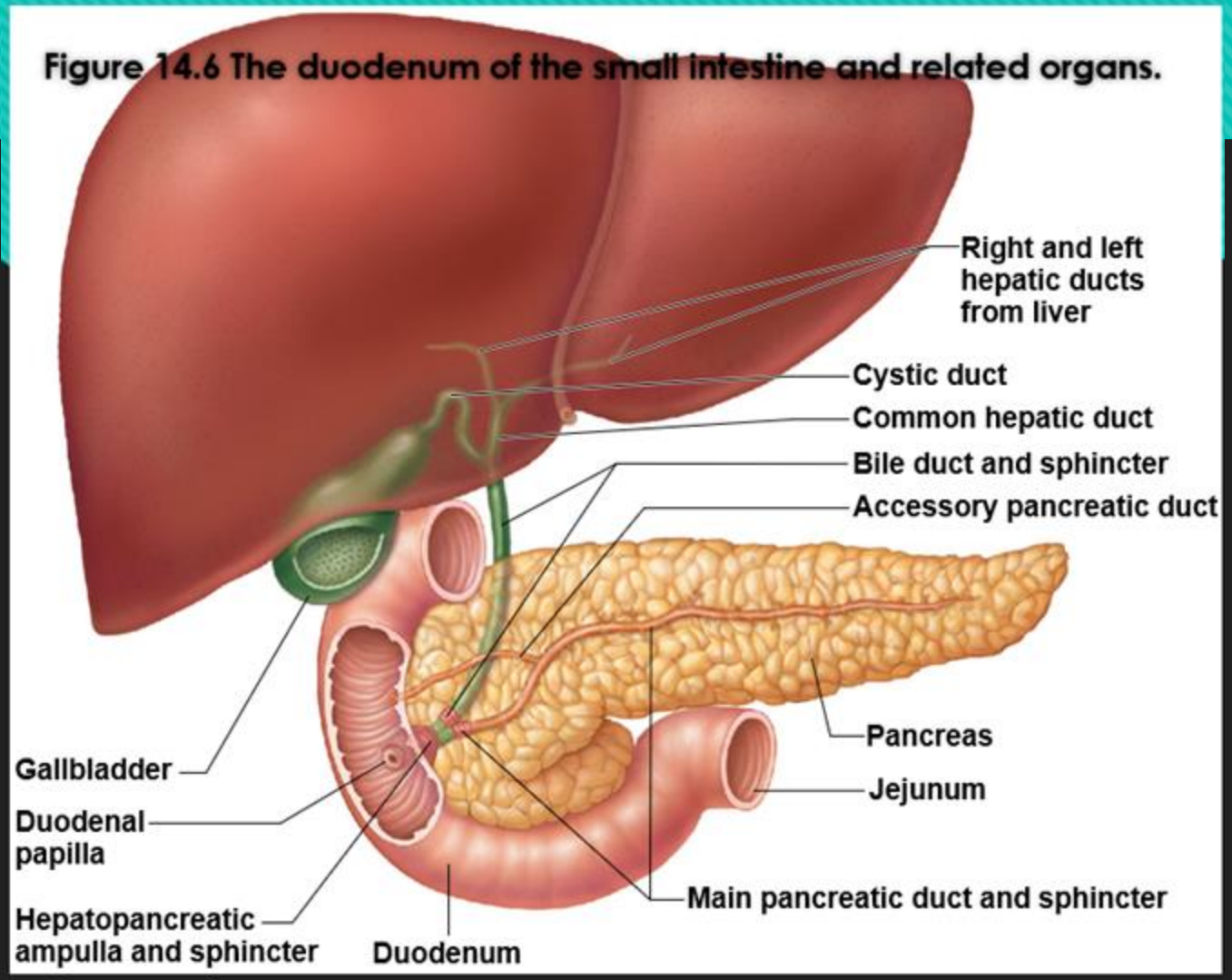
# Salivary Glands

- Saliva
  - Mixture of mucus and serous fluids
  - Helps to moisten and bind food together into a mass called a *bolus*
  - Contains:
    - Salivary amylase—begins starch digestion
    - Lysozymes and antibodies—inhibit bacteria
  - Dissolves chemicals so they can be tasted

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# Pancreas

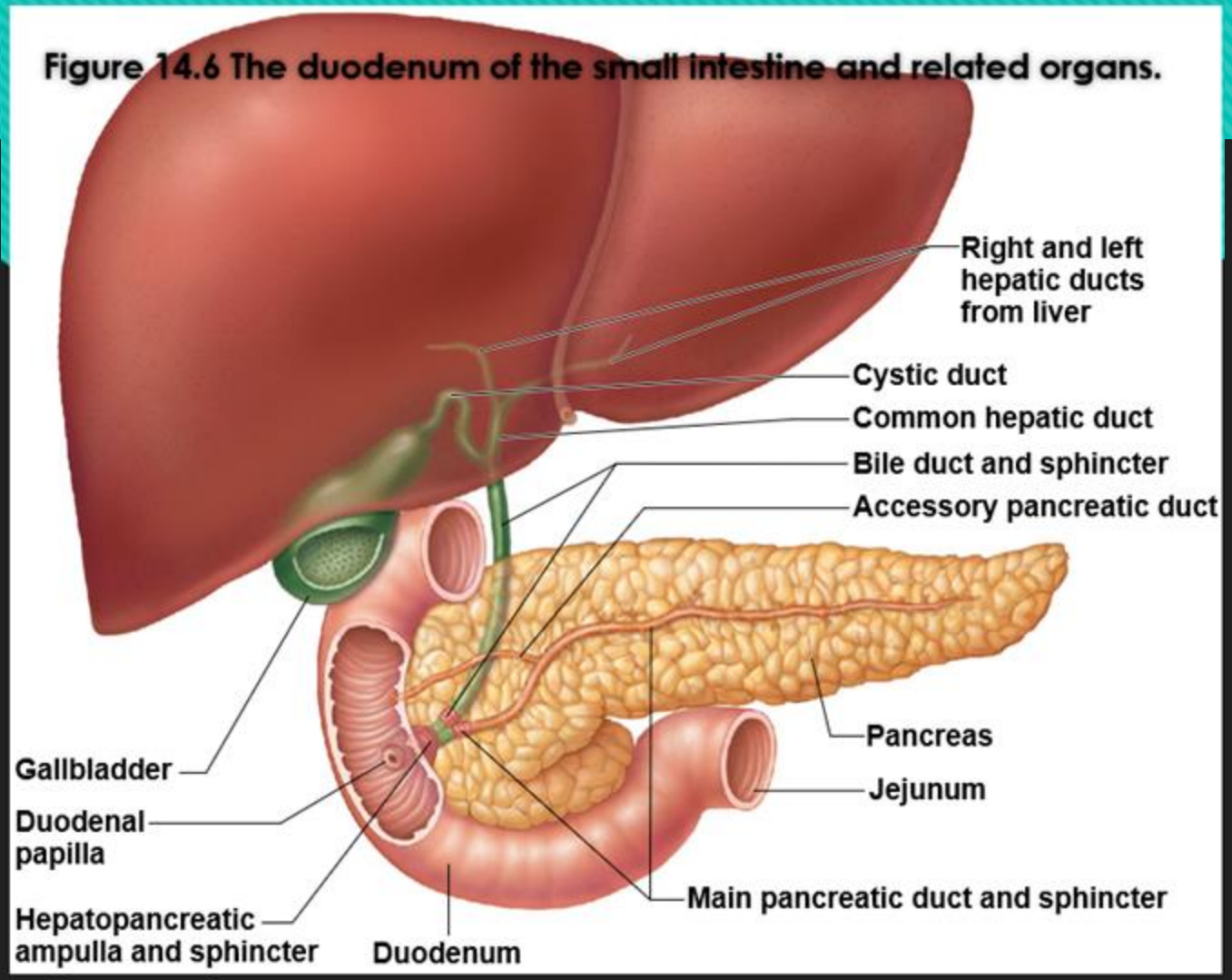
- Soft, triangular gland
- Found at the retroperitoneal
- Extends across the abdomen from spleen to duodenum




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# Pancreas

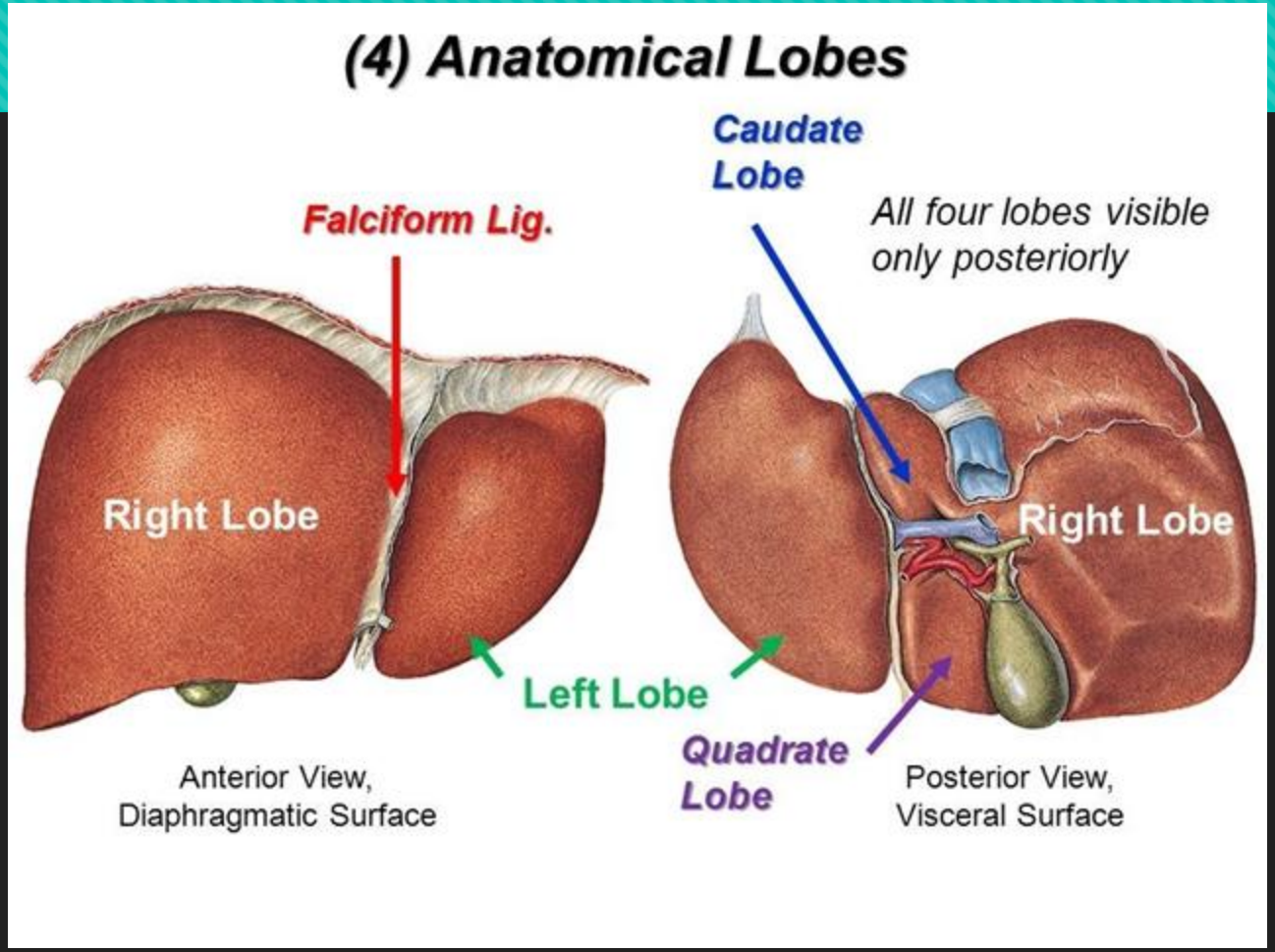
- Produces a wide spectrum of digestive enzymes that break down all categories of food
- Secretes enzymes into the duodenum
- Alkaline fluid introduced with enzymes neutralizes acidic chyme coming from stomach
- Hormones produced by the pancreas
  - Insulin
  - Glucagon




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# Liver

- Largest gland in the body
- Located on the right side of the body under the diaphragm
- Consists of four lobes suspended from the diaphragm and abdominal wall by the falciform ligament




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# Liver

**MAJOR FUNCTIONS**

(1) bile production
(2) storage of glycogen, lipids, copper, iron
(3) Interconversion of nutrients
(4) detoxification
(5) phagocytosis
(6) synthesis of blood components

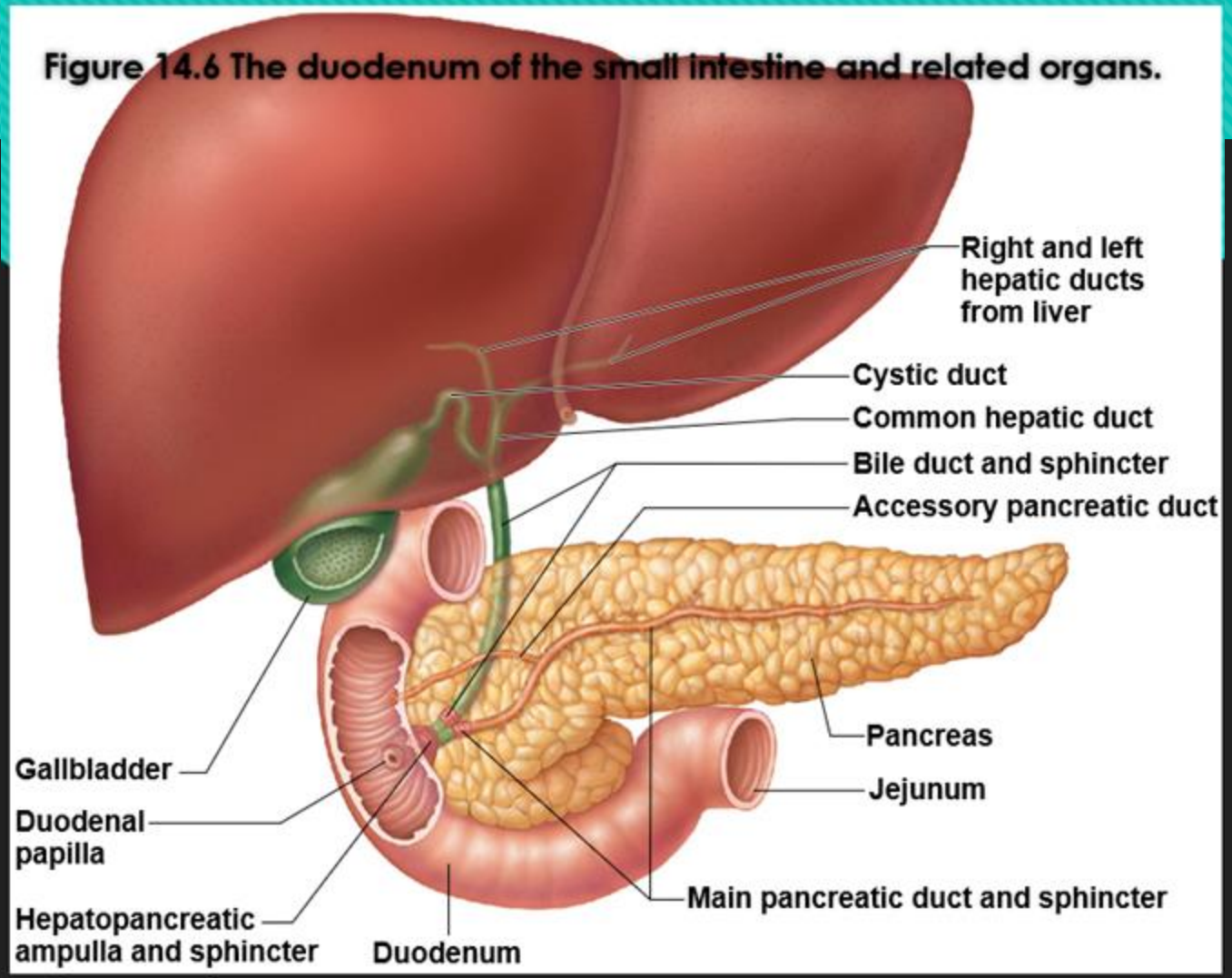
# Liver

- Digestive role is to "produce bile" (600-1000ml/ day)
  - Bile leaves the liver through the common hepatic duct and enters duodenum through the bile duct
  - Bile is yellow-green, watery solution containing:
    - Bile salts and bile pigments (mostly bilirubin from the breakdown of hemoglobin)
    - Cholesterol, phospholipids, and electrolytes
  - Bile emulsifies (breaks down) fats

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# Gallbladder

- Green sac found in a shallow fossa in the inferior surface of the liver
  - When no digestion is occurring, bile backs up the cystic duct for storage in the gallbladder
  - While in the gallbladder, bile is concentrated by the removal of water (5-10x more concentrated than that released by the liver)
  - When fatty food enters the duodenum, the gallbladder spurts out stored bile




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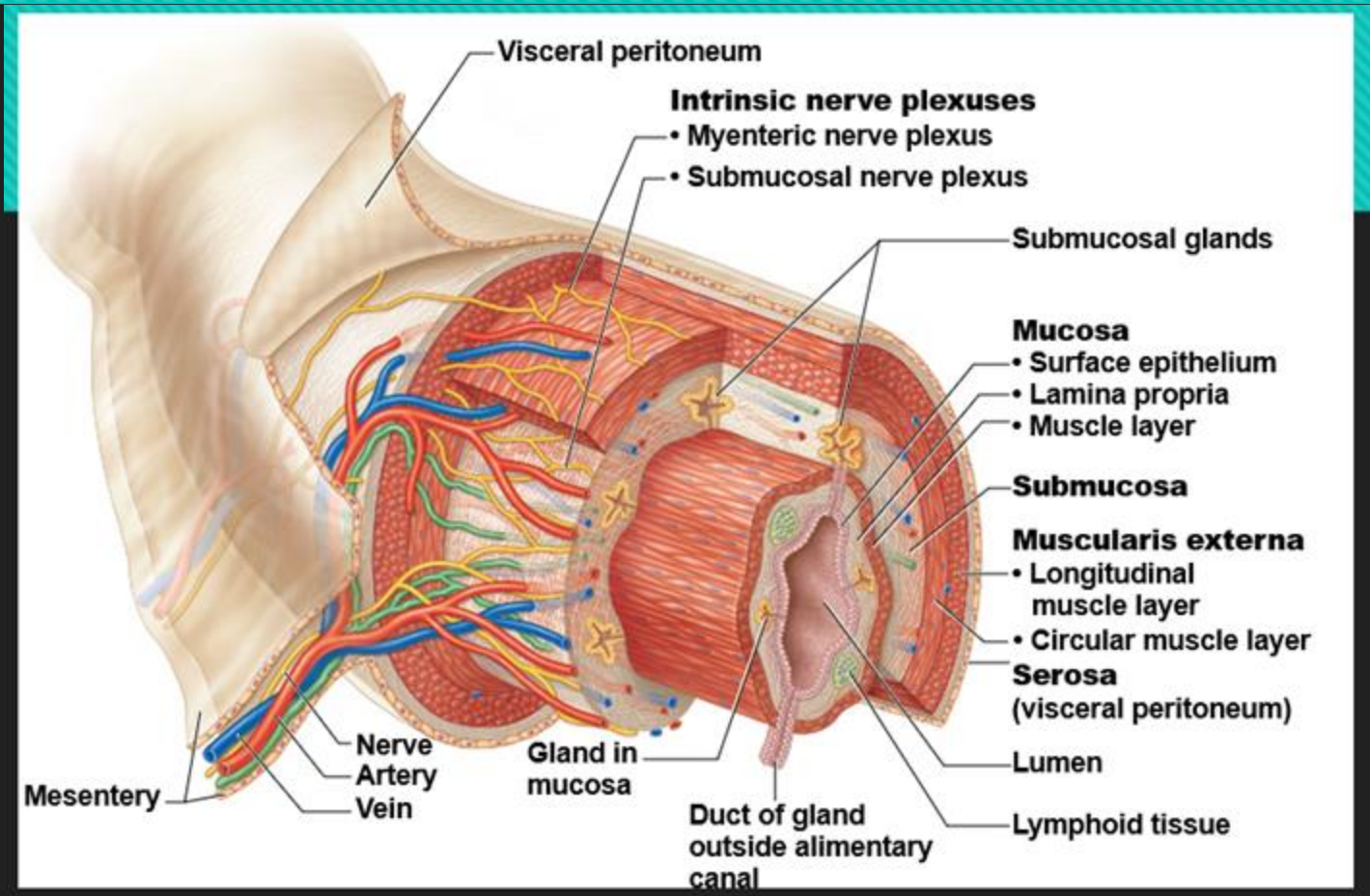
# OUTLINE OF TOPICS:



- Organs and Functions of the Digestive System
- Trunks or Layers of the Digestive Tract
- Activities and Movements in the Digestive System
- Secretions in the Digestive System
- Metabolism

# Layers of Tissue in the Alimentary Canal Organs

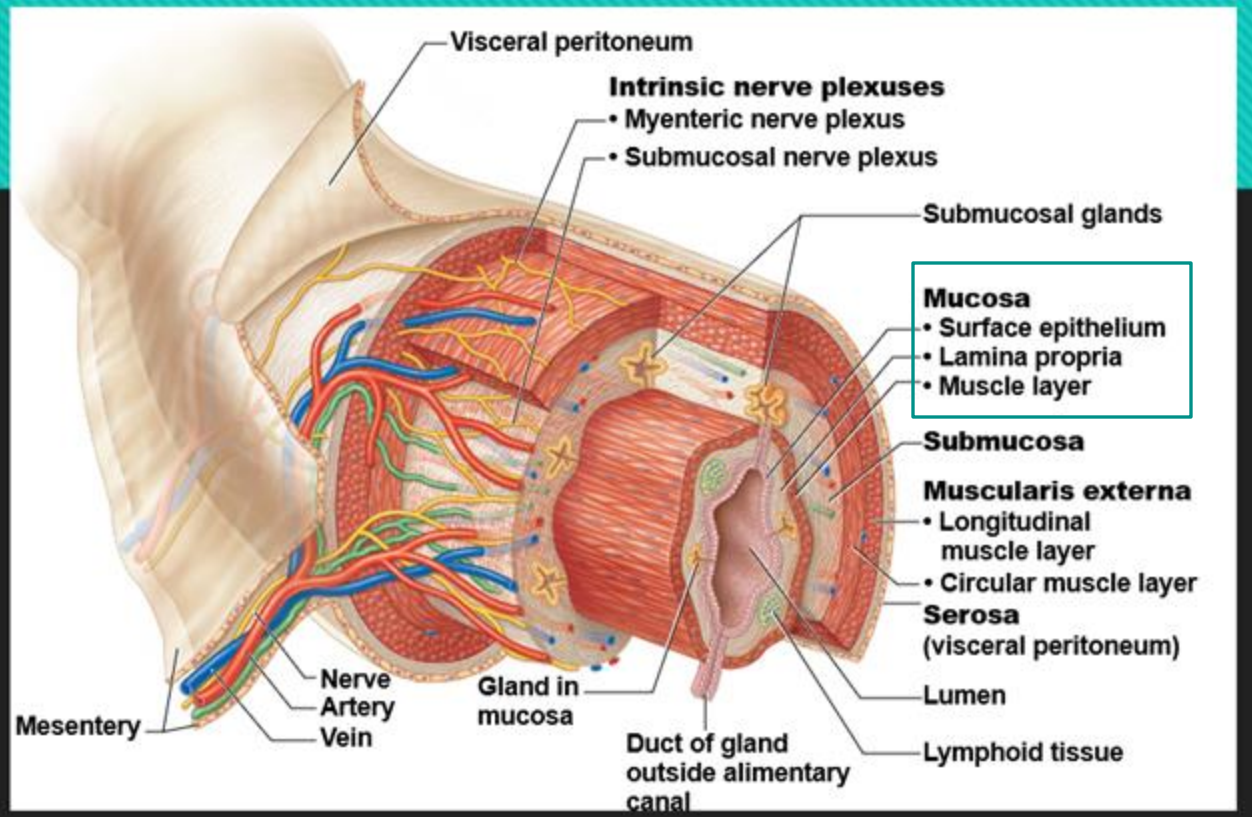
- Summary of the four layers from innermost to outermost, from esophagus to the large intestine
  1. Mucosa
  2. Submucosa
  3. Muscularis externa
  4. Serosa




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# Layers of Tissue in the Alimentary Canal Organs

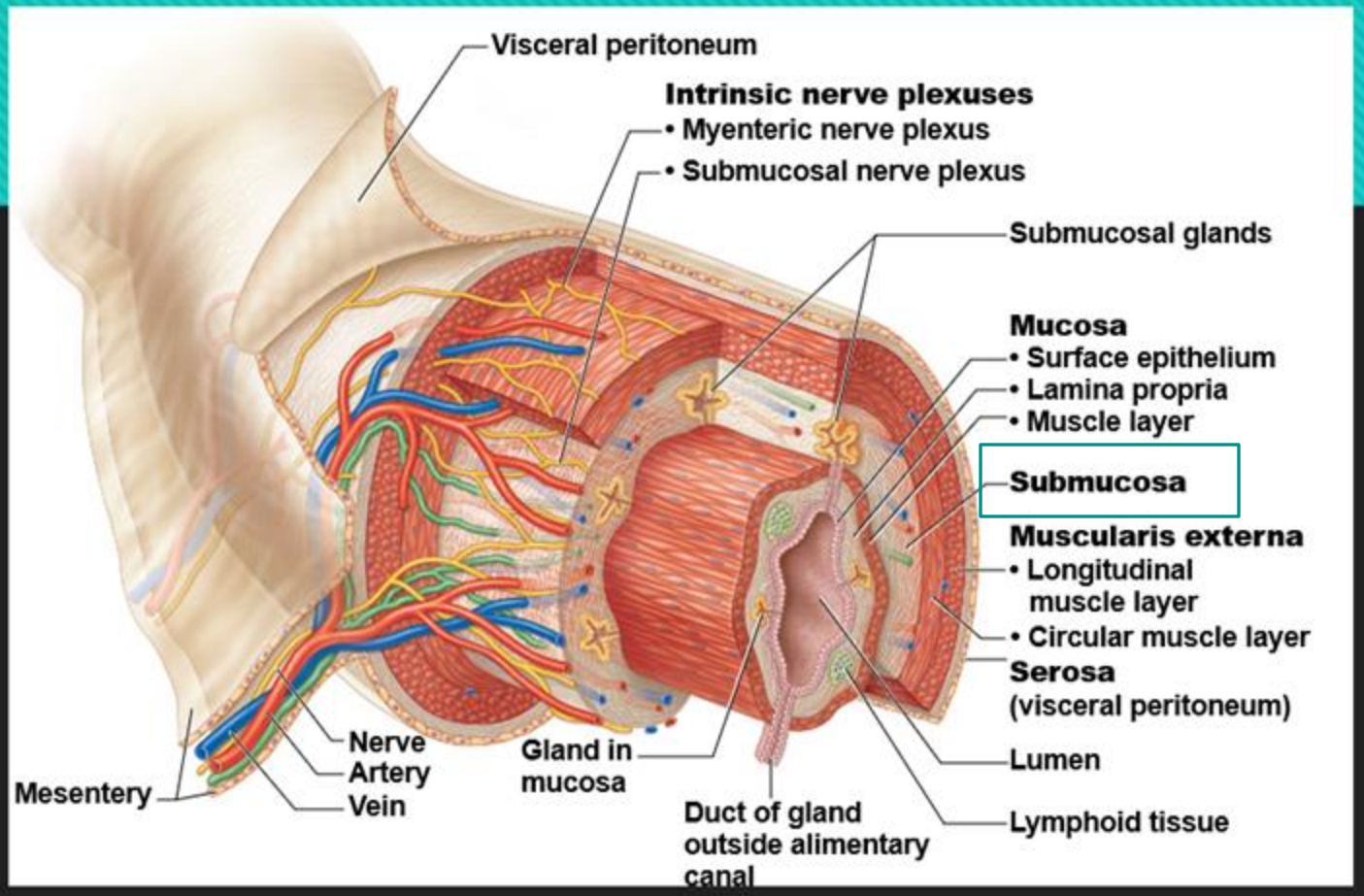
1. Mucosa
   - Innermost, moist membrane consisting of:
     - Surface epithelium that is mostly simple columnar epithelium (except for esophagus—stratified squamous epithelium)
     - Small amount of connective tissue (lamina propria)
     - Scanty smooth muscle layer
   - Lines the cavity (known as the *lumen*)




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# Layers of Tissue in the Alimentary Canal Organs

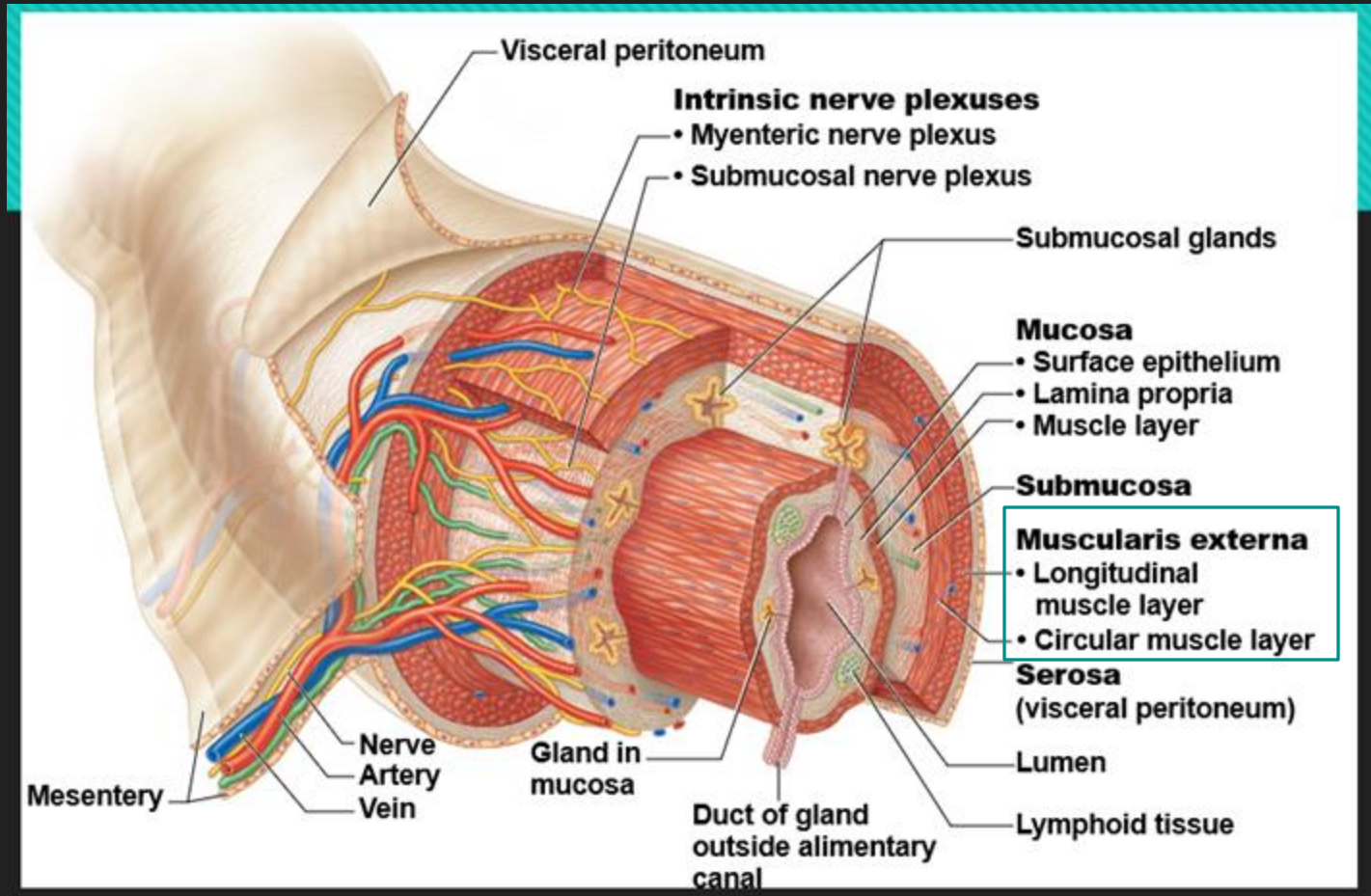
2. Submucosa
   - Just beneath the mucosa
   - Soft connective tissue with blood vessels, nerve endings, mucosa-associated lymphoid tissue, and lymphatic vessels




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# Layers of Tissue in the Alimentary Canal Organs

3. Muscularis externa—smooth muscle
   - Inner circular layer
   - Outer longitudinal layer



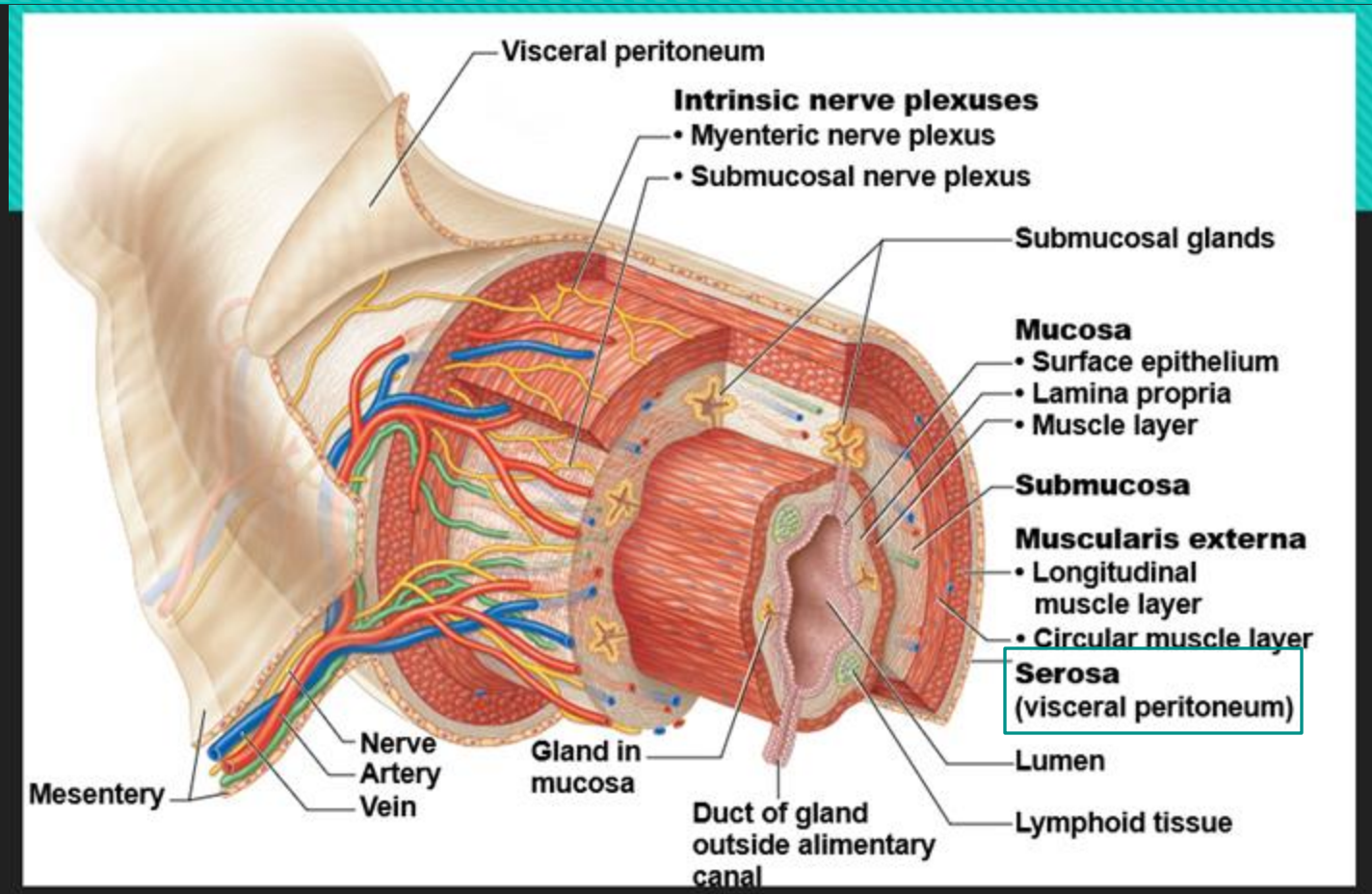

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# Layers of Tissue in the Alimentary Canal Organs

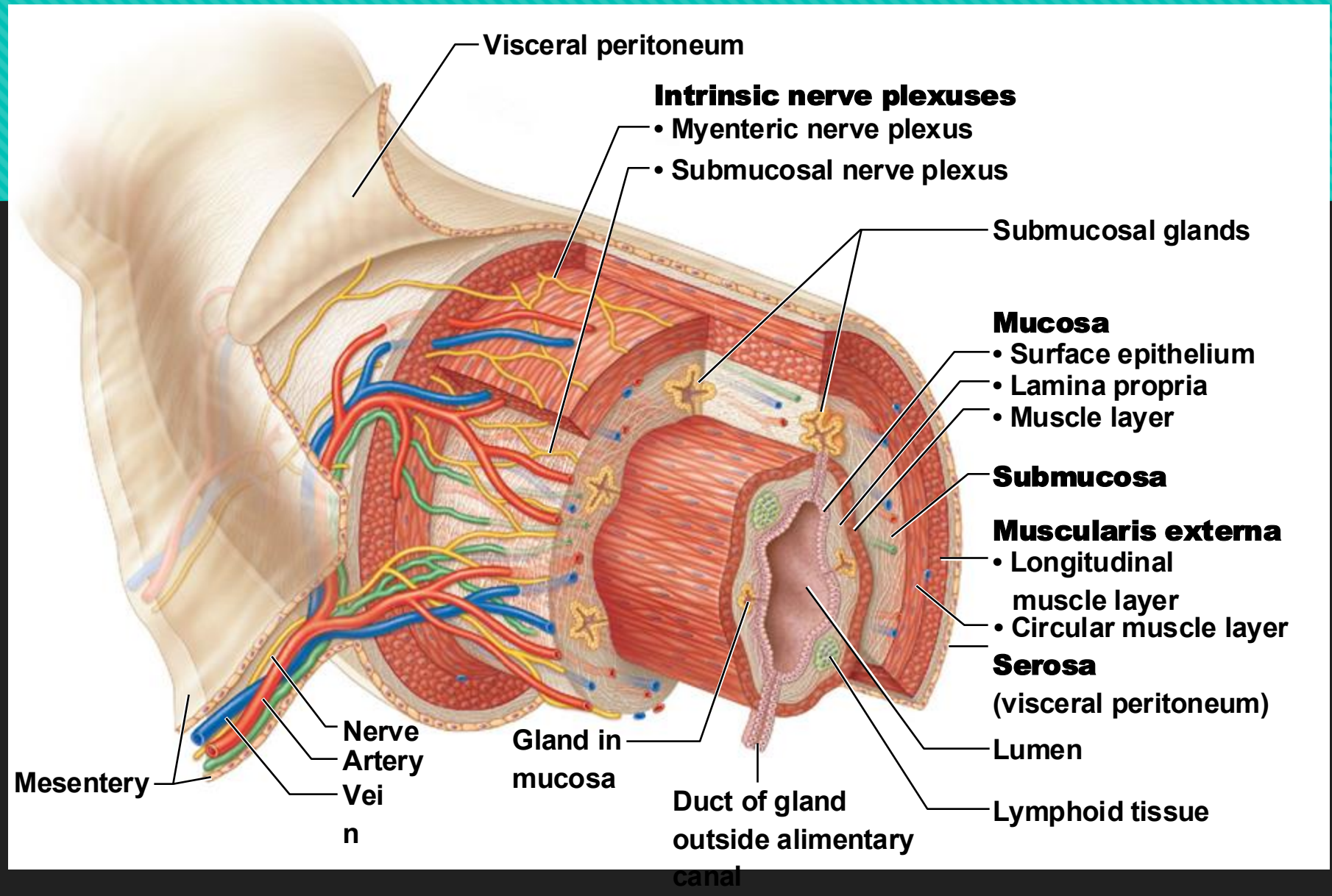
4. Serosa
   - outermost layer of the wall; contains fluid-producing cells
     - Visceral peritoneum

     —innermost layer
     - Parietal peritoneum

     —outermost layer that lines the abdominopelvic cavity by way of the mesentery




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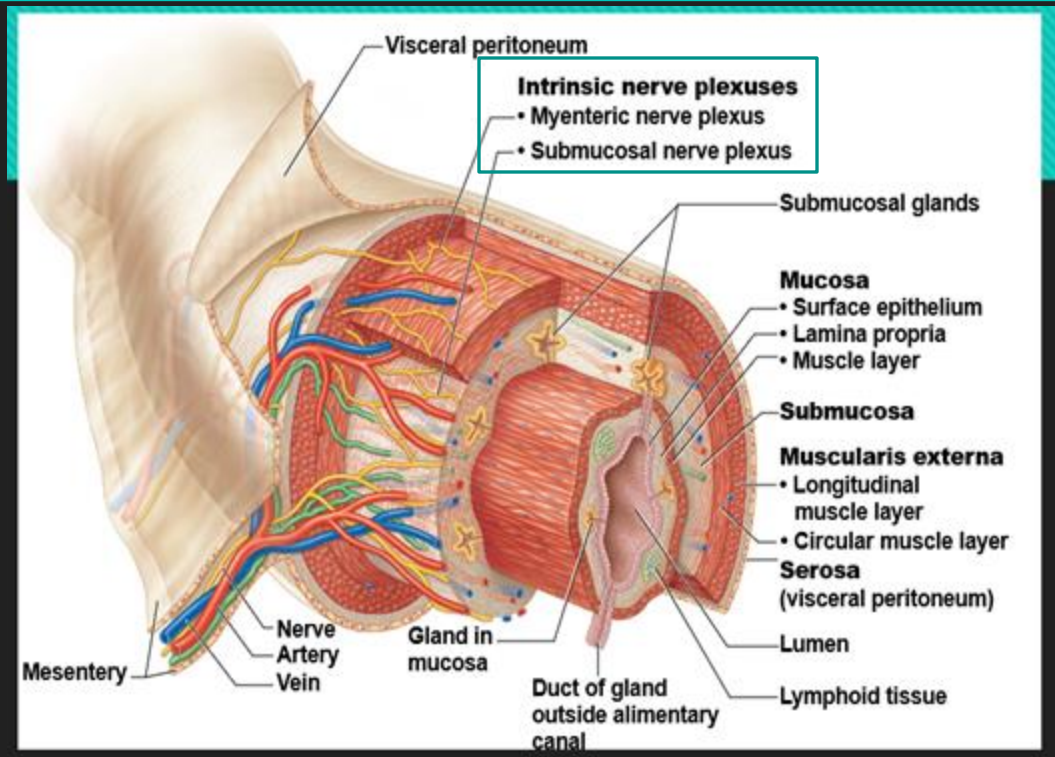

Visceral peritoneum
Intrinsic nerve plexuses
• Myenteric nerve plexus
• Submucosal nerve plexus
Submucosal glands
Mucosa
• Surface epithelium
• Lamina propria
• Muscle layer
Submucosa
Muscularis externa
• Longitudinal muscle layer
• Circular muscle layer
Serosa
(visceral peritoneum)
Lumen
Lymphoid tissue
Duct of gland outside alimentary canal
Gland in mucosa
Nerve
Artery
Vein
Mesentery

# Alimentary Canal Nerve Plexuses

The wall of the alimentary canal contains two intrinsic nerve plexuses that are part of the autonomic nervous system (ANS):

- **Submucosal (Meissner's) Nerve Plexus** – Located in the submucosa; regulates glandular secretions and local blood flow.
- **Myenteric (Auerbach's) Nerve Plexus** – Located between the circular and longitudinal layers of the muscularis externa; primarily controls gastrointestinal (GI) motility.

  Together, these plexuses form the **enteric nervous system (ENS)**, which regulates the motility and secretory activity of GI tract organs.



# OUTLINE OF TOPICS:



- Organs and Functions of the Digestive System
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- Activities and Movements in the Digestive System
- Secretions in the Digestive System
- Metabolism

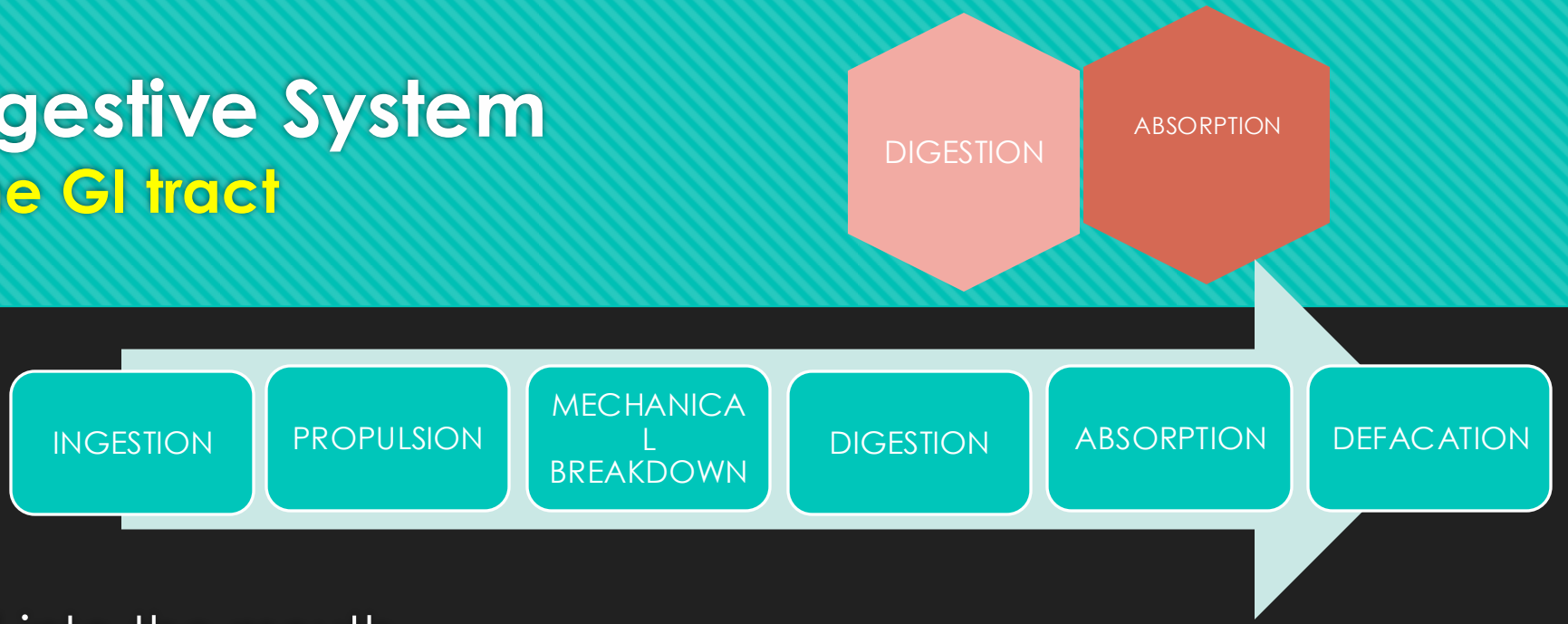
# The Digestive System Functions

- Ingestion—taking in food
- Digestion—breaking food into nutrient molecules
- Absorption—movement of nutrients into the bloodstream
- Defecation—excretes to rid the body of indigestible waste

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# Functions of the Digestive System

## Essential Processes of the GI tract




Ingestion—placing of food into the mouth

Propulsion—movement of foods from one region of the digestive system to another

- Peristalsis—alternating waves of contraction and relaxation that squeeze food along the GI tract
- Segmentation—movement of materials back and forth to foster mixing in the small intestine

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Figure 14.12a Peristaltic and segmental movements of the digestive tract.

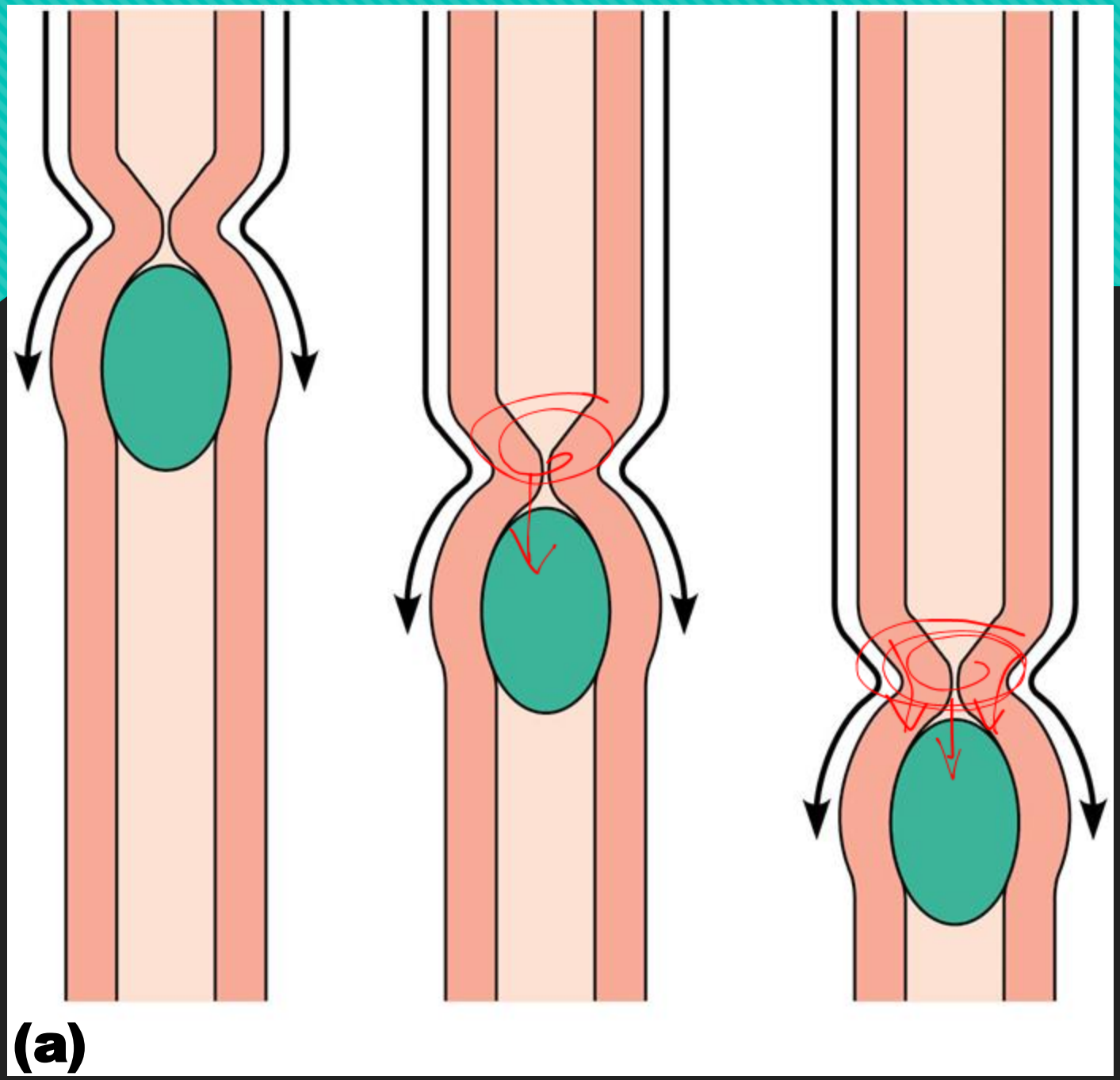
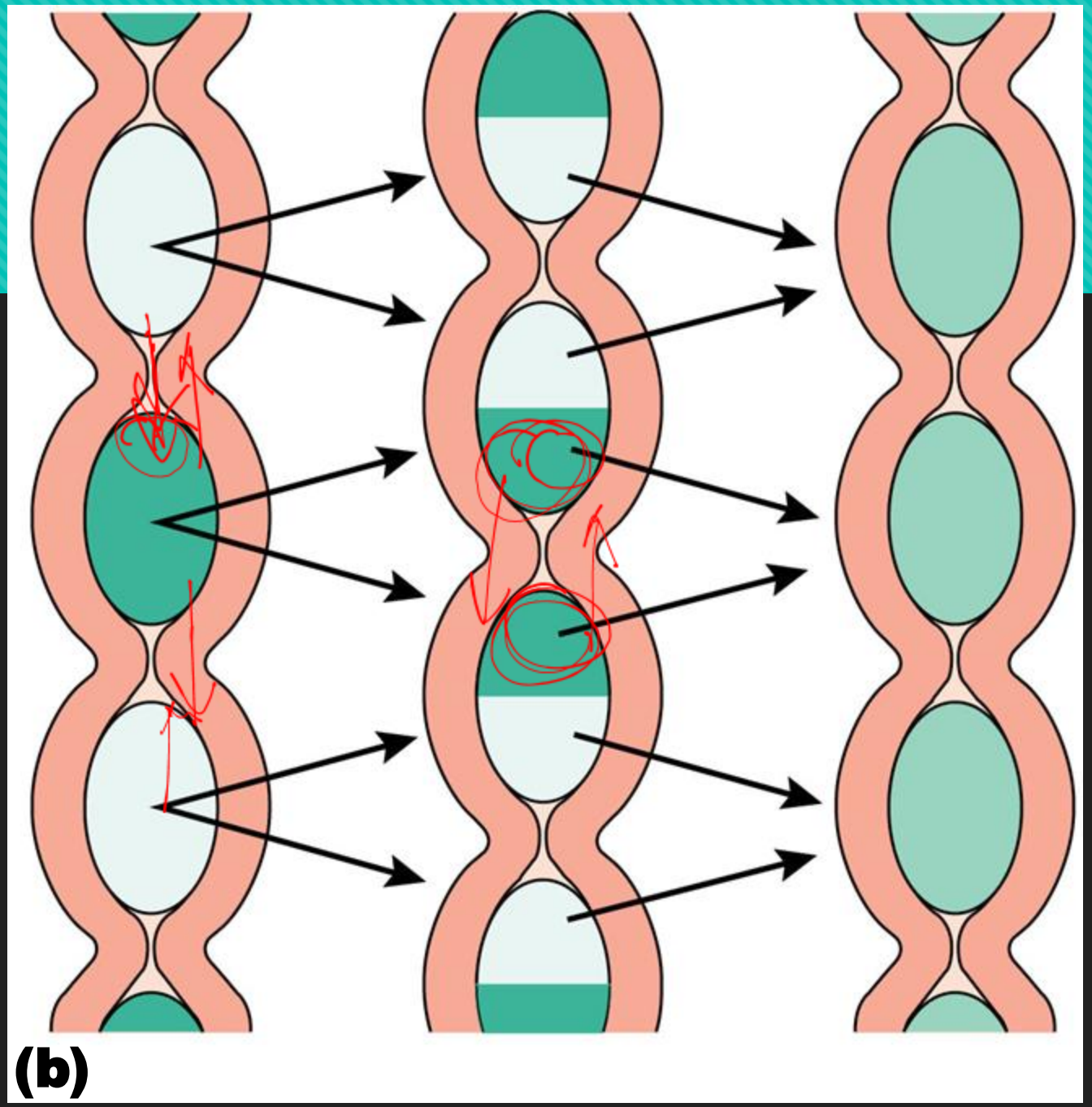
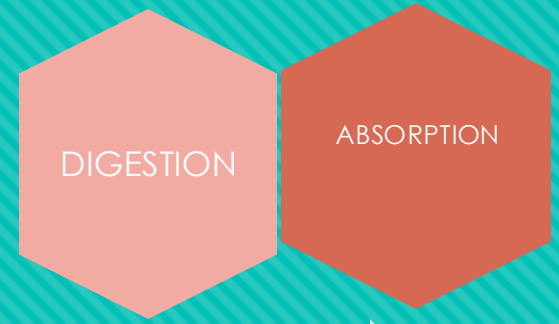


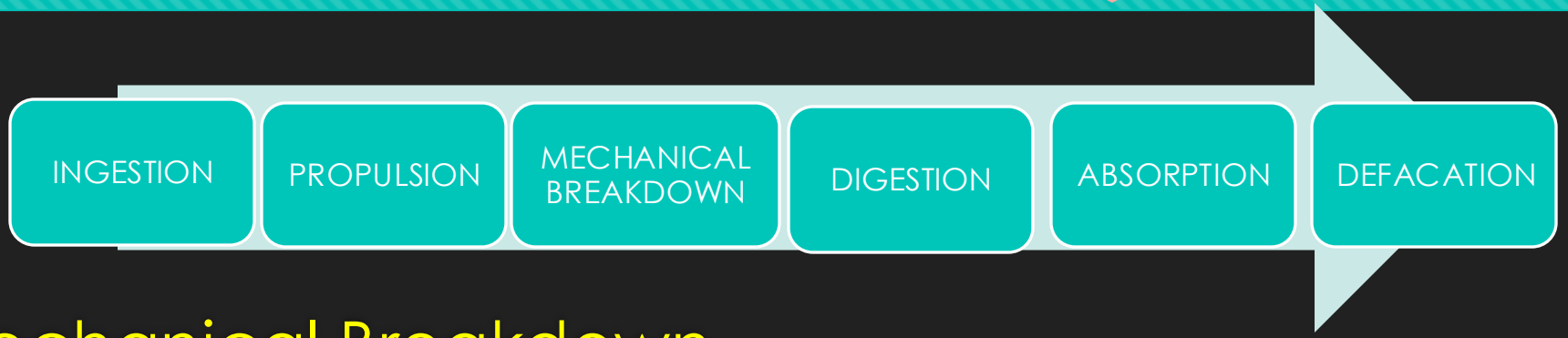
Figure 14.12b Peristaltic and segmental movements of the digestive tract.



# Functions of the Digestive System

## Essential Processes of the GI tract



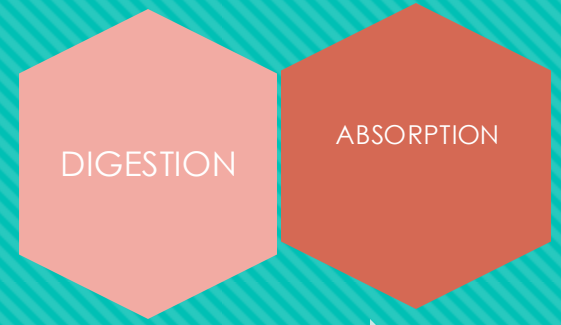



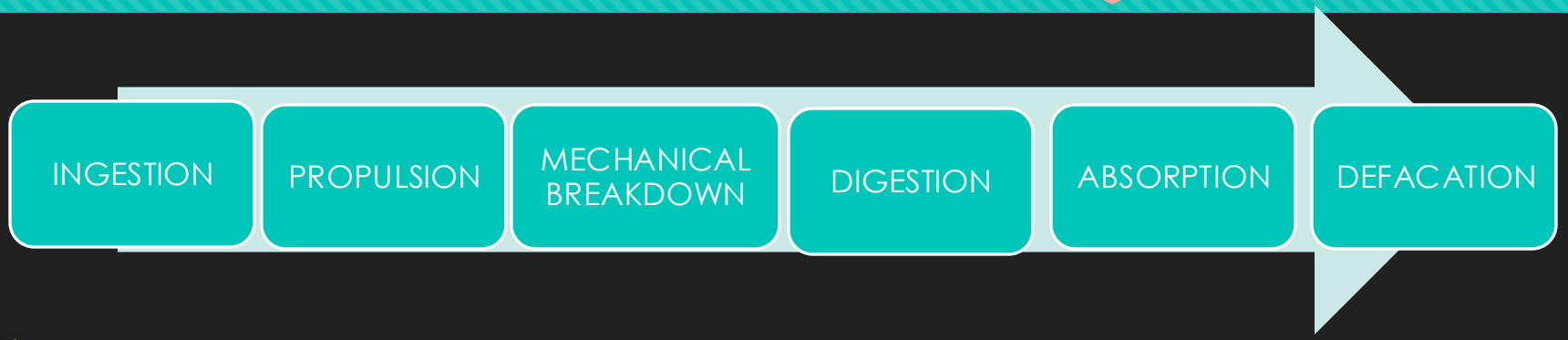

Food breakdown: Mechanical Breakdown

- Examples
  - Mixing of food in the mouth by the tongue
  - Churning of food in the stomach
  - Segmentation in the small intestine
- Mechanical digestion prepares food for further degradation by enzymes

# Functions of the Digestive System

## Essential Processes of the GI tract

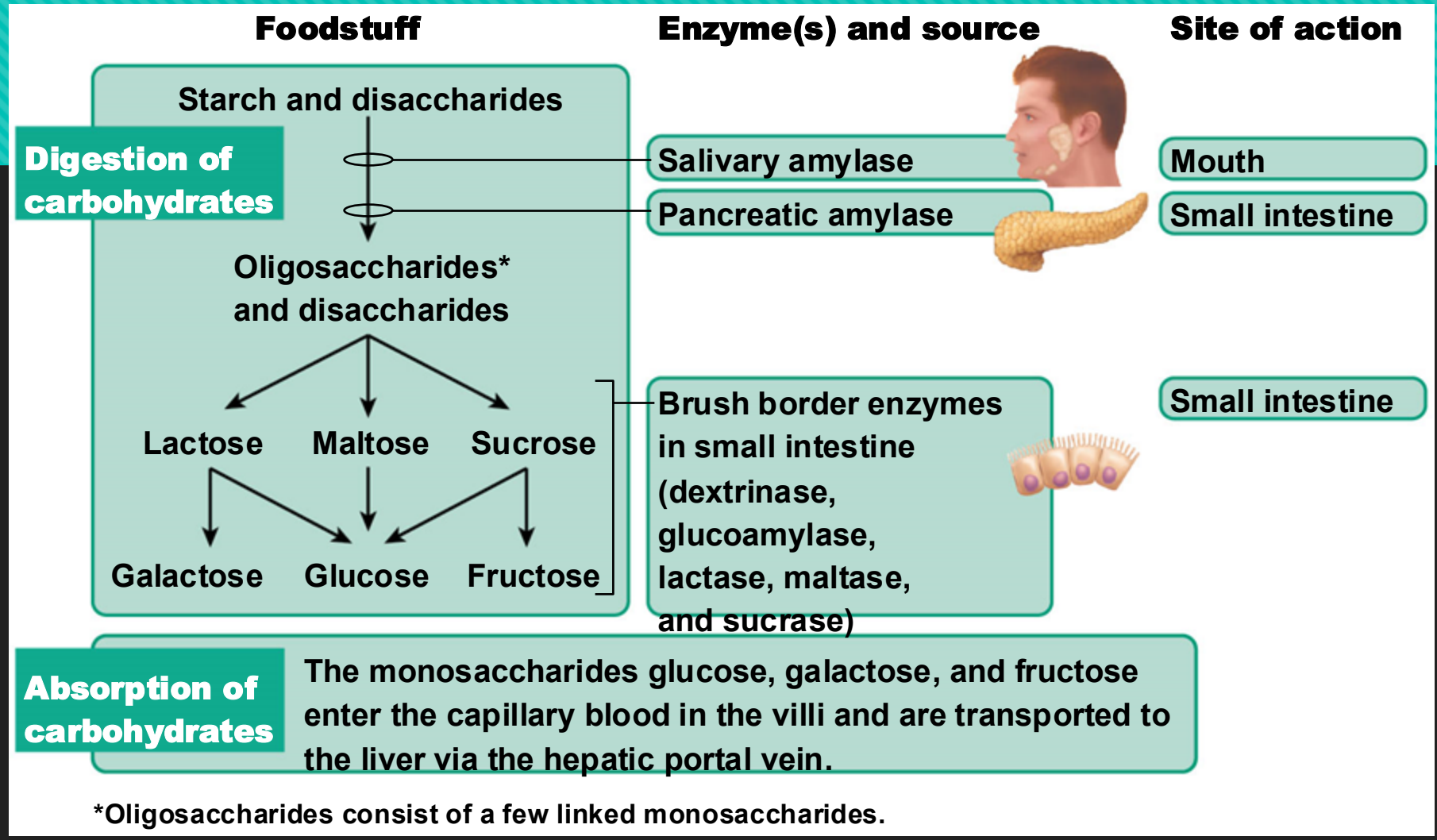


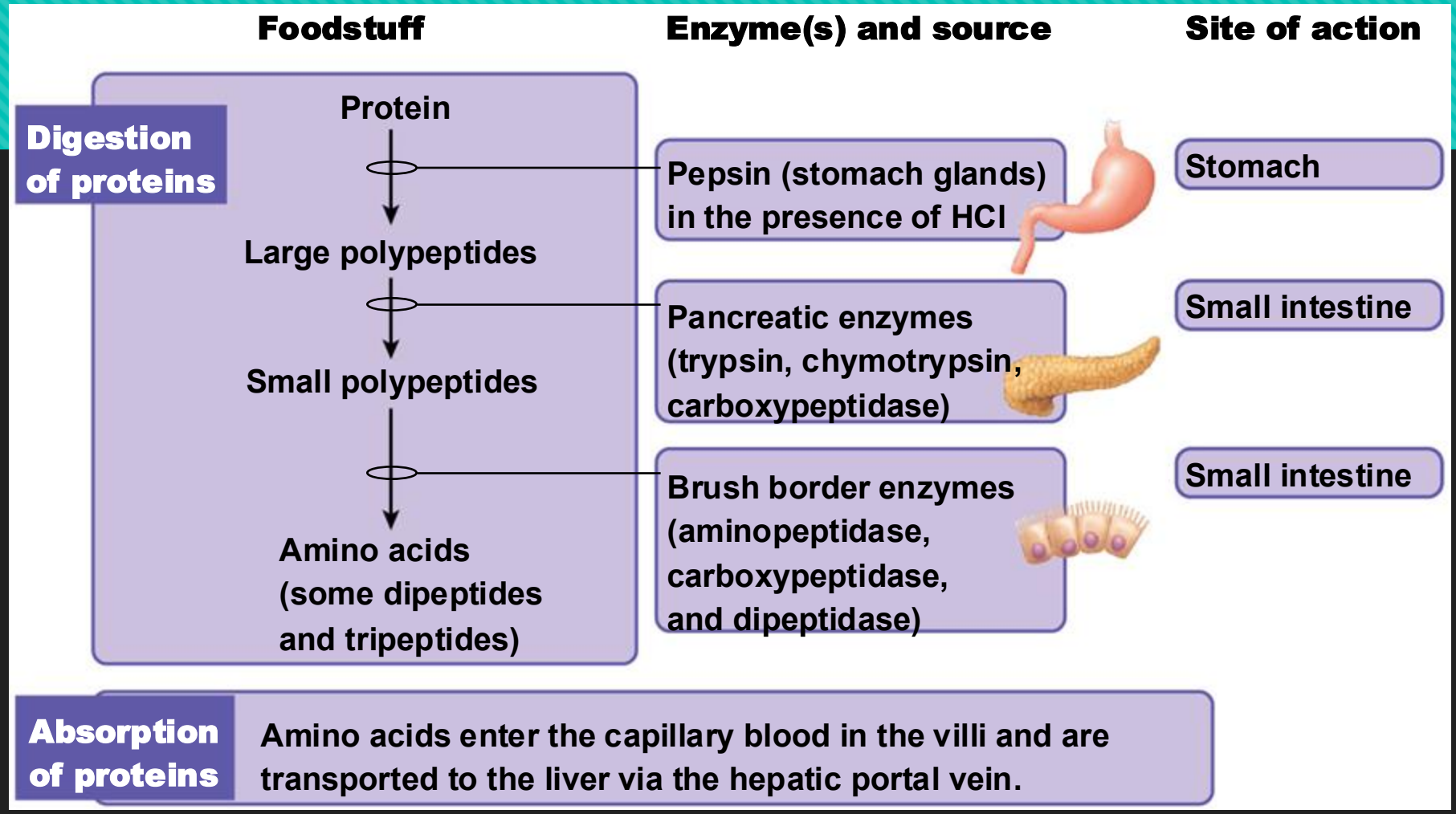

Food breakdown: Digestion

- Digestion occurs when enzymes chemically break down large molecules into their building blocks
- Each major food group uses different enzymes- HYDROLYSIS
  - Carbohydrates are broken down to monosaccharides (simple sugars)
  - Proteins are broken down to amino acids
  - Fats are broken down to fatty acids and glycerol

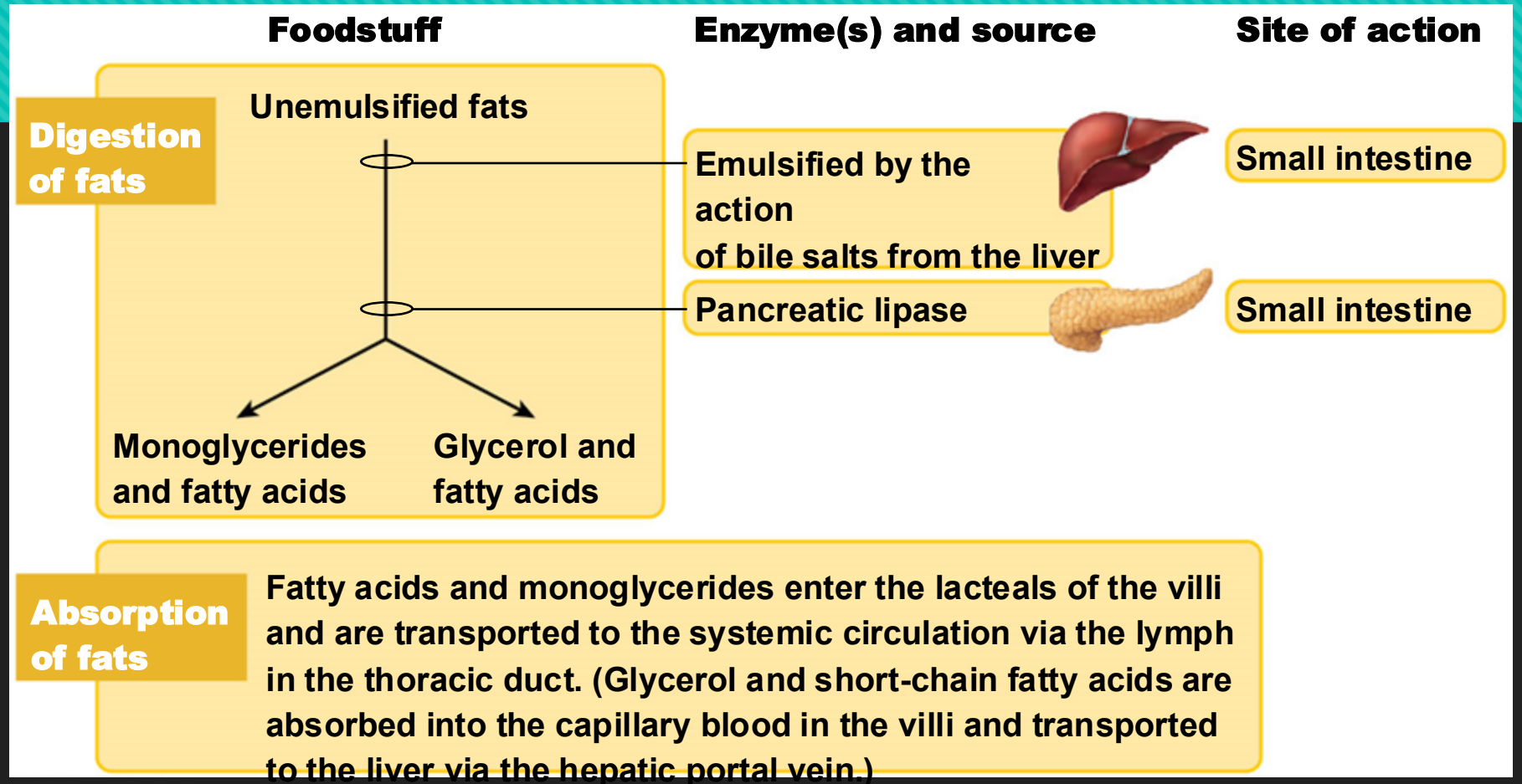
# Figure 14.13 Flowchart of digestion and absorption of foodstuffs



# Figure 14.13 Flowchart of digestion and absorption of foodstuffs

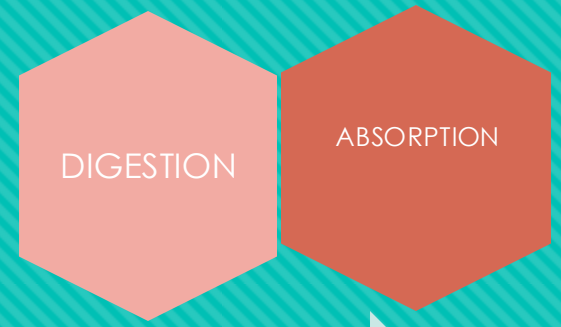


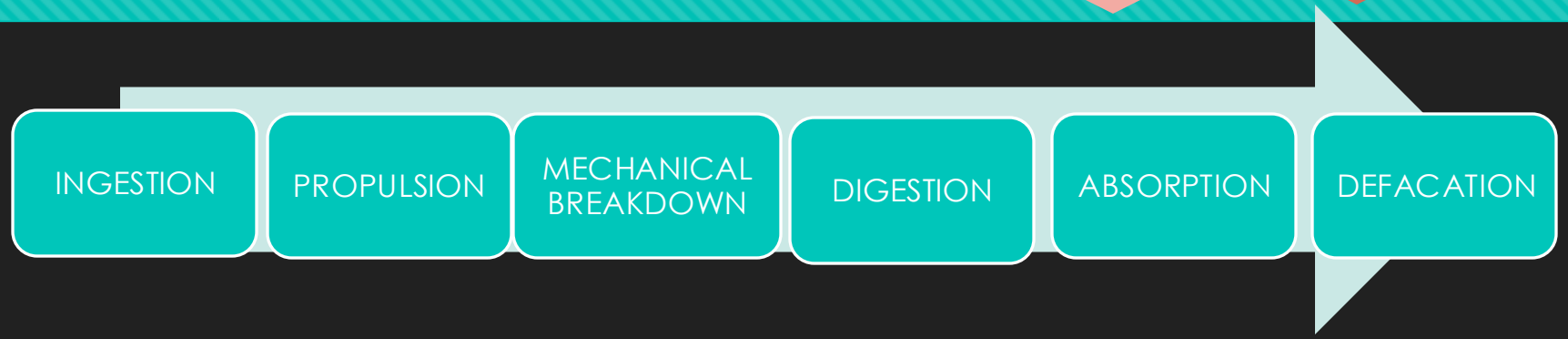
# Figure 14.13 Flowchart of digestion and absorption of foodstuffs



# Functions of the Digestive System

## Essential Processes of the GI tract




### Absorption

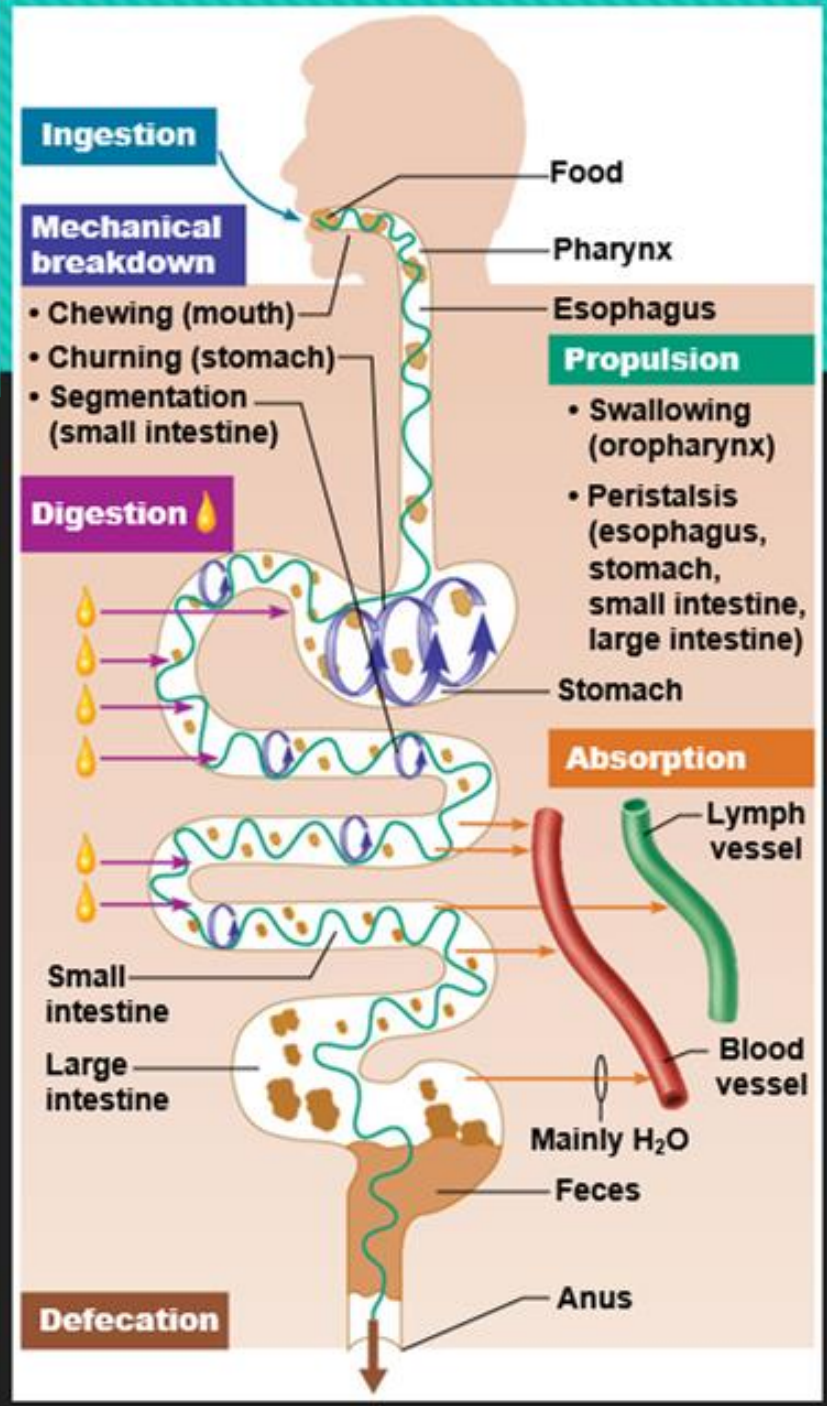
- End products of digestion are absorbed in the blood or lymph
- Food must enter mucosal cells and then move into blood or lymph capillaries

### Defecation

- Elimination of indigestible substances from the GI tract in the form of feces

# Activities Occurring in the Mouth, Pharynx, and Esophagus

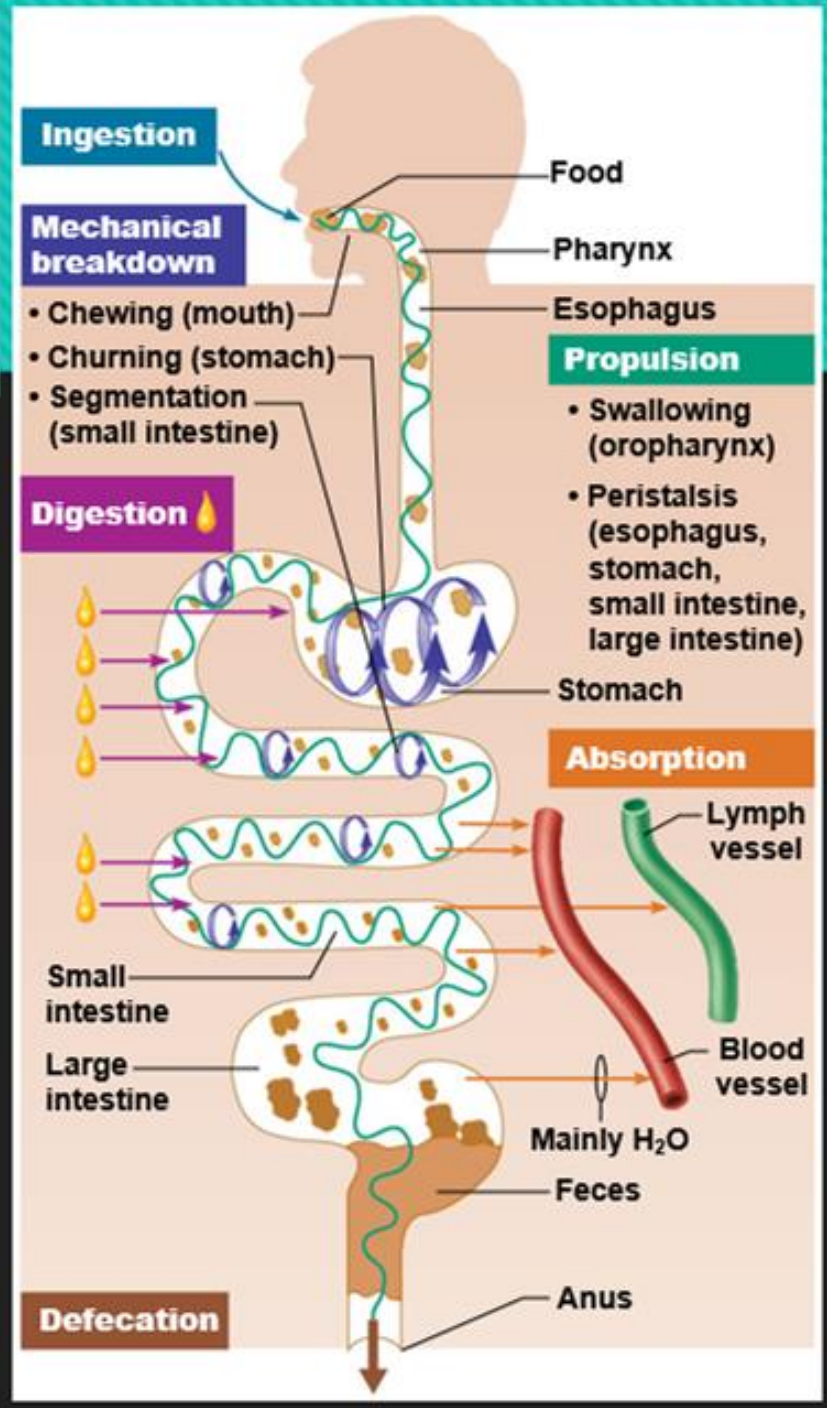
- Food ingestion and breakdown
  - Food is placed into the mouth
    - Physically broken down by chewing
    - Mixed with saliva, which is released in response to mechanical pressure and psychic stimuli
    - Salivary amylase begins starch digestion
  - Essentially, no food absorption occurs in the mouth




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# Activities Occurring in the Mouth, Pharynx, and Esophagus

- Food propulsion—swallowing and peristalsis
  - Pharynx and esophagus have no digestive function
    - Serve as passageways to the stomach
  - Pharynx functions in swallowing (deglutition)
    - Two phases of swallowing
      1. Buccal phase
      2. Pharyngeal-esophageal phase

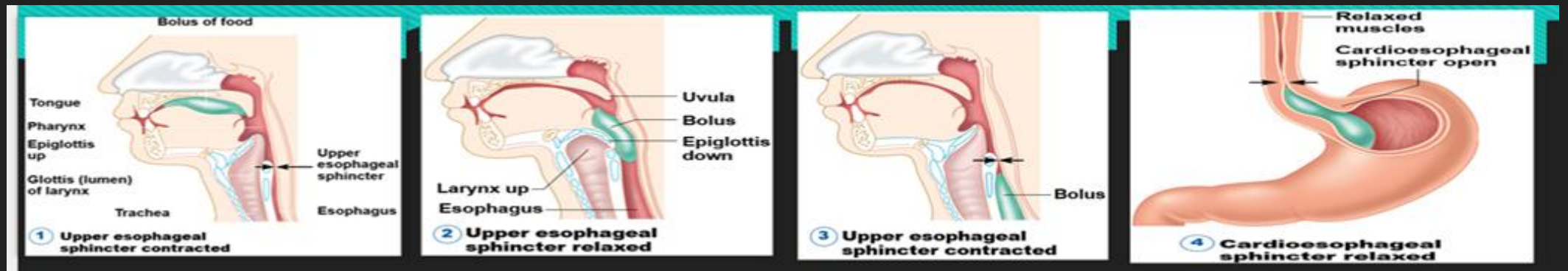



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# Activities Occurring in the Mouth, Pharynx, and Esophagus

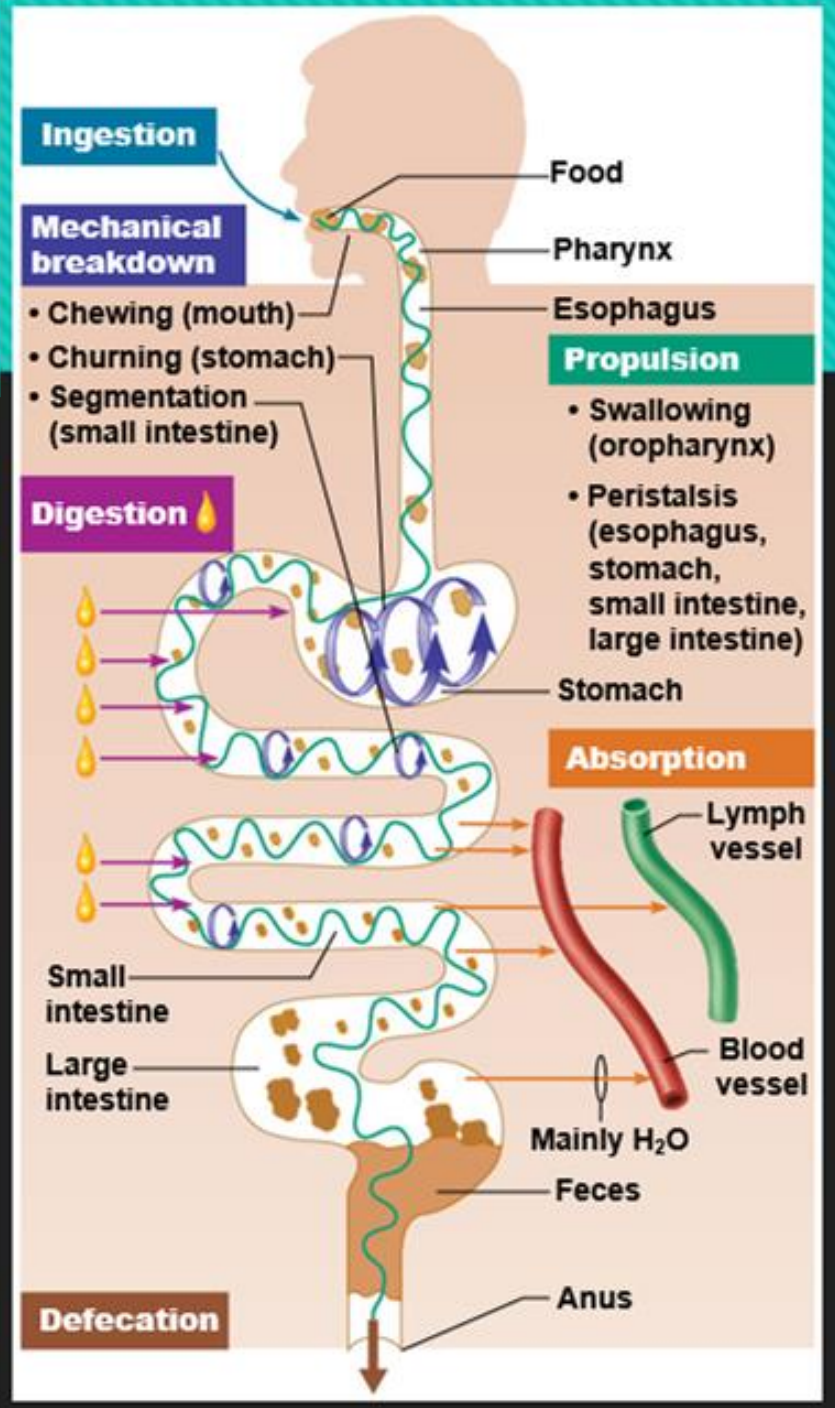
Swallowing involves two major phases:

1. **Buccal Phase (Voluntary Phase)**
   - Occurs in the mouth.
   - The tongue pushes the chewed food—now called a **bolus**—against the hard palate and into the oropharynx.
   - This phase is under **voluntary control**.
2. **Pharyngeal–Esophageal Phase (Involuntary Phase)**
   - Controlled by the **swallowing center** in the medulla oblongata and lower pons.
   - The **soft palate** rises to close off the nasopharynx, and the **epiglottis** covers the laryngeal opening to prevent aspiration.
   - **Peristaltic contractions** move the bolus through the pharynx and esophagus toward the stomach.
   - The **cardioesophageal (gastroesophageal) sphincter** relaxes to allow entry of the bolus into the stomach.



# Activities in the Stomach

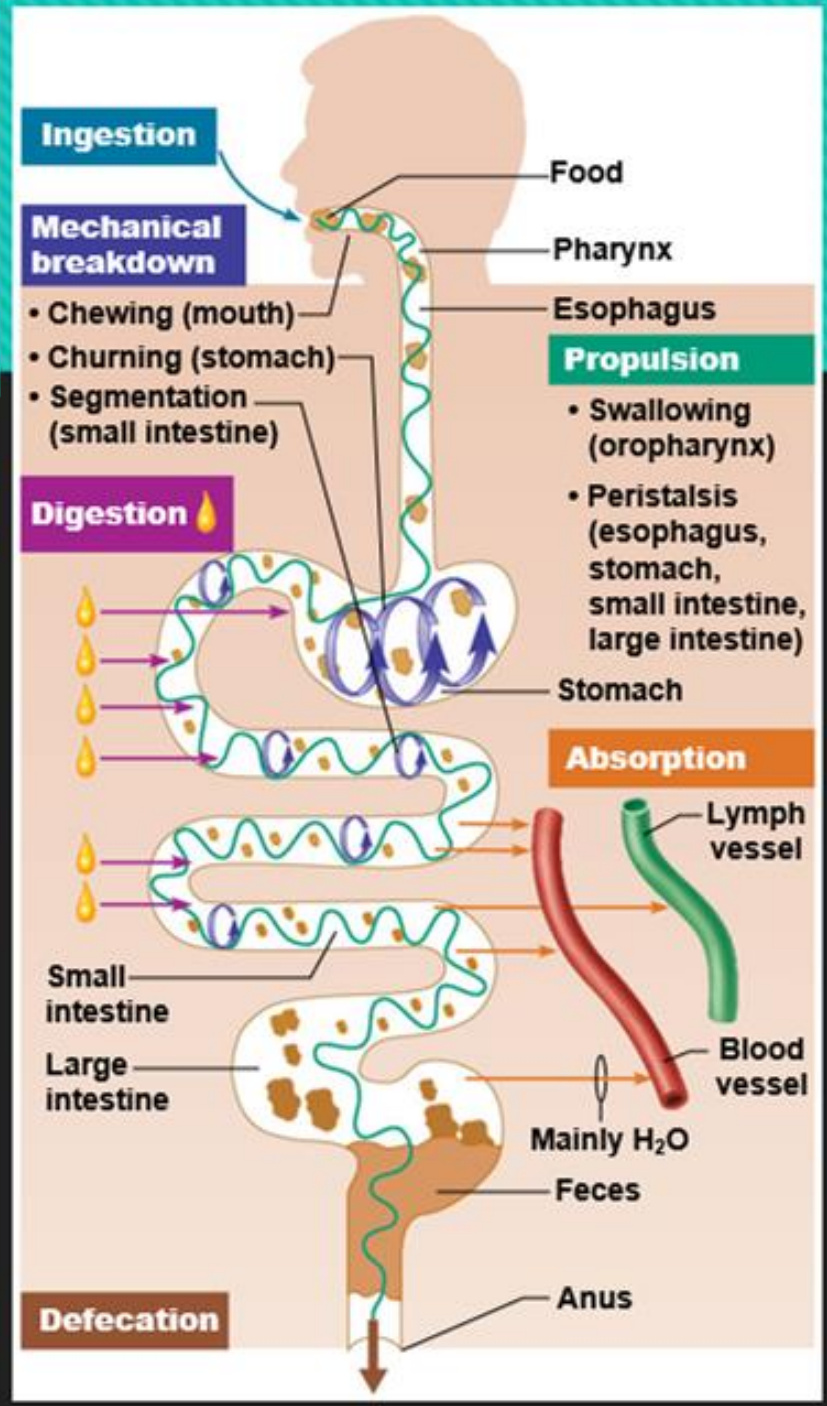
- Food breakdown
  - Presence of food or rising pH causes the release of the hormone gastrin
  - Gastrin causes stomach glands to produce:
    - Protein-digesting enzymes
    - Mucus
  - Hydrochloric acid makes the stomach contents very acidic
  - Acidic pH
    - Activates pepsinogen to pepsin
    - Provides a hostile environment for microorganisms




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# Activities in the Stomach

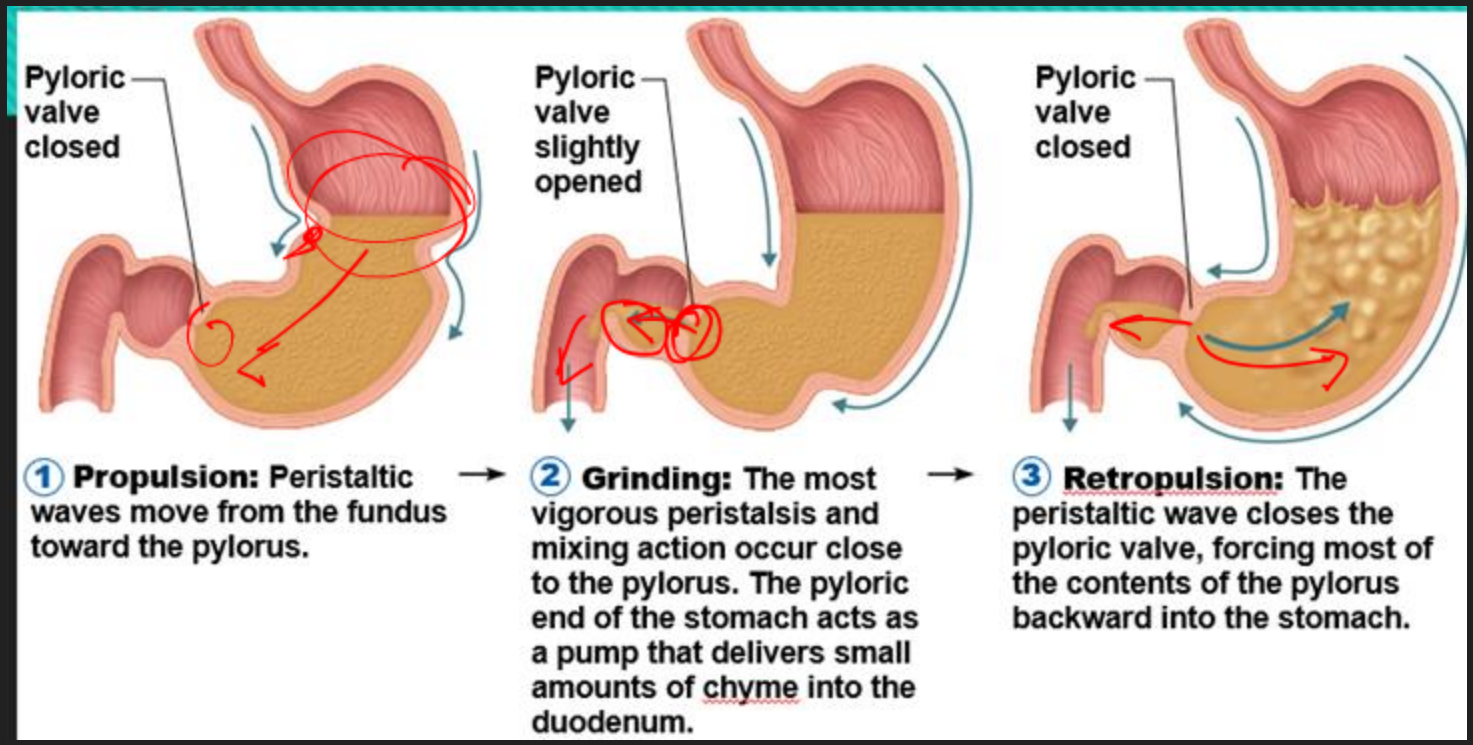
- Food breakdown
  - Protein-digestion enzymes
    - Pepsin—an active protein-digesting enzyme
    - Rennin—works on digesting milk protein in infants; not produced in adults
  - Alcohol and aspirin are virtually the only items absorbed in the stomach




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# Activities in the Stomach

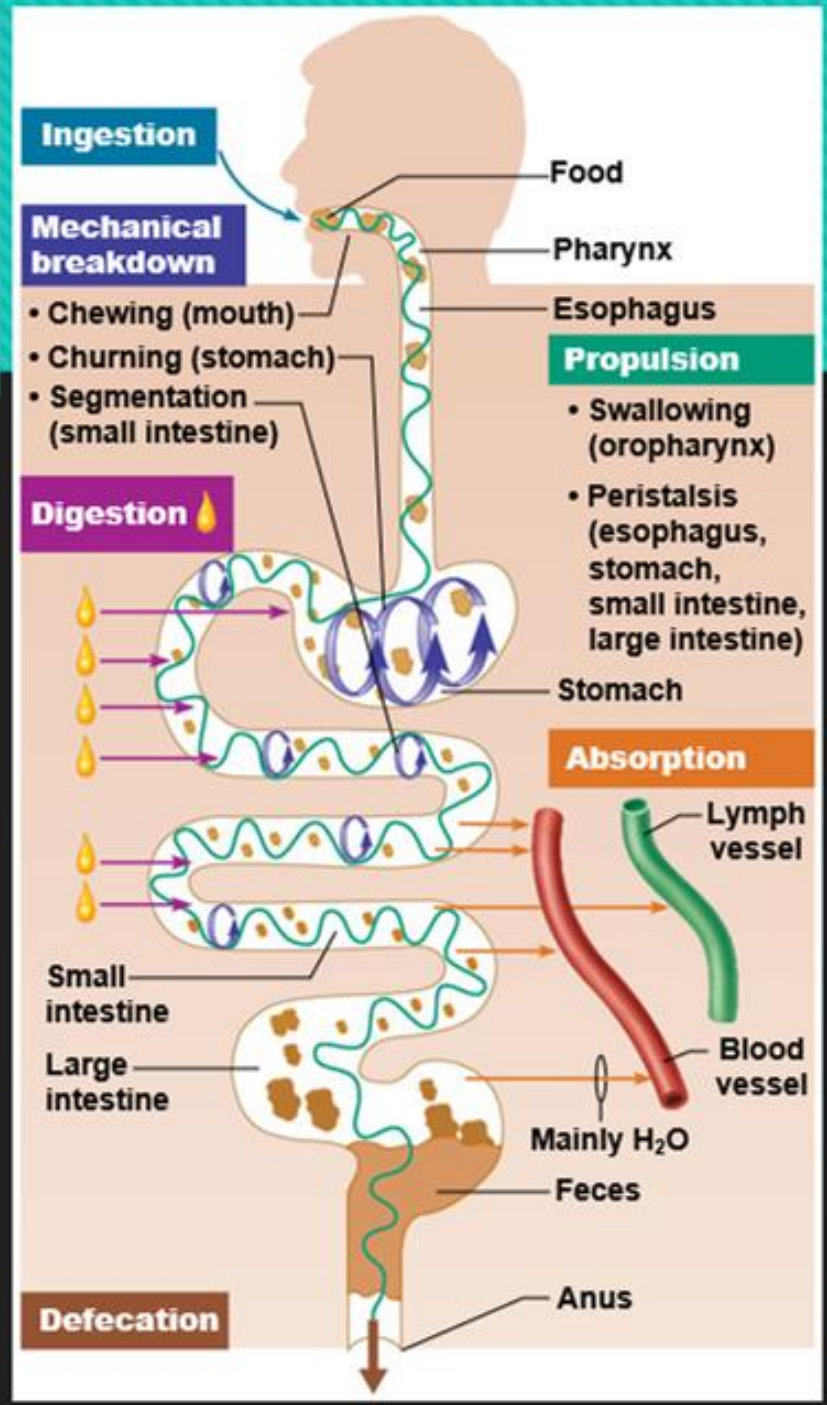
- Food propulsion
  1. Peristalsis: waves of peristalsis occur from the fundus to the pylorus, forcing food past the pyloric sphincter
  2. Grinding: the pylorus meters out chyme into the small intestine (3 ml at a time)
  3. Retropulsion: peristaltic waves close the pyloric sphincter, forcing contents back into the stomach; the stomach empties in 4–6 hours




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# Activities of the Small Intestine

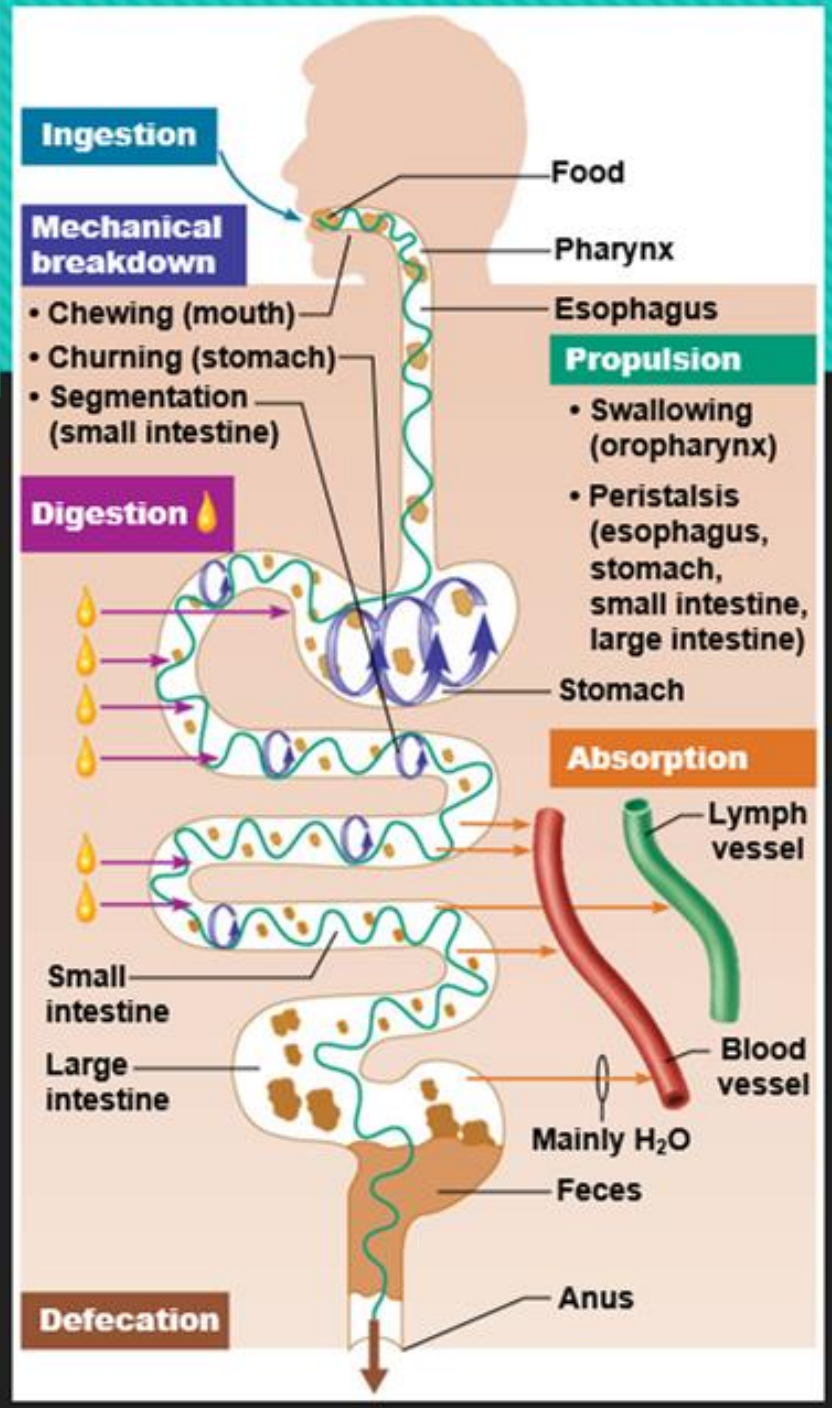
- Chyme breakdown and absorption
  - Intestinal enzymes from the brush border function to:
    - Break double sugars into simple sugars
    - Complete some protein digestion
  - Intestinal enzymes and pancreatic enzymes help to complete digestion of all food groups
  - Pancreatic enzymes play the major role in the digestion of fats, proteins, and carbohydrates
  - Alkaline content neutralizes acidic chyme and provides the proper environment for the pancreatic enzymes to operate




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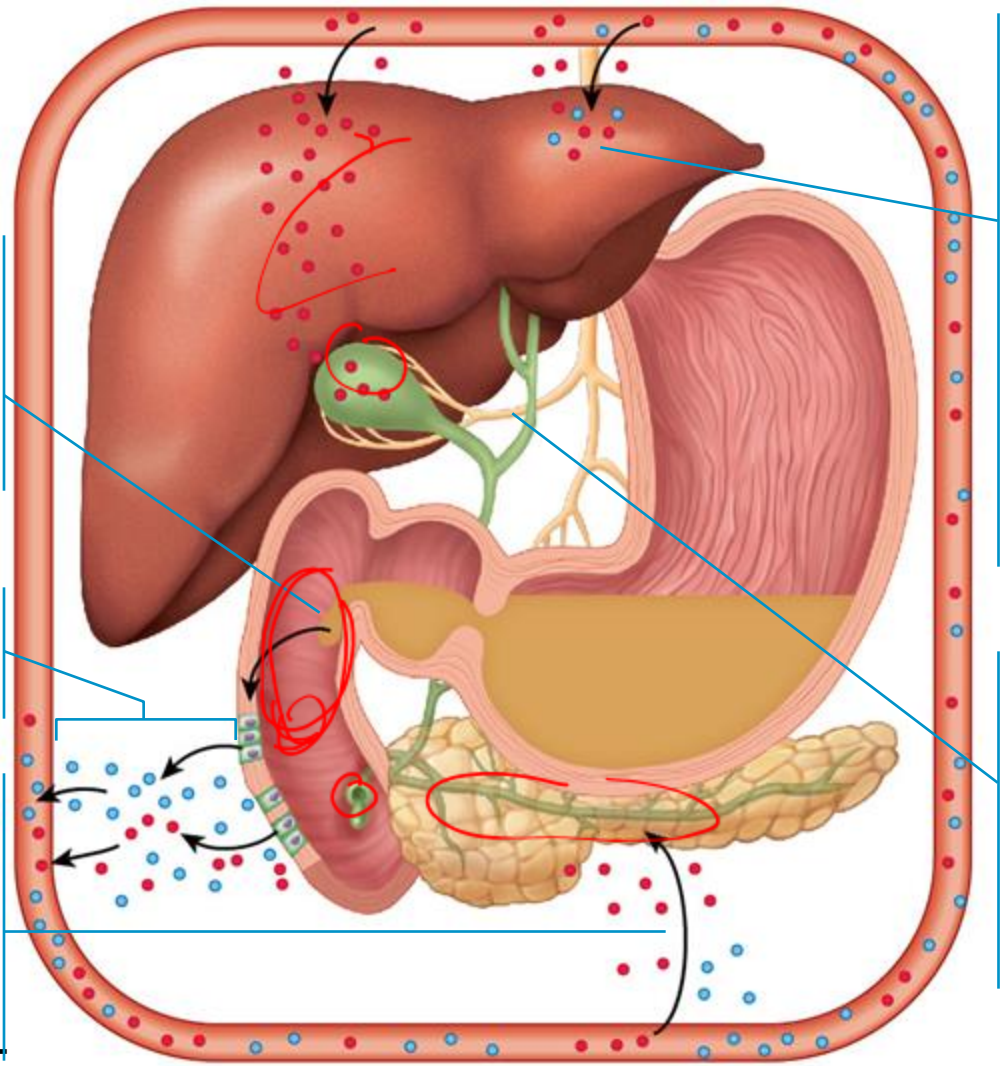
# Activities of the Small Intestine

- Chyme breakdown and absorption
  - Release of pancreatic juice from the pancreas into the duodenum is stimulated by:
    - Vagus nerves
    - Local hormones that travel via the blood to influence the release of pancreatic juice (and bile)
      - Secretin
      - Cholecystokinin (CCK)
  - Hormones (secretin and CCK) also target the liver and gallbladder to release bile




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# Figure 14.16 Regulation of pancreatic juice and bile secretion and release.



1. Chyme entering duodenum causes duodenalent eroendocrine cells to release cholecystokinin (CCK) and secretin.

2. CCK (red dots) and secretin (blue dots) enter the bloodstream.

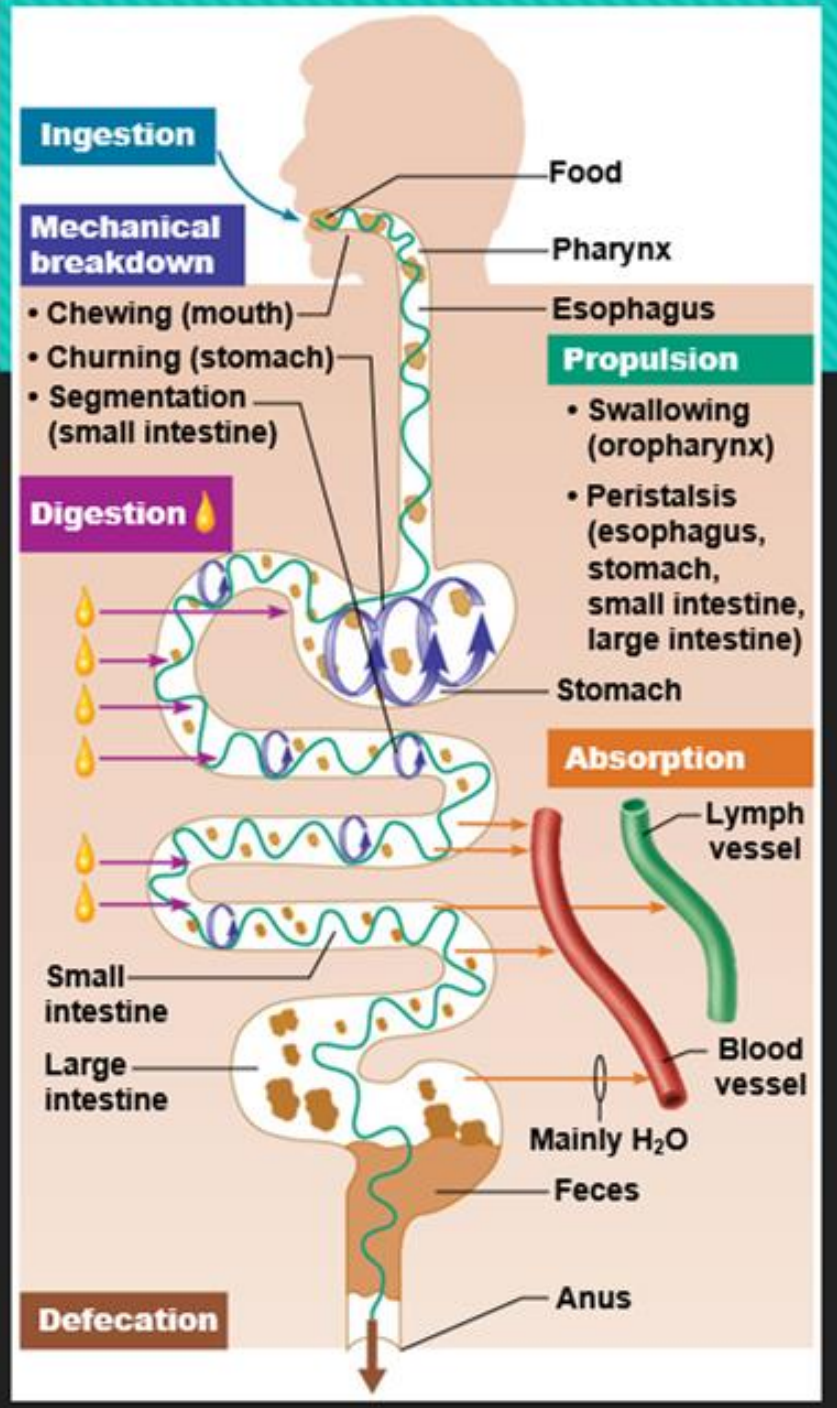
3. Upon reaching the pancreas, CCK induces secretion of enzyme-rich pancreatic juice; secretin causes secretion of bicarbonate-rich pancreatic juice.

4. Secretin causes the liver to secrete more bile; CCK stimulates the gallbladder to release stored bile and the hepatopancreatic sphincter to relax (allows bile from both sources to enter the duodenum).

5. Stimulation by vagal nerve fibers causes release of pancreatic juice and weak contractions of the gallbladder.

# Activities of the Small Intestine

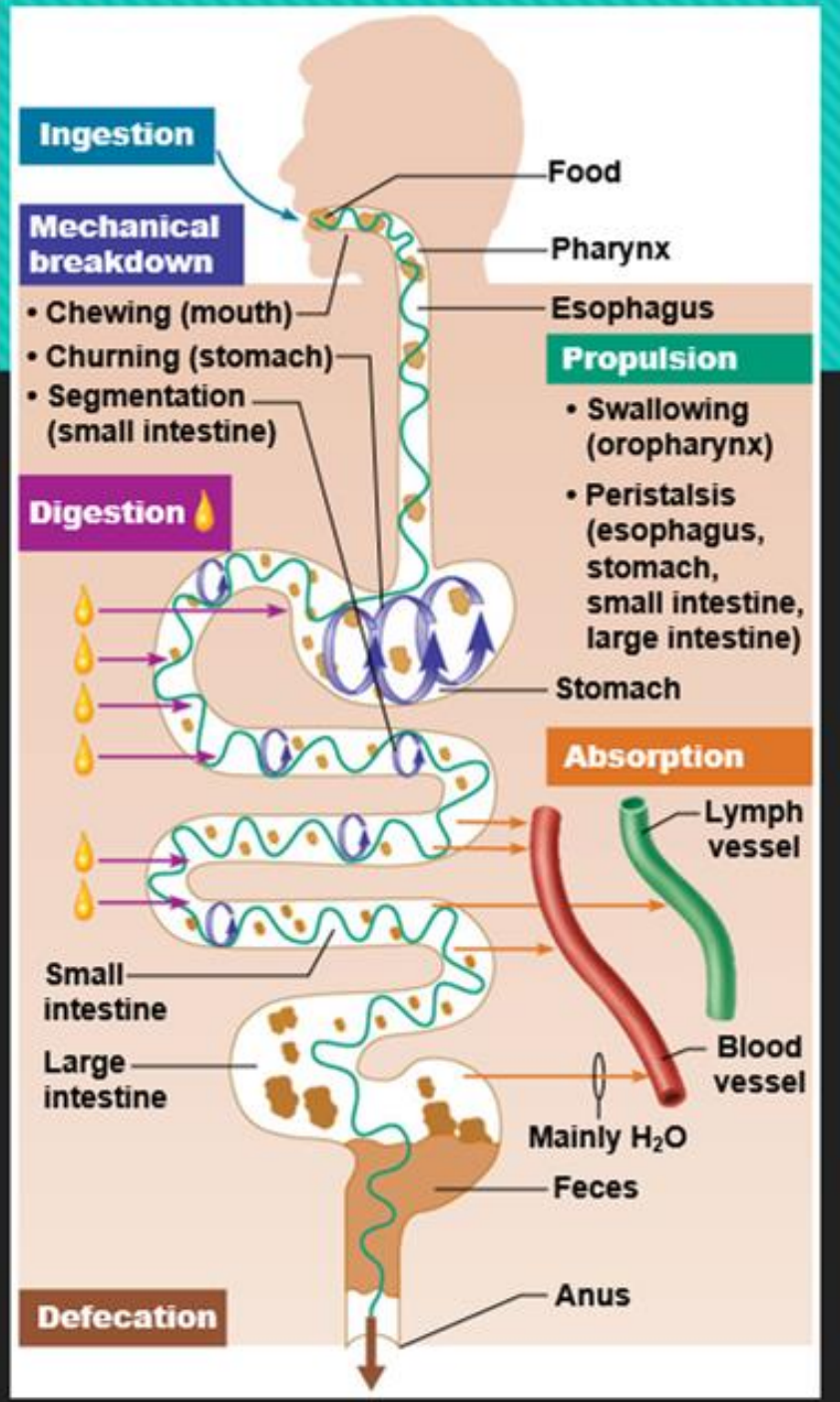
- Chyme breakdown and absorption
  - Water is absorbed along the length of the small intestine
  - End products of digestion
    - Most substances are absorbed by active transport through cell membranes
    - Lipids are absorbed by diffusion
  - Substances are transported to the liver by the hepatic portal vein or lymph




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# Activities of the Small Intestine

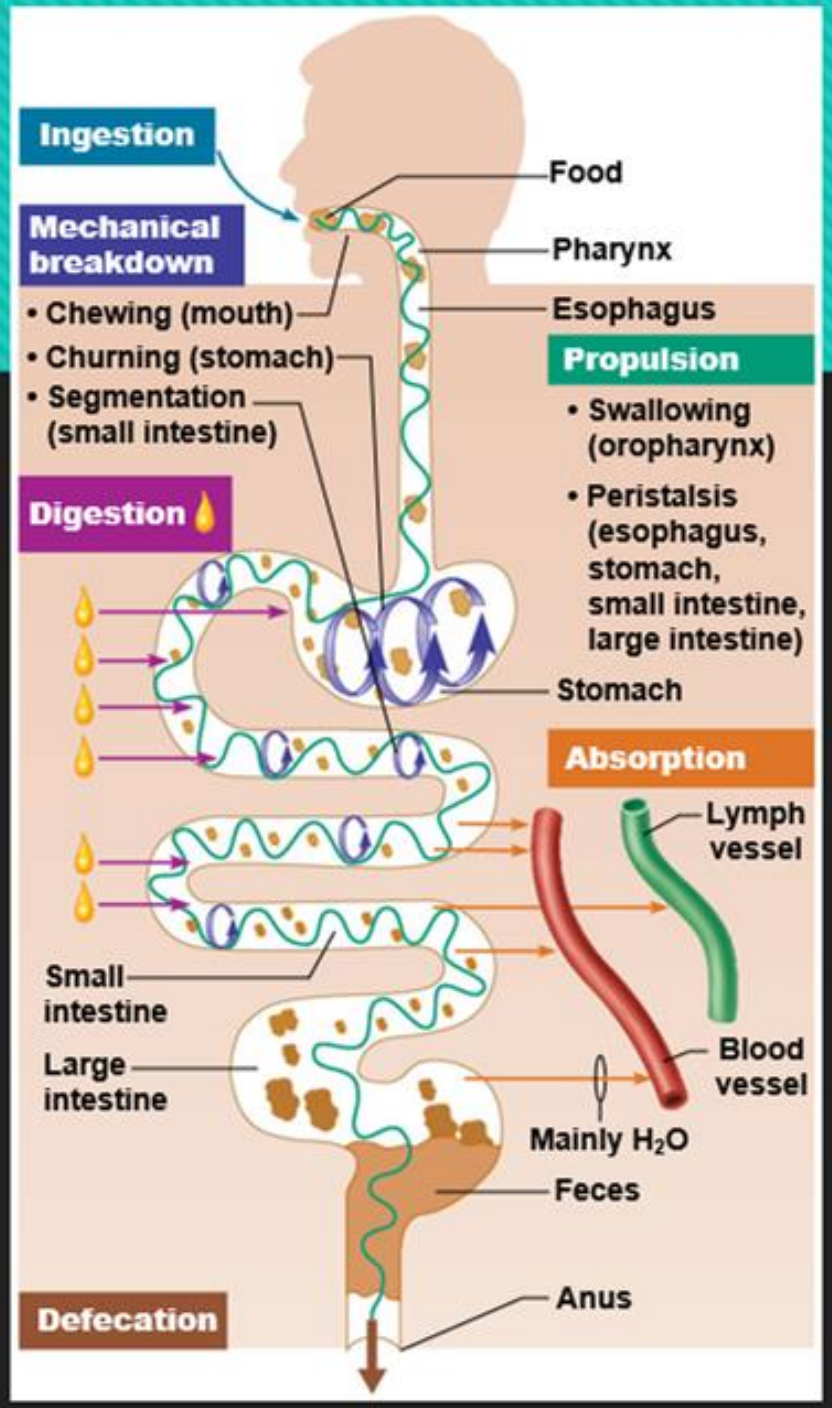
- Chyme propulsion
  - Peristalsis is the major means of moving food
  - Segmental movements
    - Mix chyme with digestive juices
    - Aid in propelling food




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# Activities of the Large Intestine

- Nutrient breakdown and absorption
  - No digestive enzymes are produced
  - Resident bacteria digest remaining nutrients
    - Produce some vitamin K and some B vitamins
    - Release gases
  - Water, vitamins, ions, and remaining water are absorbed
  - Remaining materials are eliminated via feces




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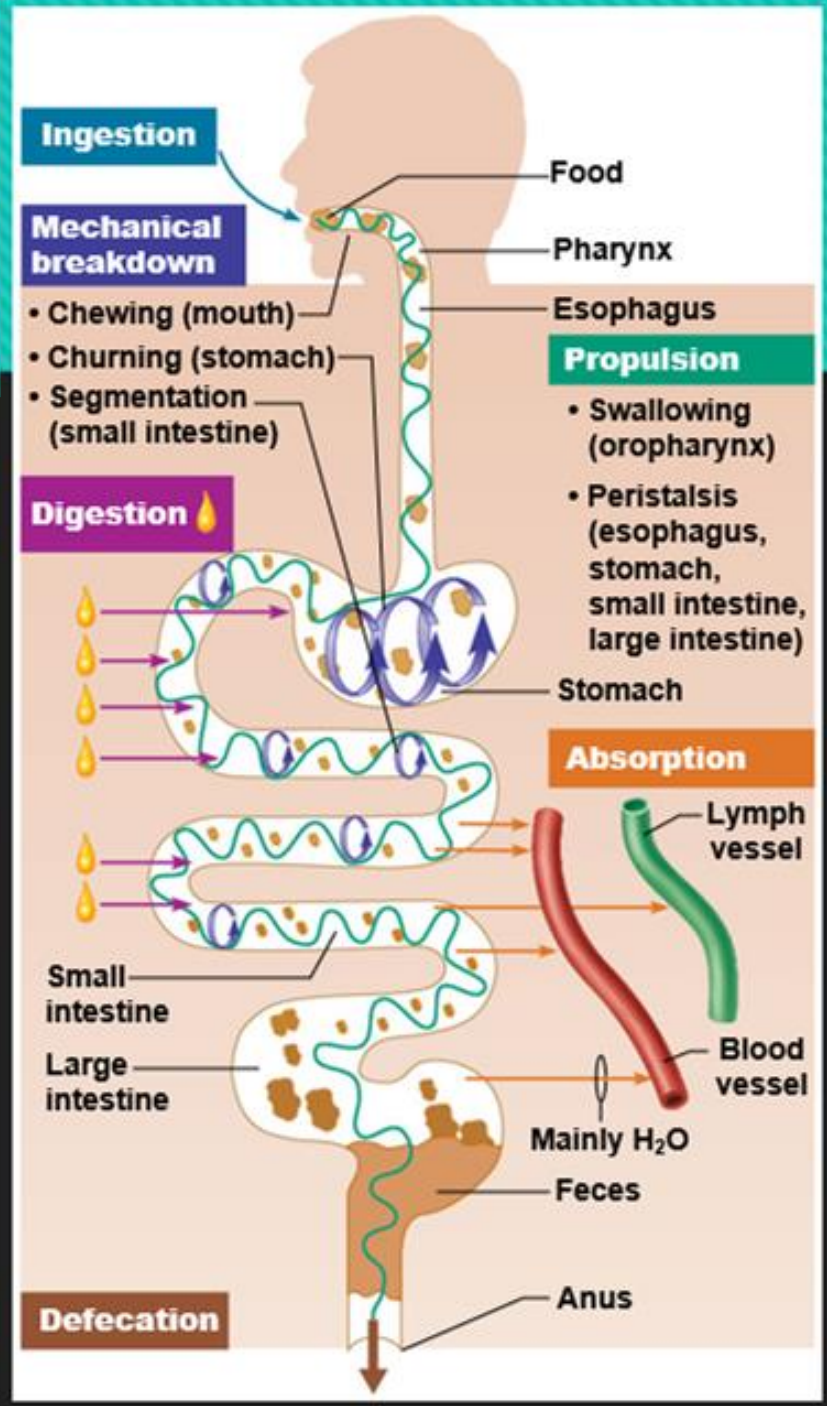
# Activities of the Large Intestine

- Nutrient breakdown and absorption
- Feces contains:
  - Undigested food residues
  - Mucus
  - Bacteria
  - Water

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# Activities of the Large Intestine

- Propulsion of food residue and defecation
  - Sluggish peristalsis begins when food residue arrives
  - Haustral contractions are the movements occurring most frequently in the large intestine
  - Mass movements are slow, powerful movements that occur three to four times per day
  - Presence of feces in the rectum causes a defecation reflex
    - Internal anal sphincter is relaxed
    - Defecation occurs with relaxation of the voluntary (external) anal sphincter




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# OUTLINE OF TOPICS:



- Organs and Functions of the Digestive System
- Trunks or Layers of the Digestive Tract
- Activities and Movements in the Digestive System
- Secretions in the Digestive System
- Metabolism

# Table 14.1 Hormones and Hormonelike Products That Act in Digestion

Table **14.1** **Hormones and Hormonelike Products That Act in Digestion**

| Hormone | Source | Stimulus for secretion | Action |
|---|---|---|---|
| Gastrin | Stomach | Food in stomach, particularly partially digested proteins; ACh released by nerve fibers | • Stimulates release of gastric juice<br>• Stimulates stomach emptying |
| Intestinal gastrin | Duodenum | Food in stomach | • Stimulates gastric secretion and emptying |
| Histamine | Stomach | Food in stomach | • Activates parietal cells to secrete hydrochloric acid |
| Somatostatin | Stomach and duodenum | Food in stomach stimulated by sympathetic nerve fibers | • Inhibits secretion of gastric juice and pancreatic juice<br>• Inhibits emptying of stomach and gallbladder |

# Table 14.1 Hormones and Hormonelike Products That Act in Digestion

Table **14.1** **Hormones and Hormonelike Products That Act in Digestion** (*continued*)

| Hormone | Source | Stimulus for secretion | Action |
|---|---|---|---|
| Secretin | Duodenum | Acidic chyme and partially digested foods in duodenum | • Increases output of pancreatic juice rich in bicarbonate ions<br>• Increases bile output by liver<br>• Inhibits gastric mobility and gastric gland secretion |
| Cholecystokinin (CCK) | Duodenum | Fatty chyme and partially digested proteins in duodenum | • Increases output of enzyme-rich pancreatic juice<br>• Stimulates gallbladder to expel stored bile<br>• Relaxes sphincter of duodenal papilla to allow bile and pancreatic juice to enter the duodenum |
| Gastric inhibitory peptide (GIP) | Duodenum | Food in duodenum | • Inhibits secretion of gastric juice<br>• Stimulates insulin release |

# OUTLINE OF TOPICS:



- Organs and Functions of the Digestive System
- Trunks or Layers of the Digestive Tract
- Activities in the Digestive System
- Movements and Secretions in the Digestive System
- Metabolism

# Nutrition and Metabolism

- Most foods are used as metabolic fuel
  - Foods are oxidized and transformed into adenosine triphosphate (ATP)
  - ATP is chemical energy that drives cellular activities
- Energy value of food is measured in kilocalories (kcal) or Calories (C)

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# Nutrition

- Nutrient—substance used by the body for growth, maintenance, and repair
- Major nutrients
  - Carbohydrates
  - Lipids
  - Proteins
  - Water
- Minor nutrients
  - Vitamins
  - Minerals

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# ADDENDUM Nutrition

- A diet consisting of foods from the five food groups normally guarantees adequate amounts of all the needed nutrients

Table 14.2 **Five Basic Food Groups and Some of Their Major Nutrients**

| Group | Example foods | Major nutrients supplied in significant amounts: By all in group | Major nutrients supplied in significant amounts: By only some in group |
|---|---|---|---|
| **Fruits**  | Apples, bananas, dates, oranges, tomatoes | Carbohydrate<br>Water | Vitamins: A, C, folic acid<br>Minerals: iron, potassium<br>Fiber |
| **Vegetables**  | Broccoli, cabbage, green beans, lettuce, potatoes | Carbohydrate<br>Water | Vitamins: A, C, E, K, and B vitamins except $B_{12}$<br>Minerals: calcium, magnesium, iodine, manganese, phosphorus<br>Fiber |
| **Grain products** (preferably whole grain; otherwise, enriched or fortified)  | Breads, rolls, bagels; cereals, dry and cooked; pasta; rice, other grains; tortillas, pancakes, waffles; crackers; popcorn | Carbohydrate Protein<br>Vitamins: thiamine ($B_1$), niacin | Water<br>Fiber<br>Minerals: iron, magnesium, selenium |

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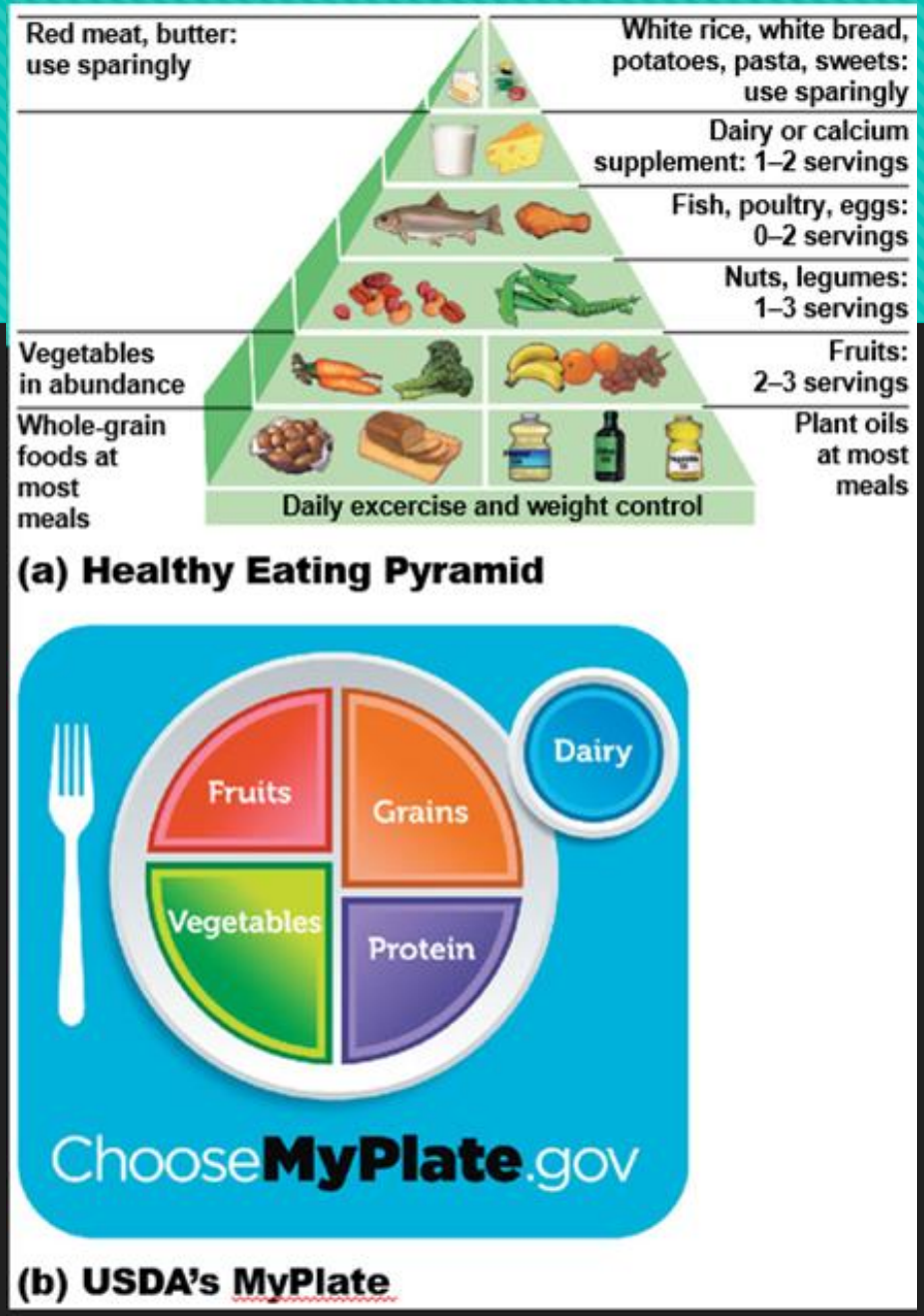
# ADDENDUM
# Table 14.2 Five Basic Food Groups and Some of Their Major Nutrients

Table **14.2** **Five Basic Food Groups and Some of Their Major Nutrients** (*continued*)

| Group | Example foods | Major nutrients supplied in significant amounts | |
|---|---|---|---|
| | | **By all in group** | **By only some in group** |
| **Milk products**  | Milk, yogurt; cheese; ice cream, ice milk, frozen yogurt | Protein<br>Fat<br>Vitamins: riboflavin, $B_{12}$<br>Minerals: calcium, phosphorus<br>Water | Carbohydrate<br>Vitamins: A, D |
| **Meats and meat alternatives**  | Meat, fish, poultry; eggs; seeds; nuts, nut butters; soybeans, tofu; other legumes (peas and beans) | Protein<br>Vitamins: niacin, $B_6$<br>Minerals: iron, zinc | Carbohydrate<br>Fat<br>Vitamins: $B_{12}$, thiamine ($B_1$)<br>Water<br>Fiber |

# ADDENDUM
# Dietary Recommendations

- Healthy Eating Pyramid
  - Issued in 1992
  - Six major food groups arranged horizontally
- MyPlate
  - Issued in 2011 by the USDA
  - Five food groups are arranged by a round plate




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# ADDENDUM Dietary Sources of the Major Nutrients

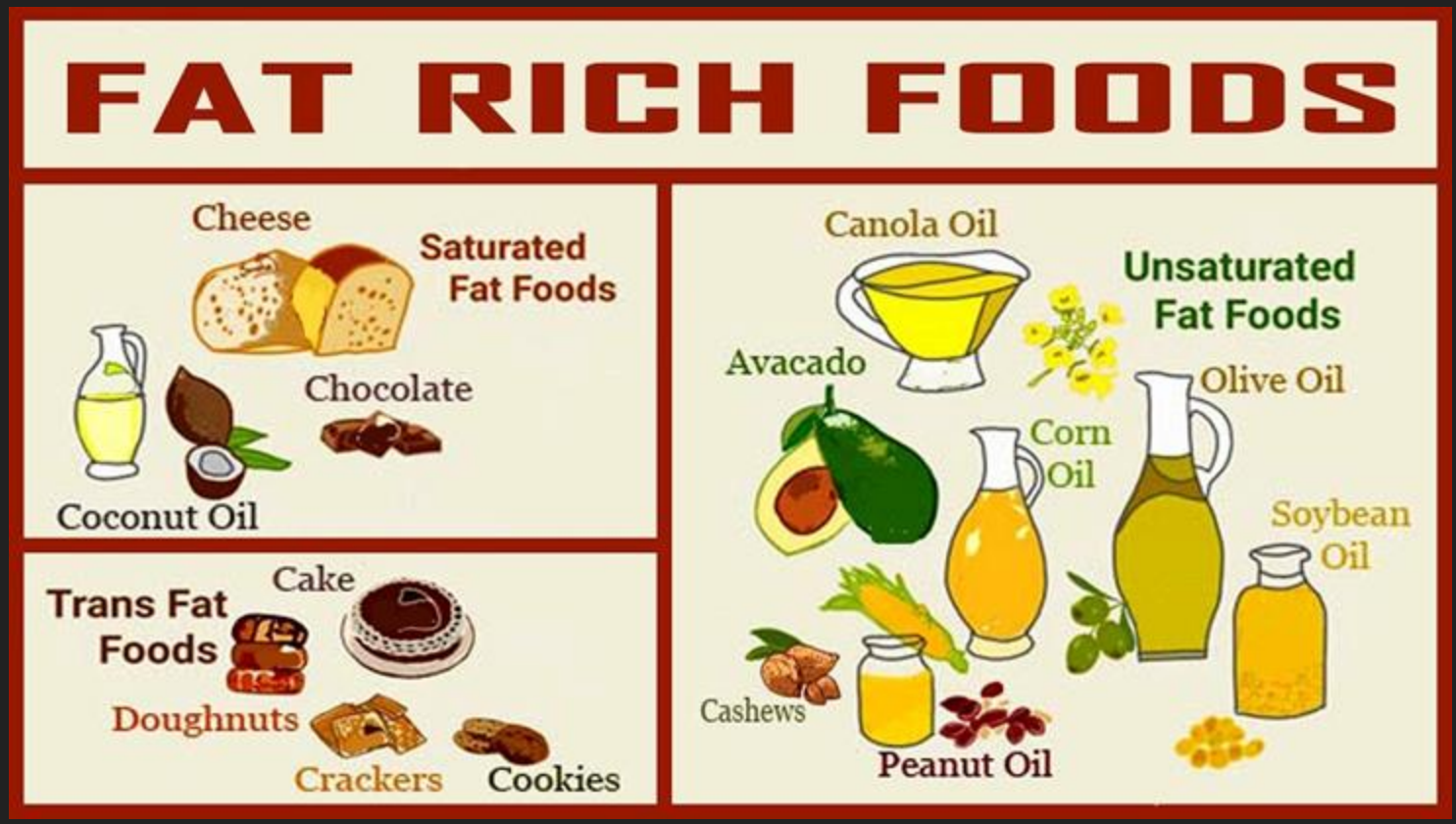
- Carbohydrates
  - Dietary carbohydrates are sugars and starches
  - Most are derived from plants such as fruits and vegetables
  - Exceptions: lactose from milk and small amounts of glycogens from meats




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# ADDENDUM
# Dietary Sources of the Major Nutrients

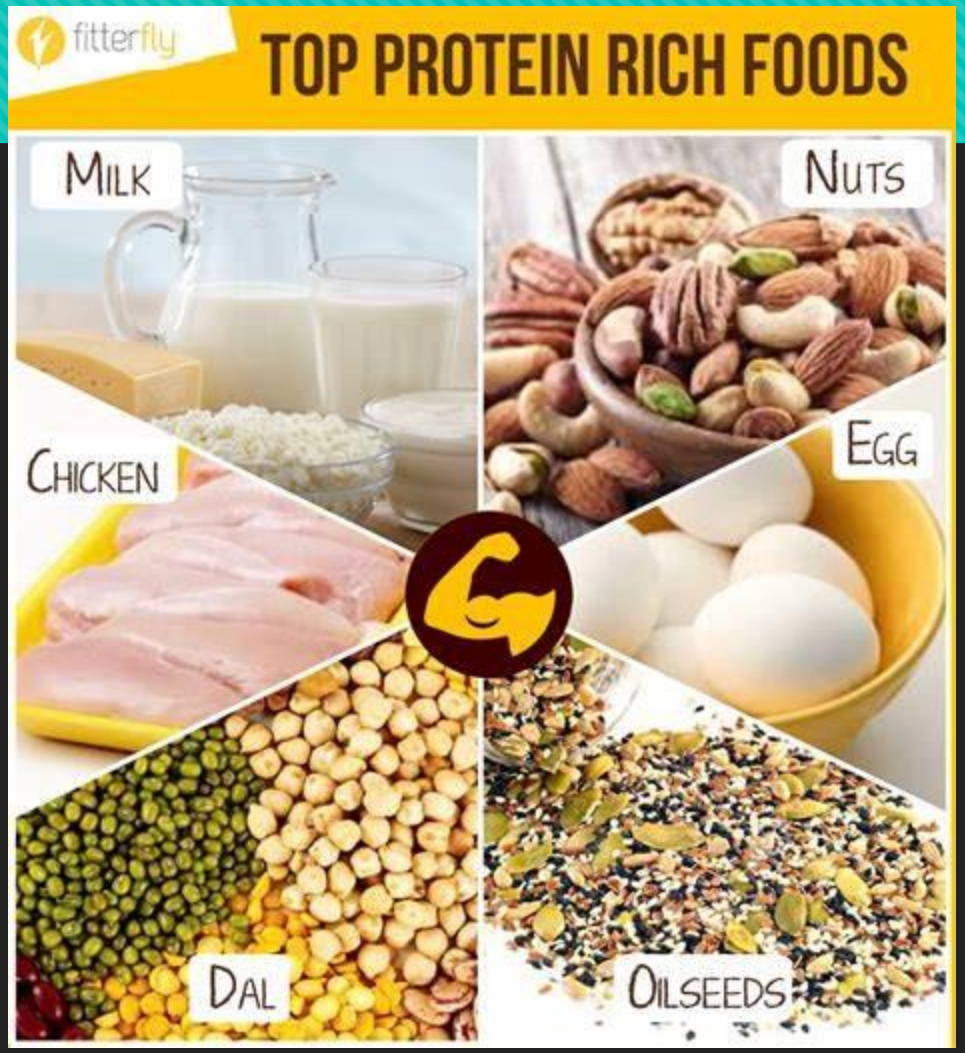
- Lipids
  - Saturated fats from animal products (meats)
  - Unsaturated fats from nuts, seeds, and vegetable oils
  - Cholesterol from egg yolk, meats, and milk products (dairy products)




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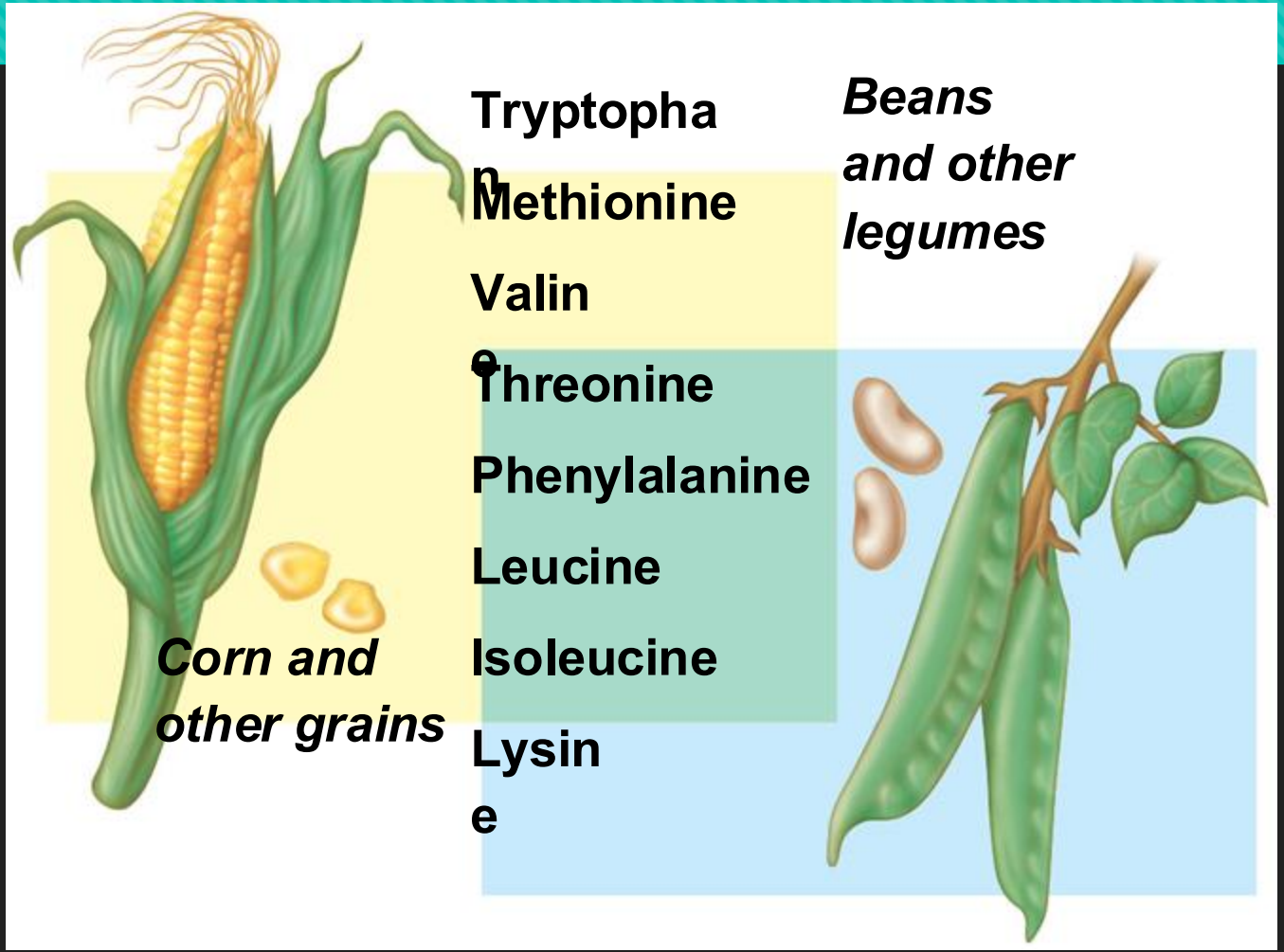
# ADDENDUM
# Dietary Sources of the Major Nutrients

- Proteins
  - Complete proteins—contain all essential amino acids
    - Most are from animal products (eggs, milk, meat, poultry, and fish)
    - Essential amino acids: those that the body cannot make and must be obtained through diet
  - Legumes and beans also have proteins, but the proteins are incomplete



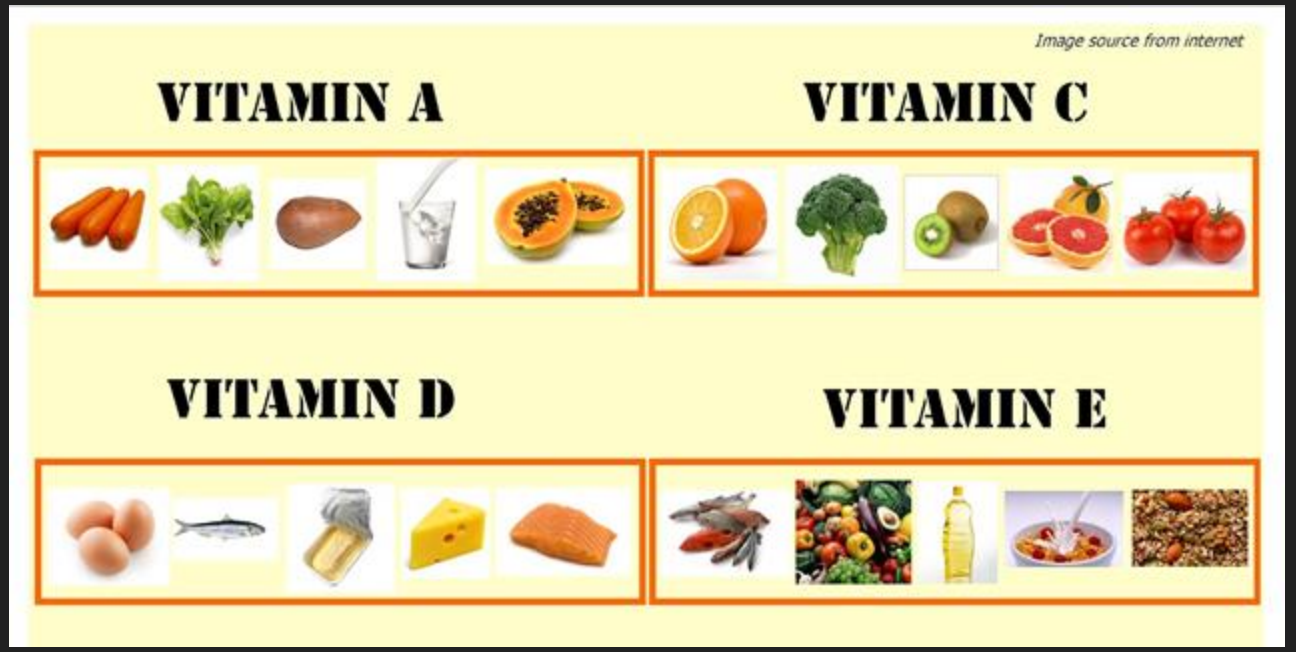

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# ADDENDUM
# Figure 14.18 The eight essential amino acids.



# ADDENDUM
# Dietary Sources of the Major Nutrients

- Vitamins
  - Most vitamins function as coenzymes
  - Found mainly in fruits and vegetables




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# ADDENDUM
# Dietary Sources of the Major Nutrients

- Minerals
  - Mainly important for enzyme activity
  - Foods richest in minerals: vegetables, legumes, milk, and some meats



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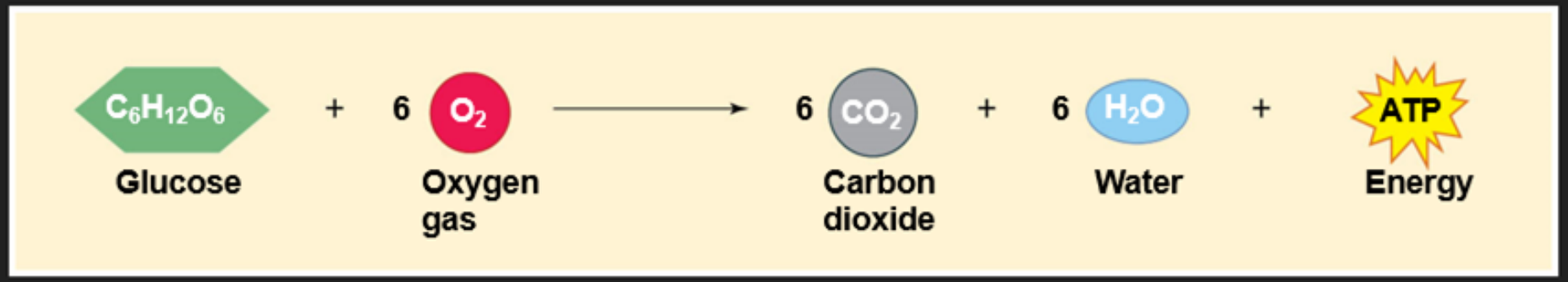
# Metabolism

- Metabolism is all of the chemical reactions necessary to maintain life
  - Catabolism—substances are broken down to simpler substances; energy is released and captured to make adenosine triphosphate (ATP)
  - Anabolism—larger molecules are built from smaller ones

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# ADDENDUM
# Carbohydrate Metabolism

- Carbohydrates are the body's preferred source to produce cellular energy (ATP)
- Glucose (blood sugar)
  - Major breakdown product of carbohydrate digestion
  - Fuel used to make ATP
- Cellular respiration
  - As glucose is oxidized, carbon dioxide, water, and ATP are formed




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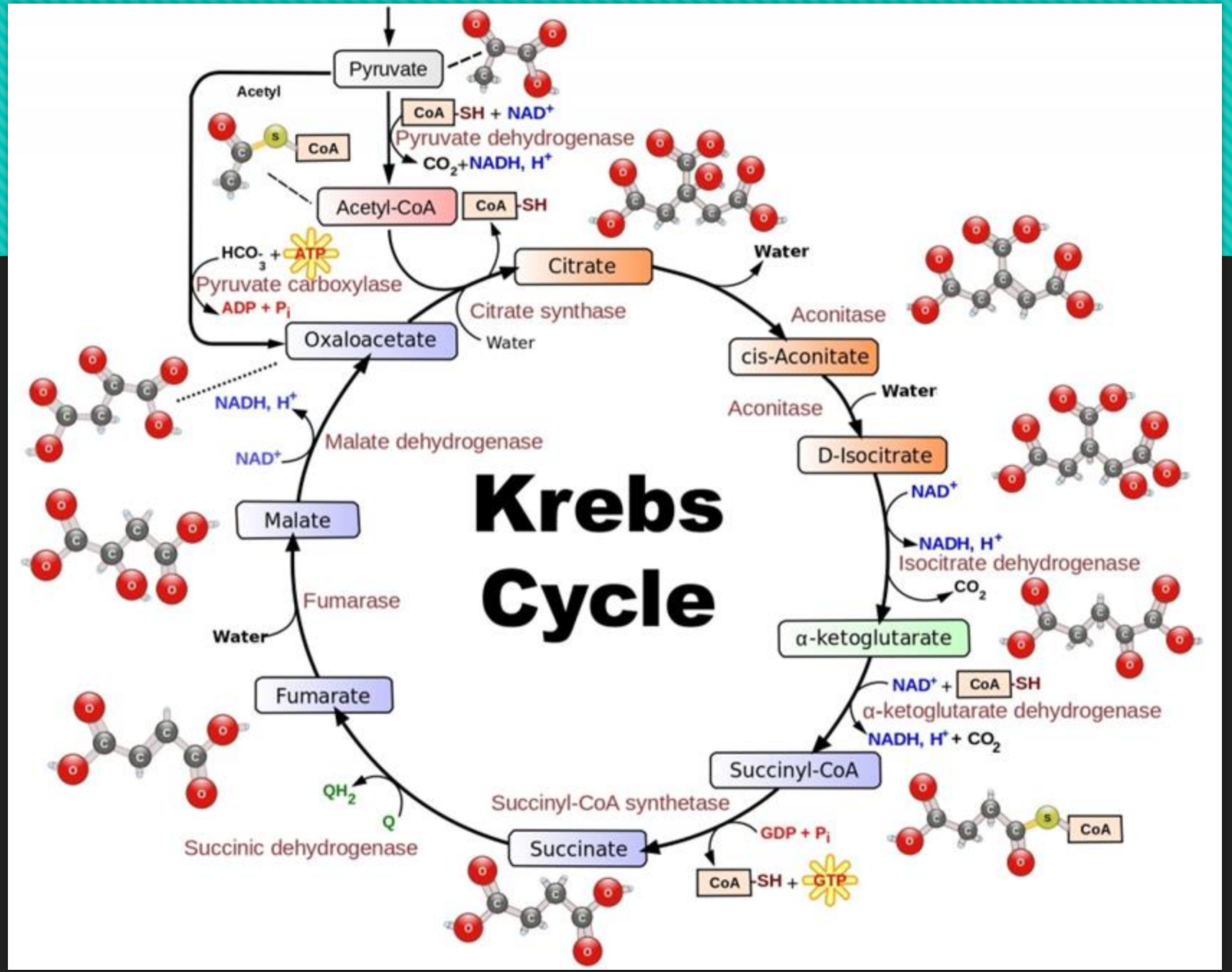
# Carbohydrate Metabolism

- EVENTS OF THREE MAIN METABOLIC PATHWAYS OF CELLULAR RESPIRATION

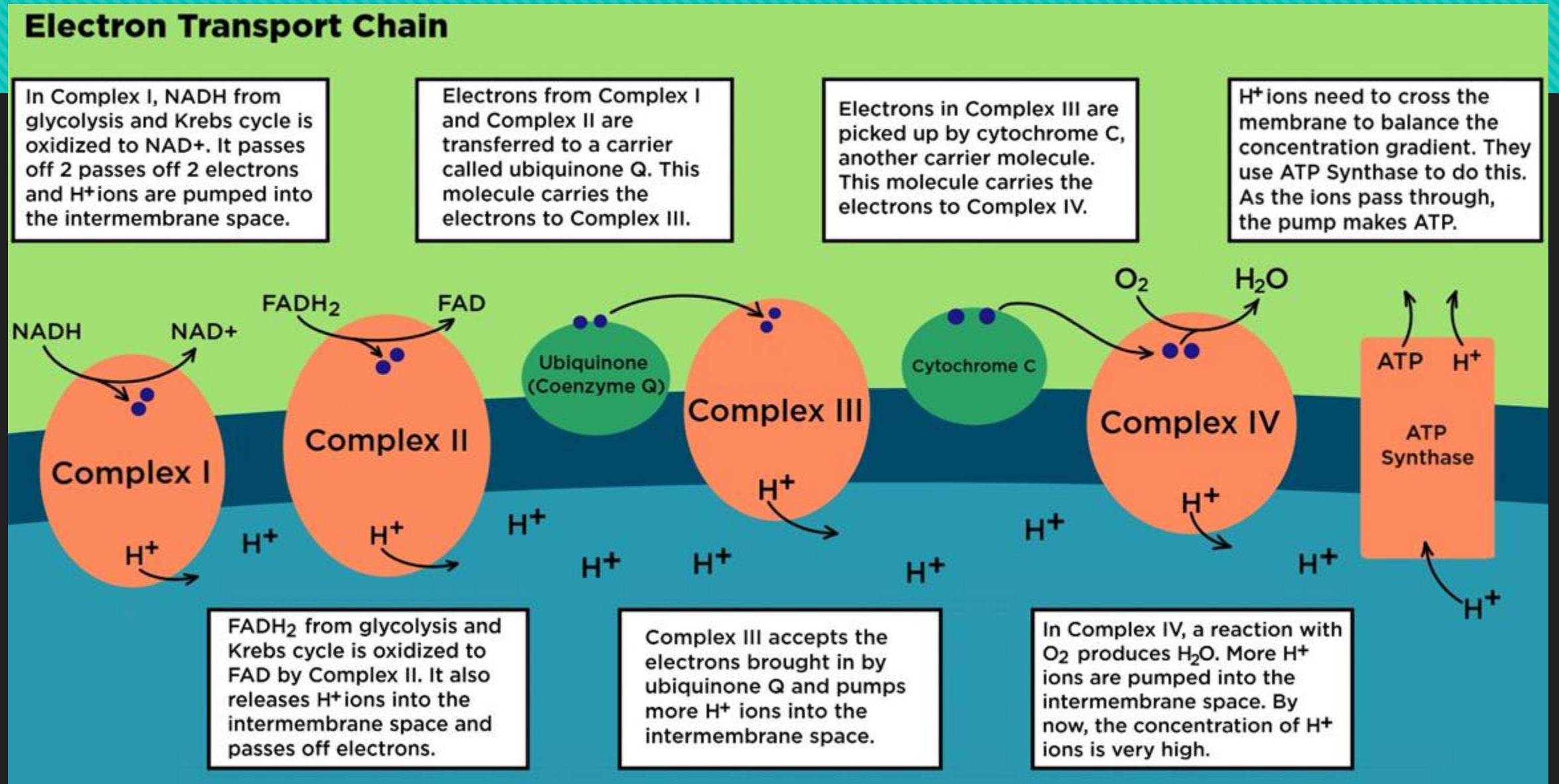
1. **Glycolysis**
   - Occurs in the cytosol
   - Energizes a glucose molecule so it can be split into two pyruvic acid molecules and yield ATP
2. **Citric acid cycle (Krebs cycle)**
   - Occurs in the mitochondrion
   - Produces virtually all the carbon dioxide and water resulting from cellular respiration
   - Yields a small amount of ATP
3. **Electron transport chain**
   - Hydrogen atoms removed during glycolysis and the citric acid cycle are delivered to protein carriers
   - Hydrogen atoms are split into hydrogen ions and electrons in the mitochondria
   - Electrons give off energy in a series of steps to enable the production of ATP

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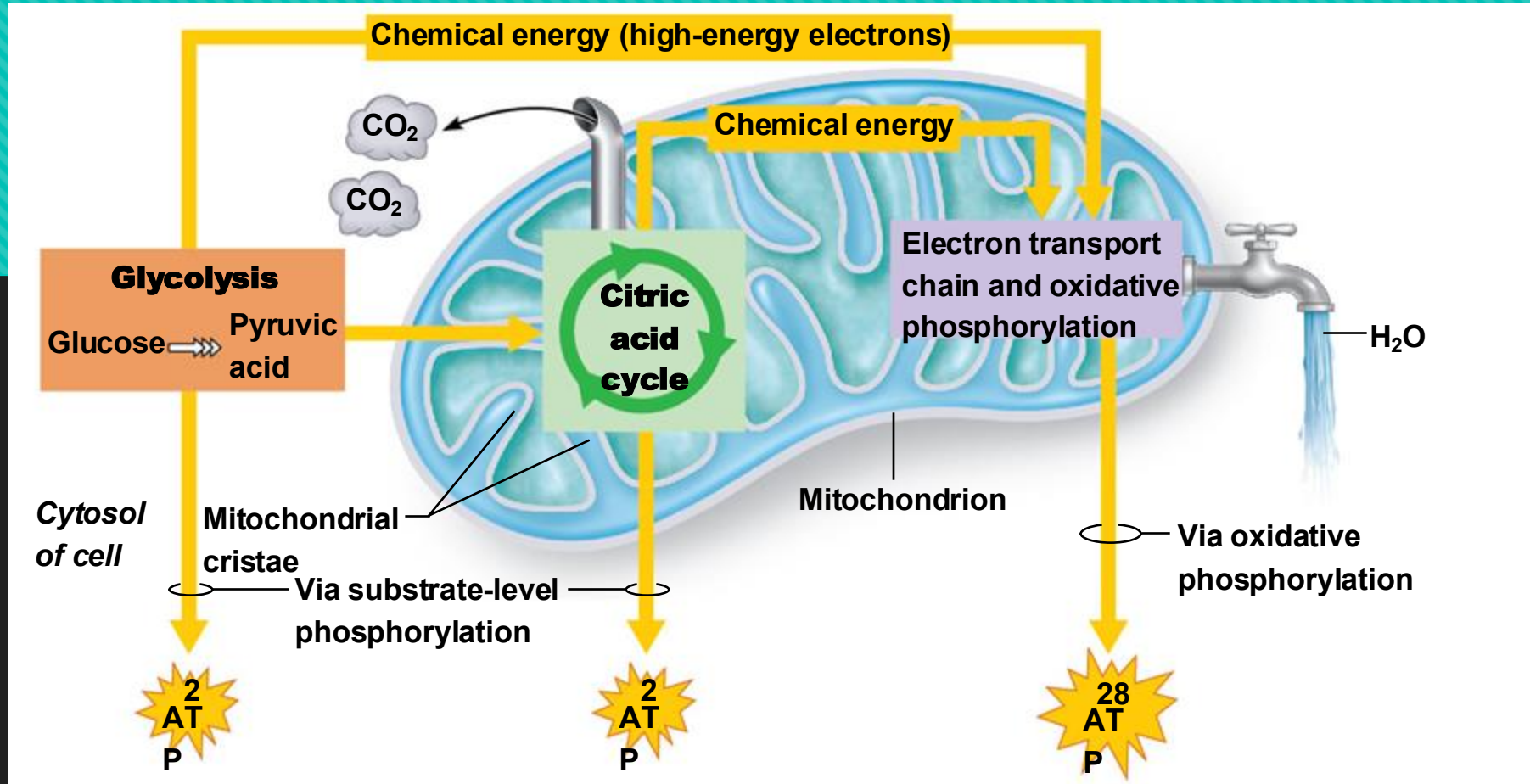
# ADDENDUM


Pyruvate
Acetyl
CoA
SH + NAD+
Pyruvate dehydrogenase
CO2+NADH, H+
Acetyl-CoA
CoA
SH
HCO3- + ATP
Pyruvate carboxylase
ADP + Pi
Citrate
Water
Citrate synthase
Water
Oxaloacetate
Aconitase
cis-Aconitate
Water
Aconitase
D-Isocitrate
NAD+
NADH, H+
Isocitrate dehydrogenase
CO2
α-ketoglutarate
NAD+ + CoA SH
α-ketoglutarate dehydrogenase
NADH, H+ + CO2
Succinyl-CoA
Succinyl-CoA synthetase
GDP + Pi
CoA SH + GTP
Succinate
QH2
Q
Succinic dehydrogenase
Fumarate
Water
Fumarase
Malate
NADH, H+
NAD+
Malate dehydrogenase
Krebs Cycle

# ADDENDUM


Electron Transport Chain
In Complex I, NADH from glycolysis and Krebs cycle is oxidized to NAD+. It passes off 2 passes off 2 electrons and H+ ions are pumped into the intermembrane space.
Electrons from Complex I and Complex II are transferred to a carrier called ubiquinone Q. This molecule carries the electrons to Complex III.
Electrons in Complex III are picked up by cytochrome C, another carrier molecule. This molecule carries the electrons to Complex IV.
H+ ions need to cross the membrane to balance the concentration gradient. They use ATP Synthase to do this. As the ions pass through, the pump makes ATP.
NADH
NAD+
FADH2
FAD
Ubiquinone (Coenzyme Q)
Cytochrome C
O2
H2O
ATP
H+
Complex I
Complex II
Complex III
Complex IV
ATP Synthase
FADH2 from glycolysis and Krebs cycle is oxidized to FAD by Complex II. It also releases H+ ions into the intermembrane space and passes off electrons.
Complex III accepts the electrons brought in by ubiquinone Q and pumps more H+ ions into the intermembrane space.
In Complex IV, a reaction with O2 produces H2O. More H+ ions are pumped into the intermembrane space. By now, the concentration of H+ ions is very high.

# Figure 14.20 The formation of ATP in the cytosol and the mitochondria during cellular respiration.




(1) During glycolysis, each glucose molecule is broken down into two molecules of pyruvic acid as hydrogen atoms containing high-energy electrons are removed.

(2) The pyruvic acid enters the mitochondrion, where citric acid cycle enzymes remove more hydrogen atoms and decompose it to $CO_2$. During glycolysis and the citric acid cycle, small amounts of ATP are formed.

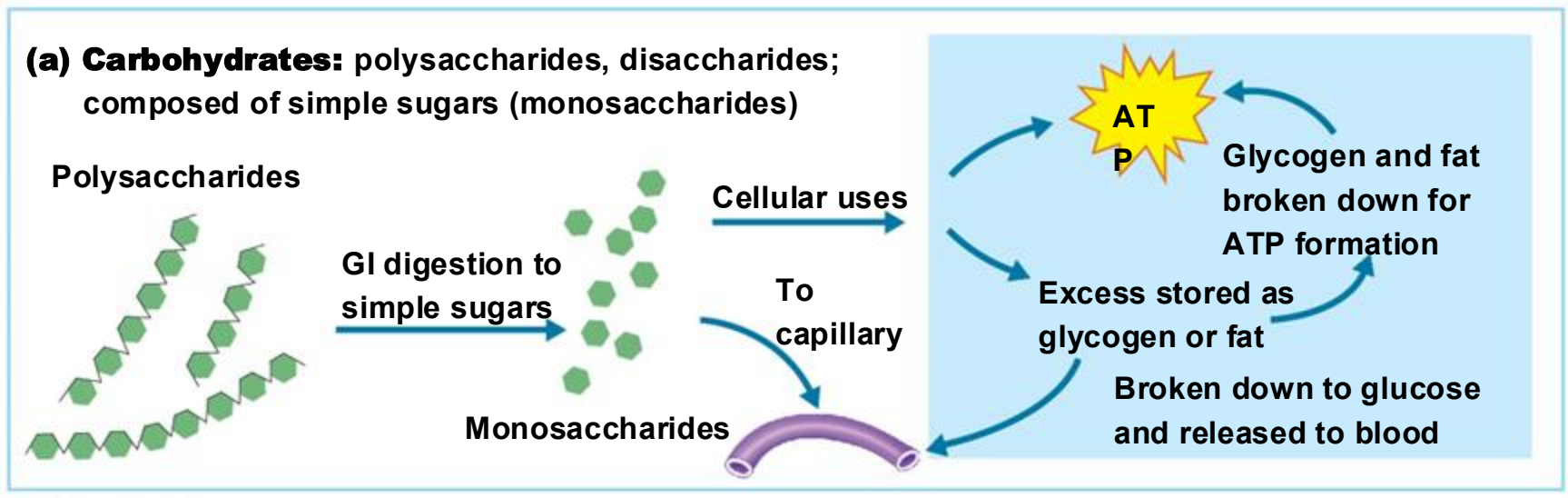
(3) Energy-rich electrons picked up by coenzymes are transferred to the electron transport chain, built into the cristae membrane. The electron transport chain carries out oxidative phosphorylation, which accounts for most of the ATP generated by cellular respiration, and finally unites the removed hydrogen with oxygen to form water.

# Carbohydrate Metabolism

- Hyperglycemia—excessively high levels of glucose in the blood
  - Excess glucose is stored in body cells as glycogen or converted to fat
- Hypoglycemia—low levels of glucose in the blood
  - Glycogenolysis, gluconeogenesis, and fat breakdown occur to restore normal blood glucose levels

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# ADDENDUM
# Figure 14.22a Metabolism by body cells.



# Fat Metabolism

- Fats
  - Insulate the body
  - Protect organs
  - Build some cell structures (membranes and myelin sheaths)
  - Provide reserve energy
- Excess dietary fat is stored in subcutaneous tissue and other fat depots

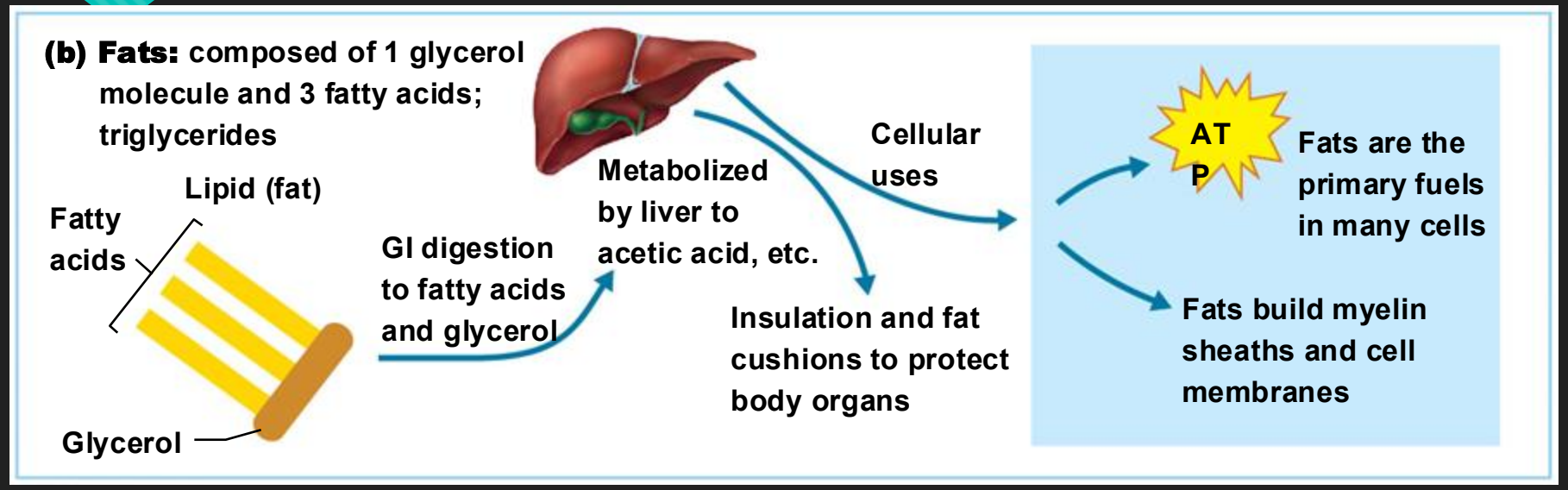
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# Fat Metabolism

- When carbohydrates are in limited supply, more fats are oxidized to produce ATP
  - Excessive fat breakdown causes blood to become acidic (acidosis or ketoacidosis)
    - Breath has a fruity odor
    - Common with:
      - "No carbohydrate" diets
      - Uncontrolled diabetes mellitus
      - Starvation

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# ADDENDUM
# Figure 14.22b Metabolism by body cells.

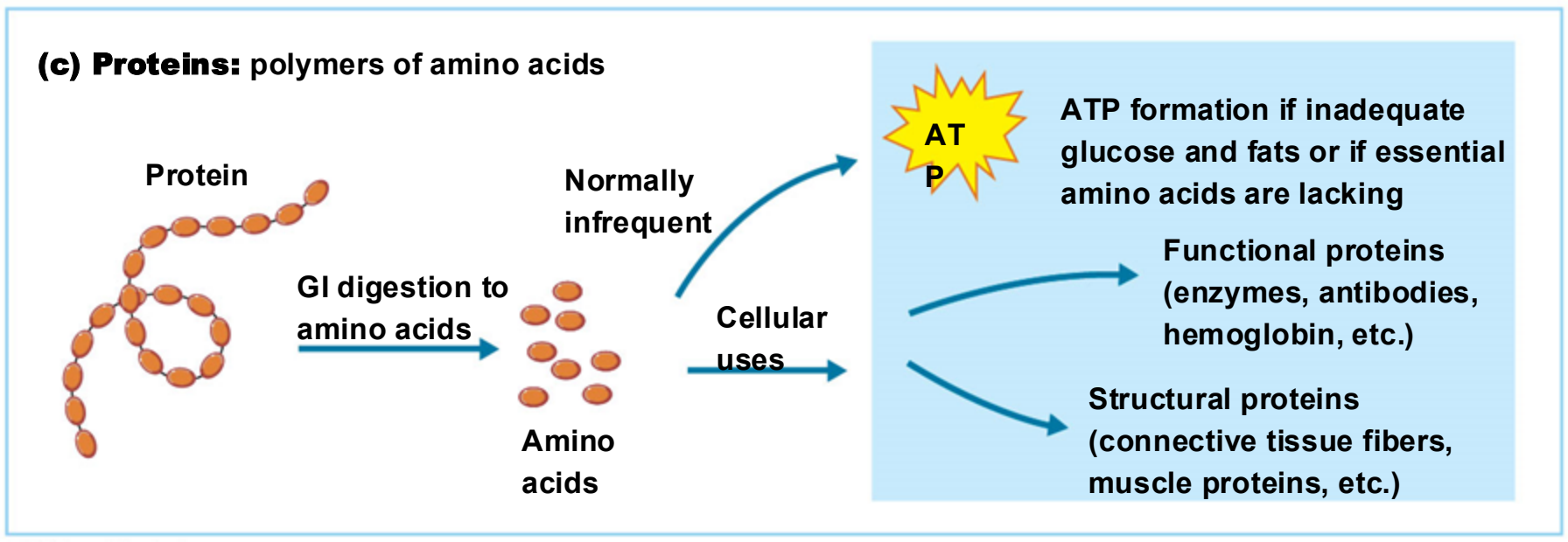


# Protein Metabolism

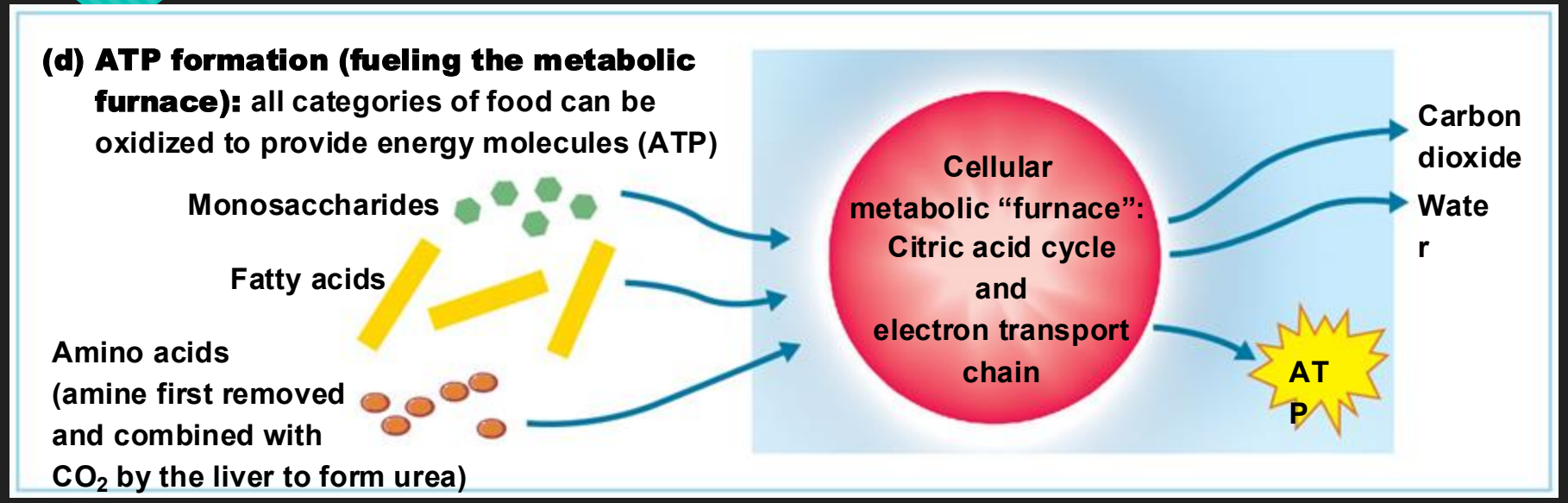
- Proteins form the bulk of cell structure and most functional molecules
- Proteins are carefully conserved by body cells
- Amino acids are actively taken up from blood by body cells
- Amino acids are oxidized to form ATP mainly when other fuel sources are not available
- Ammonia, released as amino acids are catabolized, is detoxified by liver cells that combine it with carbon dioxide to form urea

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# ADDENDUM
# Figure 14.22c Metabolism by body cells.



# ADDENDUM
# Figure 14.22d Metabolism by body cells.



# The Central Role of the Liver in Metabolism

- Liver is the body's key metabolic organ
- Roles in digestion
  - Manufactures bile
  - Detoxifies drugs and alcohol
  - Degrades hormones
  - Produces cholesterol, blood proteins (albumin and clotting proteins)
  - Plays a central role in metabolism
- Liver can regenerate if part of it is damaged or removed

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# The Central Role of the Liver in Metabolism

- To maintain homeostasis of blood glucose levels, the liver performs:
  - Glycogenesis—"glycogen formation"
    - Glucose molecules are converted to glycogen and stored in the liver
  - Glycogenolysis—"glycogen splitting"
    - Glucose is released from the liver after conversion from glycogen
  - Gluconeogenesis—"formation of new sugar"
    - Glucose is produced from fats and proteins

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# The Central Role of the Liver in Metabolism

- Fats and fatty acids are picked up by the liver
  - Some are oxidized to provide energy for liver cells
  - The rest are either stored or broken down into simpler compounds and released into the blood
- Blood proteins made by the liver are assembled from amino acids
  - Albumin is the most abundant protein in blood
- Liver cells detoxify ammonia
  - Ammonia is combined with carbon dioxide to form urea, which is flushed from the body in urine

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# The Central Role of the Liver in Metabolism

- Cholesterol metabolism and transport
  - Cholesterol is not used to make ATP
  - Functions of cholesterol:
    - Structural basis of steroid hormones and vitamin D
    - Building block of plasma membranes
  - Most cholesterol (85%) is produced in the liver; only 15% is from the diet
  - Cholesterol and fatty acids cannot freely circulate in the bloodstream
  - They are **transported by lipoproteins (lipid-protein complexes) known as LDLs and HDLs**

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# The Central Role of the Liver in Metabolism

- Cholesterol metabolism and transport
  - Low-density lipoproteins (LDLs) transport cholesterol *to* body cells
    - Rated "bad lipoproteins" since they can lead to atherosclerosis
  - High-density lipoproteins (HDLs) transport cholesterol *from* body cells to the liver
    - Rated "good lipoproteins" since cholesterol is destined for breakdown and elimination

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# Body Energy Balance

- Energy intake = Total energy output

  (heat + work + energy storage)

  - Energy intake is the energy liberated during food oxidation
    - Energy produced during glycolysis, citric acid cycle, and the electron transport chain
  - Energy output
    - Energy we lose as heat (60%)
    - Energy stored as fat or glycogen

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# Body Energy Balance

- Interference with the body's energy balance leads to:
  - Obesity
  - Malnutrition (leading to body wasting)





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# Body Energy Balance

- Metabolic rate and body heat production
  - Nutrients yield different amounts of energy
  - Energy value is measured in kilocalories (kcal)
    - Carbohydrates and proteins yield 4 kcal/gram
    - Fats yield 9 kcal/gram

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# Body Energy Balance

- Total metabolic rate (TMR)
  - total amount of kilocalories the body must consume to fuel ongoing activities
  - TMR increases dramatically with an increase in muscle activity
  - TMR must equal calories consumed to maintain homeostasis and maintain a constant weight

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# Developmental Aspects of the Digestive System and Metabolism

- The alimentary canal is present by the **fifth week of development**
- Digestive glands bud from the mucosa of the alimentary tube
- The developing fetus receives all nutrients through the placenta
- In newborns, feeding must be frequent, peristalsis is inefficient, and vomiting is common

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# Developmental Aspects of the Digestive System and Metabolism

- Newborn reflexes
  - Rooting reflex helps the infant find the nipple
  - Sucking reflex helps the infant hold on to the nipple and swallow
- Teething begins around age 6 months



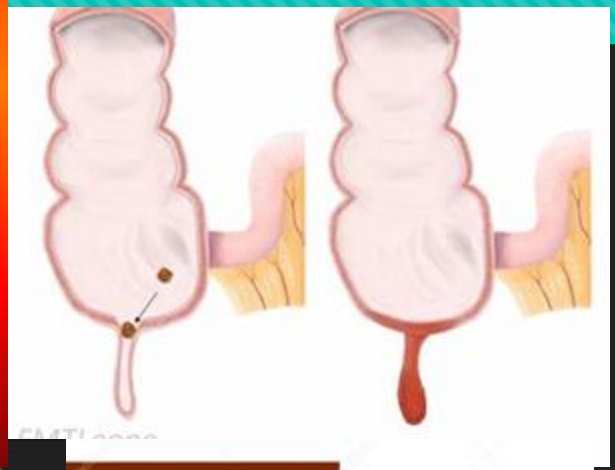


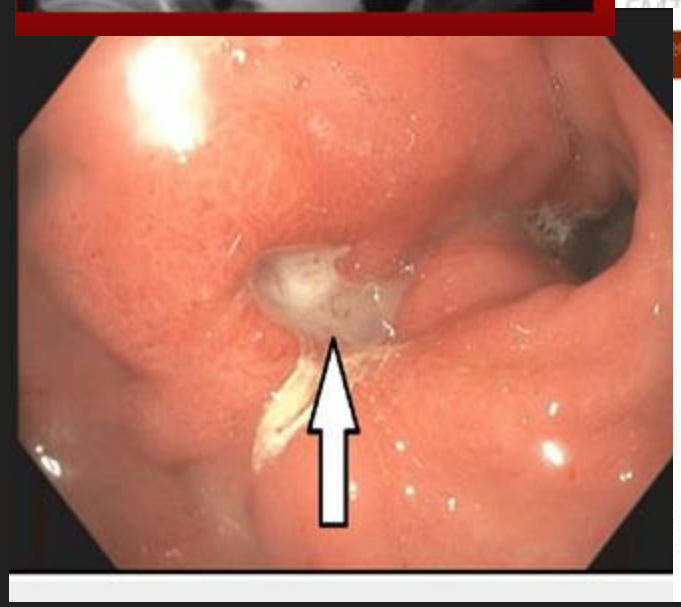

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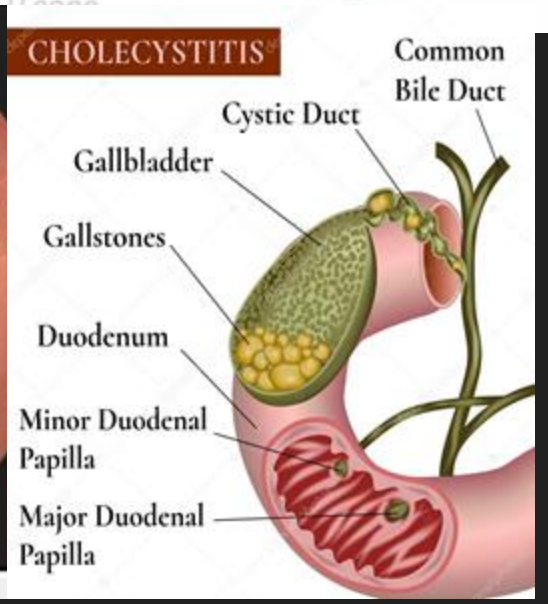
# Developmental Aspects of the Digestive System and Metabolism

- Problems of the digestive system
  - Gastroenteritis—inflammation of the gastrointestinal tract; can occur at any time
  - Appendicitis—inflammation of the appendix; common in adolescents
- Metabolism decreases with old age
- Middle-age digestive problems
  - Ulcers
  - Gallbladder problems





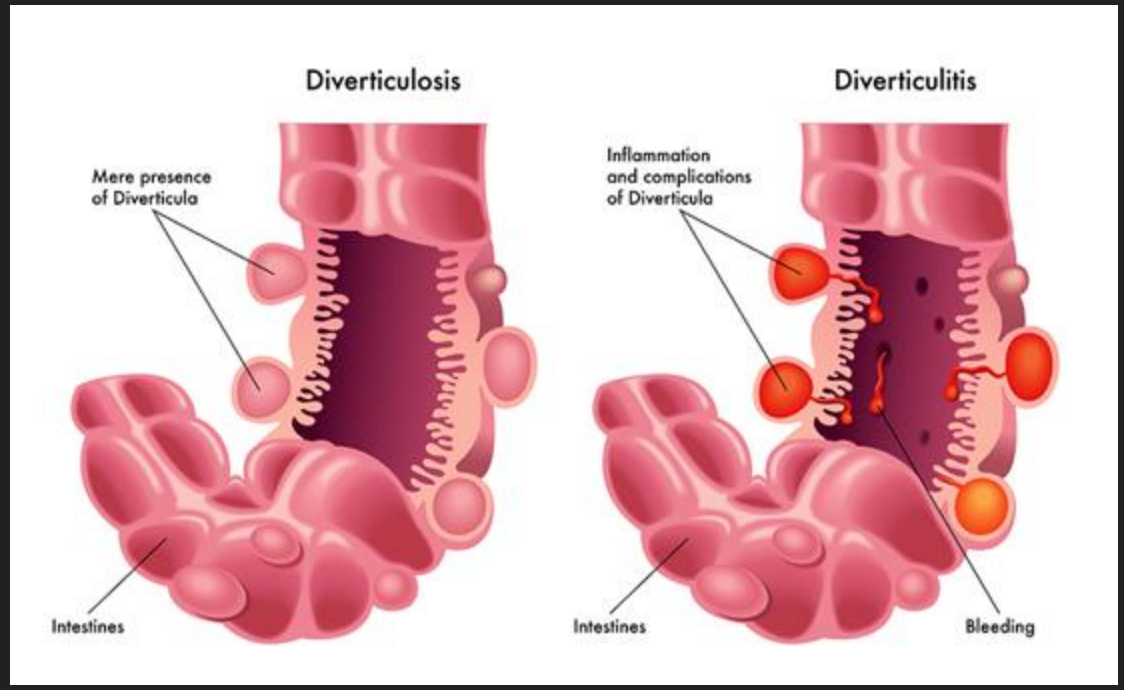





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# Developmental Aspects of the Digestive System and Metabolism

- Later middle-age problems
  - Obesity
  - Diabetes mellitus
- Activity of the digestive tract in old age
  - Fewer digestive juices
  - Peristalsis slows
  - Diverticulosis and gastrointestinal cancers are more common




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# THANK YOU FOR LISTENING!